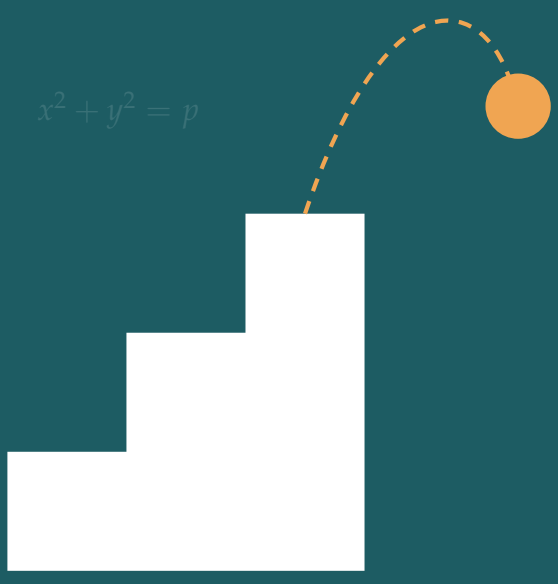
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1991 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1991 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1991 Worked Solutions

Worked solutions to all 48 questions of the 1991 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1991, Question 1

Pure • Trigonometric Identities

STEP I 1991, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q1

The key idea

The condition on the angles lets us remove one of them: since $\psi = \frac{1}{2}\pi - (\theta + \phi)$, we have $\sin\psi = \cos(\theta + \phi)$, and the identity becomes a statement about two free angles. After that, the products $\sin\theta\sin\phi$ and $\cos(\theta + \phi)$ combine to give $\cos(\theta - \phi)$, and the product $\cos(\theta + \phi)\cos(\theta - \phi)$ has a very tidy form. For the second part, the special choice of angle makes ψ equal to *half* of θ, which is where the information about $\frac{1}{10}\pi$ comes from.

Solution

The identity

Since $\theta + \phi + \psi = \frac{1}{2}\pi$,

$$\sin\psi = \sin(\tfrac{1}{2}\pi - (\theta + \phi)) = \cos(\theta + \phi).$$

So the left-hand side of the identity is

$$L = \sin^2\theta + \sin^2\phi + \cos(\theta + \phi)[\cos(\theta + \phi) + 2\sin\theta\sin\phi].$$

Inside the bracket, $\cos(\theta + \phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$, so

$$\cos(\theta + \phi) + 2\sin\theta\sin\phi = \cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta - \phi).$$

Now we need the product $\cos(\theta + \phi)\cos(\theta - \phi)$. Multiplying out the compound-angle formulae,

$$\begin{aligned}\cos(\theta + \phi)\cos(\theta - \phi) &= (\cos\theta\cos\phi - \sin\theta\sin\phi)(\cos\theta\cos\phi + \sin\theta\sin\phi)\\ &= \cos^2\theta\cos^2\phi - \sin^2\theta\sin^2\phi\\ &= \cos^2\theta\,(1 - \sin^2\phi) - (1 - \cos^2\theta)\sin^2\phi\\ &= \cos^2\theta - \sin^2\phi.\end{aligned}$$

Therefore

$$L = \sin^2\theta + \sin^2\phi + \cos^2\theta - \sin^2\phi = \sin^2\theta + \cos^2\theta = 1,$$

which is the required identity.

The cubic for $\sin\frac{1}{10}\pi$

Take $\theta = \phi = \frac{1}{5}\pi$. Then $\psi = \frac{1}{2}\pi - \frac{2}{5}\pi = \frac{1}{10}\pi$. Write

$$x = \sin\tfrac{1}{10}\pi, \qquad s = \sin\tfrac{1}{5}\pi.$$

The identity gives

$$2s^2 + x^2 + 2s^2x = 1, \qquad \text{that is,} \qquad 2s^2(1+x) = 1 - x^2. \tag{$*$}$$

Ask yourself Why not just use $\sin\psi = \cos 2\theta = 1 - 2s^2$ to eliminate s?

That relation holds for *every* θ with $\psi = \frac{1}{2}\pi - 2\theta$, so substituting $s^2 = \frac{1}{2}(1-x)$ into $(*)$ gives $(1-x)(1+x) = 1 - x^2$: true, but empty. We have to use something special about $\frac{1}{5}\pi$. The special fact is that $\frac{1}{5}\pi = 2\cdot\frac{1}{10}\pi$, so $\theta = 2\psi$, and the double-angle formula ties s to x:

$$s = \sin 2\psi = 2\sin\psi\cos\psi, \qquad \text{so} \qquad s^2 = 4x^2(1-x^2).$$

Substituting this into $(*)$,

$$8x^2(1-x^2)(1+x) = 1 - x^2.$$

Since $0 < \frac{1}{10}\pi < \frac{1}{2}\pi$, we have $0 < x < 1$, so $1 - x^2 \neq 0$ and we may divide by it:

$$8x^2(1+x) = 1, \qquad \text{that is,} \qquad 8x^3 + 8x^2 - 1 = 0.$$

So $\sin\frac{1}{10}\pi$ satisfies the given cubic, as required.

Checks

The identity can be tested with $\theta = \frac{1}{2}\pi$, $\phi = \psi = 0$: the left-hand side is $1+0+0+0 = 1$. With $\theta = \phi = \psi = \frac{1}{6}\pi$ it is $3\cdot\frac{1}{4} + 2\cdot\frac{1}{8} = 1$.

For the cubic, $\sin\frac{1}{10}\pi = \sin 18^\circ \approx 0.309017$, and $8x^3 + 8x^2 - 1$ evaluates to 0 to within rounding there. In fact the cubic factorises as $(2x+1)(4x^2+2x-1)$, with roots $-\frac{1}{2}$ and $\frac{1}{4}(-1 \pm \sqrt{5})$, and $\frac{1}{4}(\sqrt{5}-1) \approx 0.309017$ is the familiar exact value of $\sin 18^\circ$.

Common slips

- Eliminating s with $\sin\psi = \cos 2\theta$ only, and getting the identity $0 = 0$; see the box above.
- Dividing by $1 - x^2$ without saying why it is nonzero. Here $x = \sin\frac{1}{10}\pi$ lies strictly between 0 and 1.

Beyond STEP

The other two roots of the cubic are not accidents: $-\frac{1}{2} = \sin\left(-\frac{1}{6}\pi\right)$ and $\frac{1}{4}(-1-\sqrt{5}) = -\sin\frac{3}{10}\pi$. Equations like this one record the algebraic relations between values of sin and cos at rational multiples of π, which are best understood through roots of unity: $\cos\frac{1}{5}\pi$ is closely tied to the fifth roots of unity, the solutions of $z^5 = 1$. That is why constructing a regular pentagon with ruler and compasses comes down to $\sqrt{5}$. Needham's *Visual Complex Analysis* is a beautiful place to see these ideas geometrically.

WORKED SOLUTION

STEP I 1991, Question 2

Pure • Rates of Change

STEP I 1991, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q2

The key idea

Write the melting law in terms of the radius. For a sphere, $\frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t}$, and the surface area is $4\pi r^2$ too, so "volume lost at a rate proportional to surface area" says simply that the radius shrinks at a constant rate. Both snowballs share the constant, so both radii fall by the same amount in any time. Everything then becomes a function of one variable, the amount x by which each radius has shrunk.

Solution

The radii shrink at the same constant rate

Let a snowball have radius r, volume $V = \frac{4}{3}\pi r^3$ and surface area $S = 4\pi r^2$. The melting law is $\frac{\mathrm{d}V}{\mathrm{d}t} = -kS$ for a constant $k > 0$, the same for both snowballs. By the chain rule,

$$4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t} = -k \cdot 4\pi r^2 \quad \Longrightarrow \quad \frac{\mathrm{d}r}{\mathrm{d}t} = -k,$$

dividing by $4\pi r^2$, which is legitimate while the snowball still exists ($r > 0$). So after time t each radius has decreased by the same amount $x = kt$: the head has radius $2R - x$ and the body has radius $3R - x$, valid for $0 \le x \le 2R$ (the head disappears at $x = 2R$).

Volume at half height

Frosty's height is the sum of the two diameters, since the head rests on top of the body:

$$h = 2(2R - x) + 2(3R - x) = 10R - 4x.$$

Initially $h = 10R$, so half height means $10R - 4x = 5R$, that is $x = \frac{5}{4}R$. This is less than $2R$, so the head is still there, and the radii are then $\frac{3}{4}R$ and $\frac{7}{4}R$. The factor $\frac{4}{3}\pi$ cancels

in the ratio, so

$$\frac{V}{V_0} = \frac{\left(\frac{3}{4}\right)^3 + \left(\frac{7}{4}\right)^3}{2^3 + 3^3} = \frac{(27+343)/64}{35} = \frac{370}{2240} = \frac{37}{224}.$$

So at half his height Frosty has only $\frac{37}{224} \approx 0.165$ of his original volume: volume scales like length cubed.

The maximum of $\mathrm{d}V/\mathrm{d}S$

Write $a = 2R - x$ and $b = 3R - x$. For $0 \le x < 2R$,

$$V = \tfrac{4}{3}\pi(a^3 + b^3), \qquad S = 4\pi(a^2 + b^2),$$

and since $\frac{\mathrm{d}a}{\mathrm{d}x} = \frac{\mathrm{d}b}{\mathrm{d}x} = -1$,

$$\frac{\mathrm{d}V}{\mathrm{d}x} = -4\pi(a^2 + b^2), \qquad \frac{\mathrm{d}S}{\mathrm{d}x} = -8\pi(a + b).$$

Ask yourself V and S are both functions of time (or of x). What does $\mathrm{d}V/\mathrm{d}S$ mean?

As x increases, S strictly decreases (because $\frac{\mathrm{d}S}{\mathrm{d}x} < 0$), so V can be regarded as a function of S. By the chain rule $\frac{\mathrm{d}V}{\mathrm{d}S} = \frac{\mathrm{d}V}{\mathrm{d}x} \Big/ \frac{\mathrm{d}S}{\mathrm{d}x}$:

$$\frac{\mathrm{d}V}{\mathrm{d}S} = \frac{a^2 + b^2}{2(a+b)}.$$

Now use $b - a = R$, which never changes. Put $s = a + b = 5R - 2x$. Then $a^2 + b^2 = \frac{1}{2}\left((a+b)^2 + (b-a)^2\right) = \frac{1}{2}(s^2 + R^2)$, so

$$\frac{\mathrm{d}V}{\mathrm{d}S} = \frac{s^2 + R^2}{4s} = \frac{s}{4} + \frac{R^2}{4s} = f(s), \text{ say.}$$

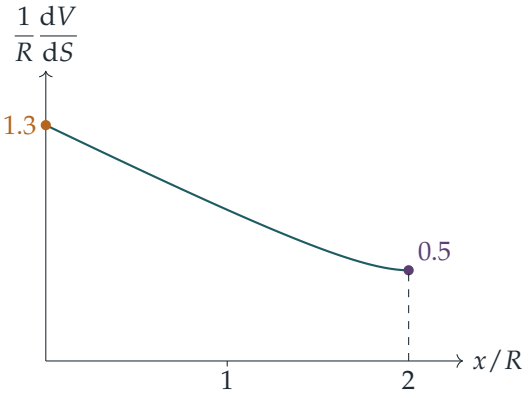
As x runs from 0 to $2R$, s runs from $5R$ down to R. Now

$$f'(s) = \frac{1}{4} - \frac{R^2}{4s^2} = \frac{s^2 - R^2}{4s^2} > 0 \quad \text{for } s > R,$$

so f is increasing on $[R, 5R]$, and its largest value is at $s = 5R$, that is, at the very start ($x = 0$):

$$\left(\frac{\mathrm{d}V}{\mathrm{d}S}\right)_{\max} = \frac{5R}{4} + \frac{R}{20} = \frac{13R}{10}.$$

(Directly: $\frac{4R^2 + 9R^2}{2 \cdot 5R} = \frac{13}{10}R$.) The value then falls steadily to $f(R) = \frac{1}{2}R$ just as the head vanishes.



Checks

- A step-by-step simulation of $\frac{dV}{dt} = -kS$ for each ball (with $R = k = 1$) reaches half height at $t \approx 1.25$, with volume ratio 0.16518, matching $\frac{37}{224} = 0.165178\ldots$.
- Finite differences of the simulated V and S agree with $\frac{a^2+b^2}{2(a+b)}$ throughout and never exceed $1.3R$.
- Units: dV/dS is a volume over an area, so a length, as $\frac{13}{10}R$ is.

Common slips

- Taking the height as the sum of the radii, not of the diameters. (Here it happens not to change x, but it is wrong.)
- Computing V/S instead of dV/dS; for a single sphere $V/S = r/3$ but $dV/dS = r/2$.
- Hunting for a stationary point: the turning point of f is at $s = R$, the endpoint, so the maximum is at the other end of the interval.

Beyond STEP

The heart of the question is that a "rate proportional to surface area" law turns into a constant rate for the radius, a first example of how geometry can make a differential equation trivial. The second half is a parametric derivative, dV/dS along a curve $(S(t), V(t))$, and the care needed to find a maximum on a closed interval, where it may sit at an endpoint rather than at a stationary point; Bartle and Sherbert's *Introduction to Real Analysis* explains why such maxima must exist. For modelling with differential equations, try Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP I 1991, Question 3

Pure • Complex Geometric Series

STEP I 1991, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q3

The key idea

In the Argand diagram, multiplying by $re^{i\phi}$ scales a vector by r and turns it anticlockwise through ϕ. Each segment of the path is the previous one scaled and turned by the same amount, so each segment vector is the previous one times a fixed complex number. The segments then form a geometric progression, and adding them up gives the position of every vertex.

Solution

The ratio of successive segments

The vector from P_n to P_{n+1} is represented by $z_{n+1} - z_n$, and the vector from P_{n-1} to P_n by $z_n - z_{n-1}$. For the quotient of two complex numbers, moduli divide and arguments subtract:

$$\left|\frac{z_{n+1} - z_n}{z_n - z_{n-1}}\right| = \frac{P_nP_{n+1}}{P_{n-1}P_n} = d, \qquad \arg\frac{z_{n+1} - z_n}{z_n - z_{n-1}} = \arg(z_{n+1} - z_n) - \arg(z_n - z_{n-1}) = \theta,$$

because the second segment is d times as long as the first, and the anticlockwise angle from the direction of one directed segment to the direction of the next is θ. (The argument is only defined up to multiples of 2π, which doesn't affect the complex number.) Hence, for every $n \geq 2$,

$$\frac{z_{n+1} - z_n}{z_n - z_{n-1}} = d\,e^{i\theta} = d(\cos\theta + i\sin\theta).$$

A formula for z_{n+1}

Write $w = de^{i\theta}$. The first segment is $z_2 - z_1 = 1$, and each later segment is w times the one before, so by induction

$$z_{k+1} - z_k = w^{k-1} \qquad (k \geq 1).$$

Adding these for $k = 1, 2, \ldots, n$, the left-hand side telescopes to $z_{n+1} - z_1 = z_{n+1}$:

$$z_{n+1} = 1 + w + w^2 + \cdots + w^{n-1} = \frac{w^n - 1}{w - 1} = \frac{d^n \mathrm{e}^{\mathrm{i}n\theta} - 1}{d\mathrm{e}^{\mathrm{i}\theta} - 1}.$$

Summing the geometric series needs $w \neq 1$, and this is where $d \neq 1$ is used, since $|w| = d$.

The case $\theta = \frac{1}{3}\pi$, $d = 2$

Now $w = 2\mathrm{e}^{\mathrm{i}\pi/3} = 1 + \mathrm{i}\sqrt{3}$, so the denominator is very simple: $w - 1 = \mathrm{i}\sqrt{3}$. Since $\frac{1}{\mathrm{i}} = -\mathrm{i}$,

$$z_{n+1} = \frac{w^n - 1}{\mathrm{i}\sqrt{3}} = -\frac{\mathrm{i}}{\sqrt{3}}\left(2^n \cos\tfrac{n\pi}{3} - 1 + \mathrm{i}\, 2^n \sin\tfrac{n\pi}{3}\right),$$

so that

$$\operatorname{Re} z_{n+1} = \frac{2^n \sin\frac{n\pi}{3}}{\sqrt{3}}, \qquad \operatorname{Im} z_{n+1} = \frac{1 - 2^n \cos\frac{n\pi}{3}}{\sqrt{3}}.$$

Ask yourself How does the sign of $\operatorname{Im} z_{n+1}$ tell us that the path crosses the real axis?

The path is joined up: it is a continuous chain of segments. If one vertex lies strictly below the real axis and a later vertex lies strictly above it (or the other way round), then the stretch of path between them must cross the real axis. So we look for vertices on opposite sides, again and again.

The value of $\cos\frac{n\pi}{3}$ repeats with period 6. Take any integer $k \geq 1$.

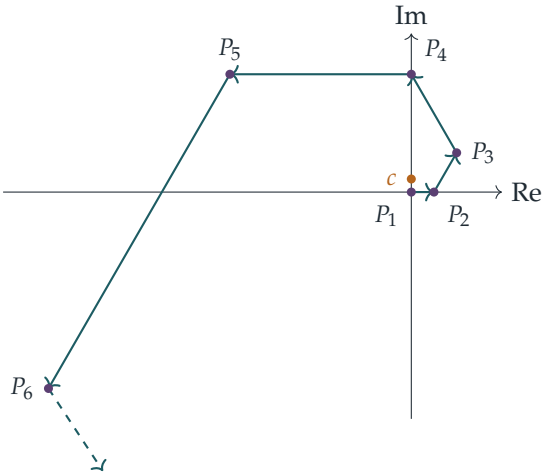
- For $n = 6k$, $\cos\frac{n\pi}{3} = \cos 2k\pi = 1$, so $\operatorname{Im} z_{6k+1} = \dfrac{1 - 2^{6k}}{\sqrt{3}} < 0$, since $2^{6k} > 1$.
- For $n = 6k + 3$, $\cos\frac{n\pi}{3} = \cos(2k\pi + \pi) = -1$, so $\operatorname{Im} z_{6k+4} = \dfrac{1 + 2^{6k+3}}{\sqrt{3}} > 0$.

Thus P_{6k+1} is below the real axis and P_{6k+4} is above it, for every $k \geq 1$. So the part of the path from P_{6k+1} to P_{6k+4} crosses the real axis, going upwards, for each $k = 1, 2, 3, \ldots$. These are different parts of the path, one for each k, so the path crosses the real axis infinitely often. (By the same argument it also crosses downwards between P_{6k+4} and P_{6k+7}.) This completes the proof.

In fact the vertices go round in blocks of three: the sign pattern of $\operatorname{Im} z_1, \operatorname{Im} z_2, \ldots$ is $0, 0, +, +, +, -, -, -, +, +, +, \ldots$. There is a neat geometric reason. Writing

$$z_{n+1} = \frac{\mathrm{i}}{\sqrt{3}} - \frac{\mathrm{i}}{\sqrt{3}}\, w^n,$$

we see that P_{n+1} is at distance $2^n/\sqrt{3}$ from the fixed point $c = \mathrm{i}/\sqrt{3}$, and each step turns it by 60° about c. The vertices lie on a spiral round c that doubles in size every step, and c sits just above the real axis, so the spiral keeps sweeping across the axis.



The figure shows $P_1, \ldots, P_6$; the next vertex is $P_7 = -21\sqrt{3}\,\mathrm{i}$, far below. The path first crosses the real axis on P_5P_6, then stays below it until it crosses upwards on P_8P_9, from $P_8 = 64 - 21\sqrt{3}\,\mathrm{i}$ to $P_9 = 128 + 43\sqrt{3}\,\mathrm{i}$.

Checks

- Directly from the geometry: $z_3 = 1 + w = 2 + \sqrt{3}\,\mathrm{i}$, and the formula gives $\operatorname{Im} z_3 = (1 - 4\cos\frac{2\pi}{3})/\sqrt{3} = 3/\sqrt{3} = \sqrt{3}$. Similarly, adding $w^2 = -2 + 2\sqrt{3}\,\mathrm{i}$ and $w^3 = -8$ gives $z_5 = -8 + 3\sqrt{3}\,\mathrm{i}$, and the formulas with $n = 4$ give $\operatorname{Re} z_5 = 16\sin\frac{4\pi}{3}/\sqrt{3} = -8$ and $\operatorname{Im} z_5 = (1+8)/\sqrt{3} = 3\sqrt{3}$.
- With $d = 1$ the vertices would lie on a regular polygon and the path would close up; the geometric-series formula fails exactly then, as it should.
- We built the path numerically from lengths and turning angles (random d, θ and starting point) and confirmed the ratio $d\mathrm{e}^{\mathrm{i}\theta}$, the closed form for z_{n+1}, and, for $\theta = \frac{1}{3}\pi$, $d = 2$, the sign pattern above: there are 13 sign changes of $\operatorname{Im} z$ among $z_1, \ldots, z_{41}$.

Common slips

- Writing the ratio as $\frac{1}{d}\mathrm{e}^{-\mathrm{i}\theta}$: the *later* segment is on top, so it is the one that is d times longer and turned through $+\theta$.
- An off-by-one error in the sum: there are n segments from P_1 to P_{n+1}, so the series runs up to w^{n-1}.
- Claiming a crossing from one vertex on the axis: you need vertices *strictly* on opposite sides (here z_1 and z_2 lie on the axis, so start from $k \geq 1$).

Beyond STEP

The rule $z_{n+1} - z_n = w(z_n - z_{n-1})$ is a linear recurrence, and the map $z \mapsto c + w(z - c)$ that carries each vertex to the next is a *spiral similarity* about its fixed point c: a rotation combined with an enlargement. Such maps are the simplest examples of Möbius transformations, and iterating them produces the equiangular (logarithmic) spirals seen in shells. Needham's *Visual Complex Analysis* develops this geometric way of thinking about complex multiplication beautifully.

WORKED SOLUTION

STEP I 1991, Question 4

Pure • Optimisation in Geometry

STEP I 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q4

The key idea

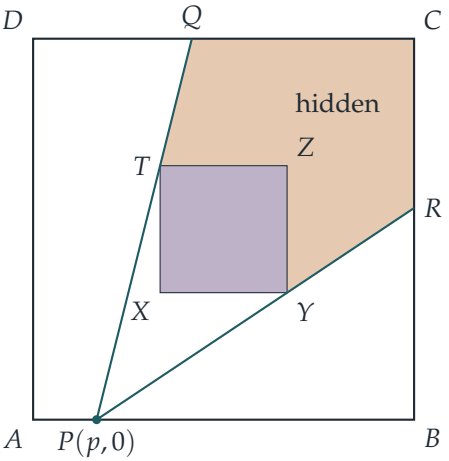
The guard sees every point of the courtyard except those hidden behind the tower, so following the hint we work out the area of the tower's "shadow" as seen from his position $P = (p, 0)$. The shadow is bounded by the two rays from P that just graze the tower. Find where those rays meet the walls, compute the shadow's area as a polygon, and the problem becomes maximising a function of one variable on $[0, 4]$. Remember to check the endpoints.

Solution

Setting up

The courtyard is the big square minus the tower, with area $144 - 16 = 128$. Let the guard be at $P = (p, 0)$ with $0 \leq p \leq 4$, and let $H(p)$ be the area he *cannot* see. The visible area is $128 - H(p)$, so the visible area is smallest when H is largest, and the other way round.

Because $p \leq 4$, P lies below the line $y = 4$ and to the left of (or on) the line $x = 4$. So from P the tower presents its bottom face XY and its left face XT, and its outline is fixed by the corners $T(4, 8)$ and $Y(8, 4)$. The corner $Z(8, 8)$ lies inside the angle TPY and is hidden. The hidden region lies beyond the path $T \to Z \to Y$, between the rays PT and PY.



Where the grazing rays meet the walls

The ray from P through $T(4,8)$ rises 8 for every $4-p$ across, so it reaches $y=12$ at

$$x = p + \tfrac{12}{8}(4-p) = 6 - \tfrac{p}{2}, \qquad \text{giving } Q = \left(6-\tfrac{p}{2},\ 12\right).$$

Since $4 \le 6 - \frac{p}{2} \le 6$, this point really is on the top wall DC.

The ray from P through $Y(8,4)$ rises 4 for every $8-p$ across, so it reaches $x=12$ at

$$y = \frac{4(12-p)}{8-p}, \qquad \text{giving } R = \left(12,\ \frac{4(12-p)}{8-p}\right).$$

This value increases from 6 to 8 as p goes from 0 to 4, so R is on the right wall BC.

The hidden area

The hidden region is the polygon $TQCRYZ$. It is easiest to find its area as

$$H = [\,PRCQ\,] - [\,PYZT\,],$$

where $[\cdots]$ denotes area: the whole wedge between the two rays, out to the walls, minus the part of the wedge in front of the far faces of the tower (the visible ground in front of the tower together with the tower itself).

Using the shoelace formula for $P(p,0)$, $R(12,q)$, $C(12,12)$, $Q(r,12)$, with $q = \frac{4(12-p)}{8-p}$ and $r = 6 - \frac{p}{2}$:

$$\begin{aligned}[\,PRCQ\,] &= \tfrac{1}{2}\big(pq + (144-12q) + (144-12r) - 12p\big) \\ &= \tfrac{1}{2}\big(288 - 12p - 12r - (12-p)q\big) = 108 - 3p - \frac{2(12-p)^2}{8-p},\end{aligned}$$

using $12r = 72 - 6p$. Similarly, for $P(p,0)$, $Y(8,4)$, $Z(8,8)$, $T(4,8)$:

$$[\,PYZT\,] = \tfrac{1}{2}\big(4p + 32 + 32 - 8p\big) = 32 - 2p.$$

Therefore

$$H(p) = 76 - p - \frac{2(12-p)^2}{8-p}.$$

Ask yourself Is there a substitution that makes this easy to optimise?

The awkward denominator suggests putting $u = 8 - p$, so that u runs over $[4, 8]$ and $12 - p = u + 4$. Then

$$H = 76 - (8-u) - \frac{2(u+4)^2}{u} = 68 + u - \frac{2u^2 + 16u + 32}{u} = 52 - \left(u + \frac{32}{u}\right).$$

Part (i): smallest visible area

We need H as large as possible, which means $u + \frac{32}{u}$ as small as possible. By the AM–GM inequality, for $u > 0$,

$$u + \frac{32}{u} \;\geq\; 2\sqrt{32} = 8\sqrt{2},$$

with equality exactly when $u = \frac{32}{u}$, that is $u = 4\sqrt{2}$. This lies in $[4, 8]$ (as $4\sqrt{2} \approx 5.66$), so it is allowed. Then $p = 8 - u$, and the visible area is smallest when the guard is

$$\boxed{p = 8 - 4\sqrt{2} \;\approx 2.34 \text{ units from } A}$$

At this point he cannot see $52 - 8\sqrt{2}$, so he sees $76 + 8\sqrt{2} \approx 87.3$ square units.

Part (ii): largest visible area

Now we need H as small as possible, so $f(u) = u + \frac{32}{u}$ as large as possible on $[4, 8]$. Since $f'(u) = 1 - \frac{32}{u^2}$, f is decreasing on $[4, 4\sqrt{2}]$ and increasing on $[4\sqrt{2}, 8]$, so its maximum is at an endpoint. We have

$$f(4) = 4 + 8 = 12, \qquad f(8) = 8 + 4 = 12,$$

which are equal. So H takes its least value, $52 - 12 = 40$, at both $u = 8$ and $u = 4$, that is at $p = 0$ and $p = 4$. The visible area is largest, $128 - 40 = 88$ square units, when the guard stands

$$\boxed{\text{at } A \ (p = 0) \text{ or at } E \ (p = 4)}$$

and at no other point of AE.

Checks

- At $p = 0$ we can do the shadow by hand: the rays hit the walls at $(6, 12)$ and $(12, 6)$, and the hidden hexagon has area 40, matching $H(0)$.

- At $p = 4$ the ray through T runs straight up the line $x = 4$, and $H(4) = 40$ again. The equal endpoint values are a good sign: the formula is not lopsided.
- We counted hidden points on a 600×600 grid, testing whether the line of sight from P passes through the tower. At $p = 8 - 4\sqrt{2}$ this gives 40.687, against $52 - 8\sqrt{2} = 40.686$. At $p = 1$ it gives 40.429, against $H(1) = \frac{283}{7} \approx 40.429$.

Common slips

- Using Z instead of T or Y as a grazing point. From below-left, the tower's outline is fixed by the two corners T and Y, not by the far corner.
- Stopping at the stationary point for part (ii). The stationary point gives the *minimum* visible area, so the maximum must come from the endpoints, and here there are two.

Beyond STEP

This is a small example of a *visibility* problem, the kind studied in computational geometry (the "art gallery problem" asks how many guards are needed to watch every point of a polygonal room). The optimisation step shows a basic fact from analysis: a continuous function on a closed interval attains its maximum and minimum, either at a stationary point or at an endpoint. That is the extreme value theorem, proved carefully in Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1991, Question 5

Pure • Regular Sets of Vectors

STEP I 1991, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q5

The key idea

Condition (ii) says the vectors add up to $\mathbf{0}$, and the cheapest way to turn a vector equation into a number is to take a dot product with one of the vectors, which condition (i) tells us how to evaluate. After that, the whole question is about counting: a vector whose dot products with some fixed vectors are all prescribed has very few possible positions, so there's only room for a small number of vectors in the set.

Solution

The identity $(n-1)\beta = -\alpha^2$

By (ii), $\mathbf{a}_1 + \mathbf{a}_2 + \cdots + \mathbf{a}_n = \mathbf{0}$. Take the dot product of both sides with $\mathbf{a}_1$. By (i), the term $\mathbf{a}_1 \cdot \mathbf{a}_1$ is α^2 and each of the other $n-1$ terms $\mathbf{a}_1 \cdot \mathbf{a}_j$ $(j \neq 1)$ is β, so

$$\alpha^2 + (n-1)\beta = 0, \qquad \text{that is,} \qquad (n-1)\beta = -\alpha^2,$$

as required. In particular $\beta = -\alpha^2/(n-1) < 0$.

Two dimensions

Now $\alpha^2 = \mathbf{a}_1 \cdot \mathbf{a}_1 = 1$, so every $\mathbf{a}_i$ is a unit vector, and

$$\beta = -\frac{1}{n-1}.$$

Write $\mathbf{a}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix}$ for $i \geq 2$. The condition $\mathbf{a}_1 \cdot \mathbf{a}_i = \beta$ says $x_i = \beta$, and then $|\mathbf{a}_i| = 1$ forces $y_i = \pm\sqrt{1-\beta^2}$. So every vector other than $\mathbf{a}_1$ is one of

$$\begin{pmatrix} \beta \\ \sqrt{1-\beta^2} \end{pmatrix}, \qquad \begin{pmatrix} \beta \\ -\sqrt{1-\beta^2} \end{pmatrix}.$$

Ask yourself How does knowing the candidates limit n?

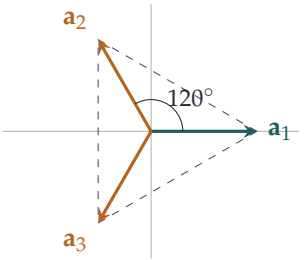
The vectors in the set are distinct, so $\mathbf{a}_2, \ldots, \mathbf{a}_n$ are $n-1$ *different* candidates. There are at most two candidates, so $n-1 \leq 2$, i.e. $n \leq 3$. Since $n \geq 2$, either $n = 2$ or $n = 3$.

$n = 2$. Then $\beta = -1$, so the two candidates coincide and $\mathbf{a}_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$. This works: $\mathbf{a}_1 \cdot \mathbf{a}_2 = -1 = \beta$ and $\mathbf{a}_1 + \mathbf{a}_2 = \mathbf{0}$.

$n = 3$. Then $\beta = -\frac{1}{2}$ and $\sqrt{1-\beta^2} = \frac{\sqrt{3}}{2}$, and $\mathbf{a}_2, \mathbf{a}_3$ must be the two different candidates:

$$\begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

(in either order). Check: $\mathbf{a}_2 \cdot \mathbf{a}_3 = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2} = \beta$, and the three vectors sum to $\mathbf{0}$. They point to the vertices of an equilateral triangle centred at the origin.



(Another way to pin down β: for $n \geq 3$ we have $-1 < \beta < 0$, so the two candidates really are different and $\mathbf{a}_2, \mathbf{a}_3$ must be both of them. Then $\mathbf{a}_2 \cdot \mathbf{a}_3 = \beta^2 - (1-\beta^2) = 2\beta^2 - 1$, and setting this equal to β gives $(2\beta+1)(\beta-1) = 0$. So $\beta = -\frac{1}{2}$, which again means $n = 3$.)

Three dimensions

Again $\alpha = 1$ and $\beta = -1/(n-1)$. The case $n = 2$ is as before: $\beta = -1$, and the only unit vector with $\mathbf{a}_1 \cdot \mathbf{a}_2 = -1$ is $\mathbf{a}_2 = -\mathbf{a}_1 = (-1, 0, 0)^{\mathrm{T}}$, which lies in the x-y plane. So $\{\mathbf{a}_1, -\mathbf{a}_1\}$ is one answer.

Now suppose $n \geq 3$, so $-\frac{1}{2} \leq \beta < 0$. Since $\mathbf{a}_2$ lies in the x-y plane, the same argument as in two dimensions gives

$$\mathbf{a}_2 = \begin{pmatrix} \beta \\ \varepsilon\sqrt{1-\beta^2} \\ 0 \end{pmatrix}, \qquad \varepsilon = \pm 1,$$

and the y-component $y_2 = \varepsilon\sqrt{1-\beta^2}$ is non-zero because $\beta \neq \pm 1$.

Ask yourself What do the conditions say about $\mathbf{a}_i$ for $i \geq 3$?

Write $\mathbf{a}_i = (x_i, y_i, z_i)^{\mathrm{T}}$ for $i \geq 3$. Then

$$\begin{aligned}
\mathbf{a}_1 \cdot \mathbf{a}_i = \beta &\implies x_i = \beta, \\
\mathbf{a}_2 \cdot \mathbf{a}_i = \beta &\implies \beta^2 + y_2 y_i = \beta \implies y_i = \frac{\beta(1-\beta)}{y_2}, \\
|\mathbf{a}_i| = 1 &\implies z_i^2 = 1 - \beta^2 - y_i^2.
\end{aligned}$$

So x_i and y_i are the same for every $i \geq 3$, and z_i can only be one of the two square roots. As the vectors are distinct, there are at most two of $\mathbf{a}_3, \ldots, \mathbf{a}_n$, so $n - 2 \leq 2$ and $n \leq 4$.

$n = 3$. By (ii), $\mathbf{a}_3 = -\mathbf{a}_1 - \mathbf{a}_2$, which lies in the x-y plane. So the set is a regular set in the plane $z = 0$, and by the two-dimensional result it is

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \\ 0 \end{pmatrix},$$

with $\mathbf{a}_2$ either of the last two. (The formulas above agree: $\beta = -\frac{1}{2}$ gives $y_3 = -y_2$ and $z_3^2 = 1 - \frac{1}{4} - \frac{3}{4} = 0$.)

$n = 4$. Now $\beta = -\frac{1}{3}$, so $\sqrt{1-\beta^2} = \frac{2\sqrt{2}}{3}$ and $y_2 = \varepsilon\frac{2\sqrt{2}}{3}$. Then

$$y_i = \frac{(-\frac{1}{3})(\frac{4}{3})}{\varepsilon\frac{2\sqrt{2}}{3}} = -\varepsilon\frac{2}{3\sqrt{2}} = -\varepsilon\frac{\sqrt{2}}{3}, \qquad z_i^2 = 1 - \frac{1}{9} - \frac{2}{9} = \frac{2}{3}.$$

Since $\mathbf{a}_3 \neq \mathbf{a}_4$, they must take the two opposite values $z = \pm\sqrt{2/3} = \pm\frac{\sqrt{6}}{3}$. This gives

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} -\frac{1}{3} \\ \varepsilon\frac{2\sqrt{2}}{3} \\ 0 \end{pmatrix}, \quad \mathbf{a}_3, \mathbf{a}_4 = \begin{pmatrix} -\frac{1}{3} \\ -\varepsilon\frac{\sqrt{2}}{3} \\ \pm\frac{\sqrt{6}}{3} \end{pmatrix},$$

one set for each choice $\varepsilon = \pm 1$. We have only used the conditions involving $\mathbf{a}_1$ and $\mathbf{a}_2$, so we must still check the rest. First, $\mathbf{a}_3 \cdot \mathbf{a}_4 = \frac{1}{9} + \frac{2}{9} - \frac{6}{9} = -\frac{1}{3} = \beta$. Second, the sum: the x-components give $1 - 3 \cdot \frac{1}{3} = 0$, the y-components give $\varepsilon\left(\frac{2\sqrt{2}}{3} - \frac{\sqrt{2}}{3} - \frac{\sqrt{2}}{3}\right) = 0$, and the z-components cancel. So both sets are regular. Geometrically, the four vectors point to the vertices of a regular tetrahedron centred at the origin.

Summary. The regular sets are: $\{\mathbf{a}_1, -\mathbf{a}_1\}$ ($n = 2$); the equilateral triangle in the x-y plane ($n = 3$); and the two regular tetrahedra above ($n = 4$, $\varepsilon = \pm 1$, mirror images in the x-z plane). No regular set has $n \geq 5$.

Checks

In each set the vectors should sum to zero and all pairwise dot products should be equal. For the tetrahedron, the angle between any two vectors is $\cos^{-1}(-\frac{1}{3}) \approx 109.47°$, the familiar bond angle of methane, and the identity gives $(4-1)(-\frac{1}{3}) = -1 = -\alpha^2$. We checked all five sets numerically (Gram matrix and centroid to 10^{-12}), and a brute-force search in the plane found regular sets only for $n = 2$ and $n = 3$, for n up to 11.

Common slips

- Forgetting that the vectors must be *distinct*: this is the condition that turns "at most two candidates" into a bound on n.
- In the $n = 4$ case, finding $\mathbf{a}_3, \mathbf{a}_4$ from their dot products with $\mathbf{a}_1, \mathbf{a}_2$ and then not checking $\mathbf{a}_3 \cdot \mathbf{a}_4 = \beta$ and the centroid condition.
- Missing the $n = 2$ case or the choice of sign ε in three dimensions.

Beyond STEP

These sets are the vertices of the regular simplex: n points in $(n-1)$-dimensional space, all the same distance apart. The matrix of dot products $G = (\mathbf{a}_i \cdot \mathbf{a}_j)$ (the *Gram matrix*) is $(\alpha^2 - \beta)I + \beta J$, where J is the all-ones matrix. Its rank is $n - 1$, and the rank of a Gram matrix can't exceed the dimension of the space, which gives $n \leq d + 1$ in d dimensions at once. Strang's *Introduction to Linear Algebra* covers rank, dot products and Gram matrices.

WORKED SOLUTION

STEP I 1991, Question 6

Pure • Stationary Points and Improper Integrals

STEP I 1991, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 3.1, *Spot the Flaw*: https://beyondstep.keithmcnulty.org/ch03/spot-the-flaw.html

The key idea

Don't trust anything: redo each calculation yourself and ask, at each "hence" or "thus", whether the step really follows. In (i) the derivative has been computed wrongly, and even with the right stationary points the reasoning about maxima, minima and global bounds would be faulty. In (ii) the antiderivative is wrong, and, more seriously, the integrand blows up inside the interval, so the fundamental theorem of calculus cannot be applied at all.

Solution

Part (i)

Step 1: the stationary points. By the quotient rule,

$$g'(x) = \frac{6x^2(x^4+4) - 4x^3(2x^3+3)}{(x^4+4)^2} = \frac{-2x^6 - 12x^3 + 24x^2}{(x^4+4)^2} = \frac{-2x^2\,(x^4+6x-12)}{(x^4+4)^2}.$$

So $g'(x) = 0$ exactly when $x = 0$ or $h(x) = 0$, where $h(x) = x^4 + 6x - 12$. Now $h(1) = -5$ and $h(-1) = -17$, so $x = \pm 1$ are *not* stationary points: in fact $g'(1) = \frac{2}{5}$ and $g'(-1) = \frac{34}{25}$. The claim fails at the first step.

How many roots does h have? Since $h'(x) = 4x^3 + 6$ vanishes only at $x_0 = -(3/2)^{1/3}$, h decreases and then increases, so it can cross zero at most twice. Also $h(0) = -12 < 0$ and $h(x) \to +\infty$ as $x \to \pm\infty$, so it crosses at least twice. Hence h has exactly two real roots, $\beta < 0 < \alpha$. Since $h(1) < 0 < h(2) = 16$ and $h(-3) = 51 > 0 > h(-2) = -8$,

$$1 < \alpha < 2, \qquad -3 < \beta < -2,$$

and numerically $\alpha \approx 1.386$, $\beta \approx -2.247$.

Step 2: the stationary values. $g(0) = \frac{3}{4}$ is correct, and $g(1) = 1$ is correct too, but $g(-1) = \frac{1}{5}$, not 1 (and neither point at ± 1 is stationary anyway). At a root t of h we have $t^4 = 12 - 6t$, so $t^3 = (12 - 6t)/t$ and

$$g(t) = \frac{2(12-6t)/t + 3}{16 - 6t} = \frac{24 - 9t}{t(16-6t)} = \frac{3(8-3t)}{2t(8-3t)} = \frac{3}{2t},$$

cancelling $8 - 3t \neq 0$ (as $t = \frac{8}{3}$ is not a root of h). So the stationary values are $g(0) = \frac{3}{4}$, $g(\alpha) = \frac{3}{2\alpha} \approx 1.083$ and $g(\beta) = \frac{3}{2\beta} \approx -0.668$.

Ask yourself If one stationary value is smaller than another, must the smaller one be a minimum?

Step 3: the nature of the stationary points. No. Comparing the *values* at stationary points tells us nothing about their nature; we must look at the sign of g' on either side (or at g''). From the factorised form, the sign of $g'(x)$ is the sign of $-h(x)$ for $x \neq 0$, since $x^2 > 0$ and $(x^4 + 4)^2 > 0$. As $h < 0$ on (β, α) and $h > 0$ outside $[\beta, \alpha]$:

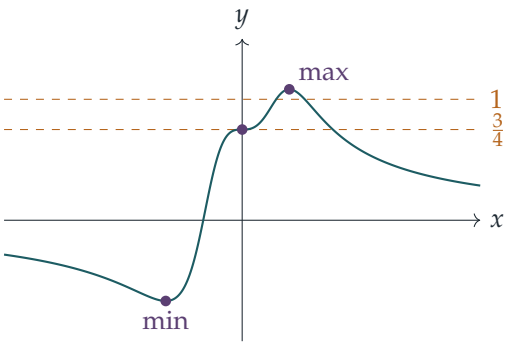
- g is decreasing for $x < \beta$, increasing on (β, α) and decreasing for $x > \alpha$;
- $x = \beta$ is a minimum and $x = \alpha$ a maximum;
- at $x = 0$, $g' > 0$ on both sides, so $x = 0$ is a stationary point of inflection, not a minimum.

Step 4: the global bounds. Even with the correct local maxima and minima, a bound "for all x" needs more: the function could exceed a local maximum somewhere else, for example as $x \to \pm\infty$. Here $g(x) \to 0$ as $x \to \pm\infty$ (the denominator has higher degree), and g is continuous everywhere since $x^4 + 4 > 0$. Combining this with the monotonicity in Step 3, g falls from values near 0 down to $g(\beta)$, rises to $g(\alpha)$, then falls back towards 0. Hence the correct statement is

$$\frac{3}{2\beta} \leq g(x) \leq \frac{3}{2\alpha} \quad \text{for all } x, \qquad \text{i.e. roughly } -0.668 \leq g(x) \leq 1.083,$$

with both bounds attained.

Verdict. The conclusion $\frac{3}{4} \leq g(x) \leq 1$ is false at both ends: $g(-1) = \frac{1}{5} < \frac{3}{4}$ (indeed $g(x) < 0$ whenever $2x^3 + 3 < 0$), and $g(\alpha) > 1$ because $\alpha < \frac{3}{2}$ (check: $h(\frac{3}{2}) = \frac{81}{16} - 3 > 0$, so $\alpha < \frac{3}{2}$ and $\frac{3}{2\alpha} > 1$).



Part (ii)

Step 1: the indefinite integral. The given answer is wrong: differentiating $-3(1-x)^{-4}$ gives $-12(1-x)^{-5}$, not $(1-x)^{-3}$. The power should go *up* by one, from -3 to -2. Since

$$\frac{\mathrm{d}}{\mathrm{d}x}(1-x)^{-2} = -2(1-x)^{-3}\cdot(-1) = 2(1-x)^{-3},$$

the correct result is

$$\int (1-x)^{-3}\,\mathrm{d}x = \tfrac{1}{2}(1-x)^{-2} + c, \qquad x \neq 1.$$

Ask yourself The correct antiderivative also gives $\frac{1}{2}\cdot\frac{1}{4} - \frac{1}{2}\cdot\frac{1}{4} = 0$ between -1 and 3. So is the answer 0 after all?

Step 2: the definite integral. No. Evaluating $[F(x)]_{-1}^{3}$ is justified only when $F' = f$ on the whole interval and f is continuous there. Here $(1-x)^{-3}$ is undefined at $x = 1$, inside $[-1, 3]$, and unbounded near it, so the integral is improper and must be defined as a limit of the two separate pieces:

$$\int_{-1}^{1-\varepsilon} (1-x)^{-3}\,\mathrm{d}x = \tfrac{1}{2}\left(\varepsilon^{-2} - \tfrac{1}{4}\right) \to +\infty \quad (\varepsilon \to 0^{+}),$$
$$\int_{1+\delta}^{3} (1-x)^{-3}\,\mathrm{d}x = \tfrac{1}{2}\left(\tfrac{1}{4} - \delta^{-2}\right) \to -\infty \quad (\delta \to 0^{+}).$$

Both pieces diverge, so the integral does not exist; "$\infty - \infty$" has no value. This also makes sense geometrically: the integrand is positive for $x < 1$ and negative for $x > 1$, and each side encloses infinite area.

Verdict. Both steps are wrong, and the conclusion is false as it stands: $\int_{-1}^{3}(1-x)^{-3}\,\mathrm{d}x$ is undefined. The "0" has a grain of truth only if we cut out a *symmetric* gap, $\varepsilon = \delta$: then the two pieces cancel exactly and the limit is 0 (the Cauchy principal value). But with $\delta = 2\varepsilon$, say, the sum is $\frac{1}{2}(\varepsilon^{-2} - \frac{1}{4}\varepsilon^{-2}) \to \infty$, so the answer depends on how we approach $x = 1$, which is exactly why the integral isn't defined.

Checks

A numerical derivative of g agrees with the factorised formula at random points; $g'(1) = 0.4 \neq 0$. Bisection gives $\alpha \approx 1.3856$, $\beta \approx -2.2467$, and $g(\alpha) = 1.0825 = 3/(2\alpha)$, $g(\beta) = -0.6676 = 3/(2\beta)$; sampling g on $[-100, 100]$ confirms these are the global maximum and minimum. For (ii), midpoint quadrature on $[-1, 0.99]$ gives about 4999.87, matching $\frac{1}{2}(10^4 - \frac{1}{4})$, and the matching piece on $[1.01, 3]$ gives -4999.87.

Common slips

- Deciding the nature of stationary points by comparing their values.
- Forgetting the behaviour as $x \to \pm\infty$ when claiming a global bound.
- Applying $[F(x)]_a^b$ across a point where the integrand is infinite; always check the integrand is continuous on $[a, b]$ first.

Beyond STEP

Part (ii) is about improper integrals: an integral over an interval containing a singularity is defined as a limit, and it converges only if each one-sided limit exists separately. The symmetric limit, the Cauchy principal value, is a genuinely useful idea in complex analysis and physics, but it's a different object. Part (i) is about the gap between local and global extrema, which analysis makes precise via the extreme value theorem. Abbott's *Understanding Analysis* treats both carefully.

WORKED SOLUTION

STEP I 1991, Question 7

Pure • Separable Differential Equations

STEP I 1991, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q7

The key idea

Both rates are constant multiples of $(1+V/V_0)^2$, so the natural variable is $u = 1+V/V_0$, which satisfies $\frac{\mathrm{d}u}{\mathrm{d}t} = \pm c\,u^2$. Separating the variables shows that the *reciprocal* $w = 1/u$ simply changes linearly: it falls at rate a in a growth year and rises at rate b in a decline year. Everything then reduces to tracking one straight-line graph and asking two questions: does w hit 0 (the money becomes infinite), or does it hit 1 (the Treasury is empty)?

Solution

Setting up

Measure t in years and write $V = kV_0$, so $u = 1+k$ and $\mathrm{d}V/\mathrm{d}t = V_0\,\mathrm{d}u/\mathrm{d}t$. In a growth year

$$V_0\frac{\mathrm{d}u}{\mathrm{d}t} = aV_0u^2 \quad\Longrightarrow\quad \frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}t} = a \quad\Longrightarrow\quad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{u}\right) = -a,$$

and in the same way, in a decline year (while $V > 0$), $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{u}\right) = +b$. So with

$$w = \frac{1}{1+V/V_0},$$

w decreases linearly at rate a during growth and increases linearly at rate b during decline. Note that $V \geq 0$ corresponds to $0 < w \leq 1$: the Treasury is empty exactly when $w = 1$, and $V \to \infty$ exactly when $w \to 0^+$.

The meaning of the inequalities

At the start $w = w_0 = \frac{1}{1+k_0}$, and during the first year $w = w_0 - at$. Then:

- $k_0 \geq 0$ simply says the Treasury cannot hold a negative amount of money.

- The money stays finite throughout the first year exactly when w stays positive up to $t = 1$, i.e. $w_0 - a > 0$, which is

$$\frac{1}{1+k_0} > a \iff 1 + k_0 < \frac{1}{a} \iff k_0 < a^{-1} - 1.$$

  If this failed, V would become infinite at time $t = w_0/a \leq 1$, within the very first year, which is absurd for any model of a real treasury.

Together the inequalities also force $a^{-1} - 1 > 0$, that is $a < 1$; we shall use this at the end.

Surviving the second year

After the first year $w_1 = \frac{1}{1+k_1}$ (and indeed $w_1 = w_0 - a$). During the decline year $w = w_1 + bt$, as long as $w < 1$. The Treasury is still non-empty at the end of the year exactly when w has not reached 1 by $t = 1$:

$$\frac{1}{1+k_1} + b < 1 \iff b < \frac{k_1}{1+k_1}.$$

In particular we need $b < 1$, and then this rearranges to

$$k_1 > \frac{b}{1-b}.$$

If instead $b \geq \frac{k_1}{1+k_1}$, the money runs out at time $t = (1 - w_1)/b \leq 1$ and stays at zero for the rest of the year.

Unlimited growth in the third year

> **Ask yourself** What if the Treasury empties in year two? Could it still blow up in year three?

If the Treasury is empty at the end of year two, year three starts with $w = 1$ and $w = 1 - at$, which stays positive throughout the year because $a < 1$. So an empty Treasury cannot blow up in year three, and we only need to consider the case where money survives year two.

In that case $w_2 = w_1 + b = \frac{1}{1+k_0} - a + b$, and in year three $w = w_2 - at$. The money becomes unlimited during the third year exactly when w reaches 0 by $t = 1$, i.e. $w_2 \leq a$:

$$\frac{1}{1+k_0} - a + b \leq a \iff b \leq 2a - \frac{1}{1+k_0}.$$

We must check that this is consistent with "but not before".

- No blow-up in year one is guaranteed by $k_0 < a^{-1} - 1$, and none can happen in year two, when the money is decreasing.

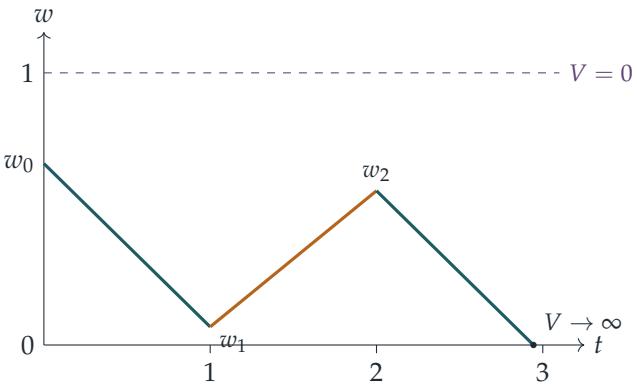
- Money survives year two: if $w_2 \le a$ then $w_2 < 1$ since $a < 1$, so the survival condition $w_2 < 1$ is automatic.

Hence the model predicts unlimited growth in the third year, but not before, exactly when

$$b \le 2a - \frac{1}{1+k_0}.$$

With strict inequality V becomes infinite at time $t = w_2/a < 1$ into the third year; in the equality case $V \to \infty$ just as the third year ends. (Since $b > 0$, this can only happen if $2a > \frac{1}{1+k_0}$, that is, $k_0 > \frac{1}{2a} - 1$.)

The figure shows w against t for $a = 0.6$, $b = 0.5$, $k_0 = 0.5$, a case where the condition holds.



Checks

In the example above, $k_1 = 14$ and the survival condition $14 > \frac{0.5}{0.5} = 1$ holds; then $k_2 = \frac{13}{17} \approx 0.765$, and the third year's blow-up happens at $t = w_2/a \approx 0.944$, consistent with $2a - \frac{1}{1+k_0} - b = \frac{1}{30} > 0$. We also integrated the differential equations numerically (fourth-order Runge–Kutta) for 300 random choices of a, b and k_0: the closed forms for k_1 and k_2 agreed to about 10^{-11}, and both the survival and blow-up criteria predicted the numerical behaviour correctly in every case.

Common slips

- Forgetting that the Treasury can empty in year two. Here it does no harm, because $a < 1$, but it needs to be said.
- Getting the blow-up condition backwards: blow-up corresponds to $1/(1+k)$ reaching 0, not k reaching 0.
- Ignoring the requirement $b < 1$ when rearranging $\frac{1}{1+k_1} + b < 1$ into a condition on k_1.

Beyond STEP

The equation $\dot{u} = au^2$ is the standard example of *finite-time blow-up*: unlike exponential growth, a solution can become infinite at a finite time, so solutions of differential equations need not exist for all time. In a first ODE course this is where the distinction between local and global existence of solutions appears. Boyce and DiPrima, *Elementary Differential Equations*, discusses this when it treats nonlinear first-order equations.

WORKED SOLUTION

STEP I 1991, Question 8

Pure • Integration by Substitution

STEP I 1991, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q8

The key idea

The substitution $y = \pi - x$ maps $[0, \pi]$ to itself and leaves $\sin x$ unchanged, but it turns the awkward factor x into $\pi - y$. Adding the integral to its reflected copy makes the x disappear altogether. In part (ii) the integrand is built from a function and its derivative, so one well-chosen substitution does all the work.

Solution

Part (i)

Write $I = \int_0^\pi x\,\mathrm{f}(\sin x)\,\mathrm{d}x$. We want a substitution that swaps the ends of $[0, \pi]$ and keeps $\sin x$ the same, so take $k = \pi$: put $x = \pi - y$, so that $\mathrm{d}x = -\mathrm{d}y$, $\sin x = \sin(\pi - y) = \sin y$, and $x = 0, \pi$ correspond to $y = \pi, 0$. Then

$$\begin{aligned} I &= \int_\pi^0 (\pi - y)\,\mathrm{f}(\sin y)\,(-\mathrm{d}y) = \int_0^\pi (\pi - y)\,\mathrm{f}(\sin y)\,\mathrm{d}y \\ &= \pi \int_0^\pi \mathrm{f}(\sin y)\,\mathrm{d}y - I, \end{aligned}$$

since y is only a dummy variable, so the last integral is I again. Hence

$$I = \frac{\pi}{2} \int_0^\pi \mathrm{f}(\sin x)\,\mathrm{d}x.$$

? **Ask yourself** How do we get from $\frac{\pi}{2}\int_0^\pi$ to $\pi \int_0^{\pi/2}$?

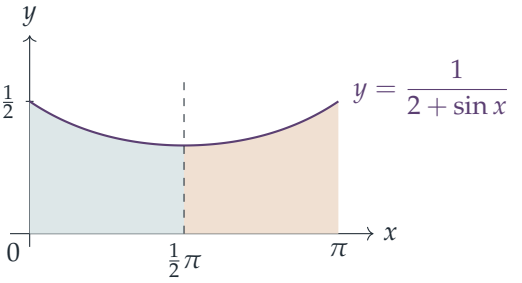
The graph of $\mathrm{f}(\sin x)$ on $[0, \pi]$ is symmetric about $x = \frac{1}{2}\pi$, because $\sin(\pi - x) = \sin x$. To make this precise, split the integral at $\frac{1}{2}\pi$ and use the same substitution $x = \pi - y$ on the second half:

$$\int_{\pi/2}^\pi \mathrm{f}(\sin x)\,\mathrm{d}x = \int_{\pi/2}^0 \mathrm{f}(\sin y)\,(-\mathrm{d}y) = \int_0^{\pi/2} \mathrm{f}(\sin y)\,\mathrm{d}y.$$

So $\int_0^{\pi} \mathrm{f}(\sin x)\,\mathrm{d}x = 2\int_0^{\pi/2} \mathrm{f}(\sin x)\,\mathrm{d}x$, and therefore

$$\int_0^{\pi} x\,\mathrm{f}(\sin x)\,\mathrm{d}x = \pi \int_0^{\frac{1}{2}\pi} \mathrm{f}(\sin x)\,\mathrm{d}x,$$

as required.



Now take $\mathrm{f}(u) = \dfrac{1}{2+u}$, which is continuous for $u \in [-1,1]$. Then

$$\int_0^{\pi} \frac{x}{2+\sin x}\,\mathrm{d}x = \pi \int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{2+\sin x}.$$

For the remaining integral use $t = \tan \frac{1}{2}x$, so that

$$\sin x = \frac{2t}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2},$$

and $x = 0, \frac{1}{2}\pi$ correspond to $t = 0, 1$. Then

$$\begin{aligned}
\int_0^{\frac{1}{2}\pi} \frac{\mathrm{d}x}{2+\sin x} &= \int_0^1 \frac{1}{2+\frac{2t}{1+t^2}} \cdot \frac{2\,\mathrm{d}t}{1+t^2} = \int_0^1 \frac{2\,\mathrm{d}t}{2(1+t^2)+2t} = \int_0^1 \frac{\mathrm{d}t}{t^2+t+1} \\
&= \int_0^1 \frac{\mathrm{d}t}{\left(t+\frac{1}{2}\right)^2 + \frac{3}{4}} = \frac{2}{\sqrt{3}}\left[\tan^{-1}\frac{2t+1}{\sqrt{3}}\right]_0^1 \\
&= \frac{2}{\sqrt{3}}\left(\tan^{-1}\sqrt{3} - \tan^{-1}\frac{1}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}}\left(\frac{\pi}{3} - \frac{\pi}{6}\right) = \frac{\pi}{3\sqrt{3}}.
\end{aligned}$$

Hence

$$\int_0^{\pi} \frac{x}{2+\sin x}\,\mathrm{d}x = \frac{\pi^2}{3\sqrt{3}} = \frac{\sqrt{3}\,\pi^2}{9}.$$

Part (ii)

? **Ask yourself** What is the derivative of $(\sin^{-1} t)^2$?

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\sin^{-1} t\right)^2 = \frac{2\sin^{-1} t}{\sqrt{1-t^2}},$$

which is (twice) exactly the factor multiplying the cosine. So substitute $u = (\sin^{-1} t)^2$, with $\mathrm{d}u = \dfrac{2\sin^{-1} t}{\sqrt{1-t^2}}\,\mathrm{d}t$. When $t = 0$, $u = 0$; when $t = 1$, $\sin^{-1} 1 = \frac{1}{2}\pi$, so $u = \frac{1}{4}\pi^2$. Therefore

$$\int_0^1 \frac{(\sin^{-1} t)\cos\left[(\sin^{-1} t)^2\right]}{\sqrt{1-t^2}}\,\mathrm{d}t = \frac{1}{2}\int_0^{\pi^2/4} \cos u\,\mathrm{d}u = \frac{1}{2}\sin\frac{\pi^2}{4}.$$

The integrand blows up at $t = 1$, so strictly this is an improper integral. The substitution handles it cleanly: in terms of $\theta = \sin^{-1} t$ the integral is $\int_0^{\pi/2} \theta\cos\theta^2\,\mathrm{d}\theta$, whose integrand is continuous, so the integral converges and the value above is correct. (Doing it in two steps, $\theta = \sin^{-1} t$ then $u = \theta^2$, is a perfectly good alternative route.)

The answer $\frac{1}{2}\sin\frac{1}{4}\pi^2$ is exact and doesn't simplify further: $\frac{1}{4}\pi^2$ is not a 'nice' angle, which is presumably why the examiners warned against reaching for a calculator.

Checks

- Simpson's rule gives $\int_0^{\pi} x/(2+\sin x)\,\mathrm{d}x \approx 1.899406$, and $\pi^2/(3\sqrt{3}) \approx 1.899406$.
- A quick sanity check: $\frac{1}{3} \le \frac{1}{2+\sin x} \le \frac{1}{2}$, so the integral lies between $\frac{1}{3}\cdot\frac{1}{2}\pi^2 \approx 1.64$ and $\frac{1}{2}\cdot\frac{1}{2}\pi^2 \approx 2.47$.
- The identity in (i) can be tested with $\mathrm{f}(u) = 1$: both sides equal $\frac{1}{2}\pi^2$.
- For (ii), $\frac{1}{2}\sin(\frac{1}{4}\pi^2) \approx 0.312133$, which matches a numerical evaluation of $\int_0^{\pi/2}\theta\cos\theta^2\,\mathrm{d}\theta$.

Common slips

- Stopping at $I = \frac{\pi}{2}\int_0^{\pi}\mathrm{f}(\sin x)\,\mathrm{d}x$ without justifying the halving of the range.
- Forgetting to change the limits when substituting $t = \tan\frac{1}{2}x$ (the upper limit becomes 1, not ∞).
- Losing the factor $\frac{1}{2}$ in part (ii): $\mathrm{d}u$ carries a 2.

Beyond STEP

The trick in part (i), exploiting a symmetry of the interval to evaluate $\int x\,g(x)\,\mathrm{d}x$ without ever finding an antiderivative, is a small example of a big theme: symmetry often tells you the value of an integral more cheaply than calculation. Part (ii) touches the theory of improper integrals, where a change of variables can turn an unbounded

integrand into a continuous one. Abbott's *Understanding Analysis* makes the change-of-variables theorem rigorous, and Siklos's *Advanced Problems in Mathematics* has many more integrals of this type.

WORKED SOLUTION

STEP I 1991, Question 9

Pure • Inequalities and Harmonic Series

STEP I 1991, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q9

The key idea

In part (i), each inequality traps x between two kth roots, and the trick is to find one lower bound and one upper bound that clash. To compare a cube root with a fifth root without a calculator, raise both sides to the fifteenth power. In part (ii) we bound a long sum below by (number of terms) $\times$ (smallest term). Then we chain disjoint blocks of that shape together to make the harmonic series as big as we like.

Solution

Part (i)

Since $x > 1$, every power of x is positive, and raising both sides of an inequality between positive numbers to a positive power keeps its direction. We use this throughout.

Suppose, for a contradiction, that $n \geq 5$. Then in particular the third and fifth inequalities hold:

$$x^3 > 3 \qquad \text{and} \qquad x^5 < 6.$$

Raise the first to the fifth power and the second to the cube:

$$x^{15} > 3^5 = 243 \qquad \text{and} \qquad x^{15} < 6^3 = 216.$$

These give $243 < x^{15} < 216$, which is impossible. So $n \geq 5$ can't happen, and therefore $n \leq 4$.

? **Ask yourself** How would you know to pick the third and fifth inequalities?

Each inequality says $k^{1/k} < x < (k+1)^{1/k}$. The lower bounds $k^{1/k}$ go 1, 1.41, 1.44, 1.41, 1.38, …, largest at $k = 3$. The upper bounds $(k+1)^{1/k}$ decrease, and $6^{1/5} \approx 1.43$ is the first to drop below $3^{1/3} \approx 1.44$. A rough mental estimate points to the pair, and the fifteenth powers then prove it exactly. (The value $n = 4$ really can occur. For example $x = 1.45$ gives $x^2 = 2.1025$, $x^3 \approx 3.05$, $x^4 \approx 4.42$.)

Part (ii)

Let $n \geq 2$. The sum $\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n^2}$ has $n^2 - n$ terms, and each is at least $\frac{1}{n^2}$, the last one. Since $n \geq 2$ there are $n^2 - n \geq 2$ terms, so the first term $\frac{1}{n+1}$ is strictly bigger than $\frac{1}{n^2}$. Hence

$$\frac{1}{n+1} + \cdots + \frac{1}{n^2} > (n^2 - n) \cdot \frac{1}{n^2} = 1 - \frac{1}{n}.$$

Adding $\frac{1}{n}$ to both sides,

$$\frac{1}{n} + \frac{1}{n+1} + \cdots + \frac{1}{n^2} > 1,$$

as required.

Making the harmonic sum exceed 10 Each block $\frac{1}{m} + \cdots + \frac{1}{m^2}$ with $m \geq 2$ contributes more than 1. To use several of them we need blocks that don't overlap, so we start each new block just after the last one ends. Define

$$a_1 = 2, \qquad a_{k+1} = a_k^2 + 1,$$

so $a_1 = 2$, $a_2 = 5$, $a_3 = 26$, $a_4 = 677, \ldots$. Block k runs from $\frac{1}{a_k}$ to $\frac{1}{a_k^2}$, and the next block starts at $\frac{1}{a_k^2+1} = \frac{1}{a_{k+1}}$. So the blocks for $k = 1, \ldots, 9$ are disjoint and consecutive, and together they cover $\frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{a_9^2}$. Therefore

$$\sum_{n=1}^{a_9^2} \frac{1}{n} = 1 + \sum_{k=1}^{9} \left(\frac{1}{a_k} + \cdots + \frac{1}{a_k^2} \right) > 1 + 9 = 10.$$

Ask yourself a_9^2 is a gigantic number. Can we name an explicit N instead?

Yes, by bounding the a_k. We claim $a_k < 2^{2^k}$. This holds for $k = 1$ since $2 < 4$. If $a_k < 2^{2^k}$, then $a_k \leq 2^{2^k} - 1$, so

$$a_{k+1} = a_k^2 + 1 \leq \left(2^{2^k} - 1\right)^2 + 1 = 2^{2^{k+1}} - 2^{2^k+1} + 2 < 2^{2^{k+1}}.$$

So by induction $a_9 < 2^{512}$ and $a_9^2 < 2^{1024}$. Adding more positive terms only makes the sum bigger, so

$$N = 2^{1024} \quad \text{gives} \quad \sum_{n=1}^{N} \frac{1}{n} > 10.$$

Any larger N works too.

Otherwise: grouping into blocks $\frac{1}{2^j+1} + \cdots + \frac{1}{2^{j+1}}$, each of 2^j terms that are all at least $\frac{1}{2^{j+1}}$, shows that each block exceeds $\frac{1}{2}$ for $j \geq 1$ and equals $\frac{1}{2}$ for $j = 0$. So $\sum_{n=1}^{2^{18}} \frac{1}{n} > 1 + \frac{18}{2} = 10$, and $N = 2^{18} = 262\,144$ is a much smaller answer.

Checks

- Part (i): numerically, the inequalities leave $x \in (3^{1/3}, 5^{1/4}) \approx (1.442, 1.495)$ when $n = 4$. Adding the fifth inequality forces $x < 6^{1/5} \approx 1.431$, so nothing is left.
- Part (ii): with exact fractions, $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12} > 1$, and the block sum exceeds 1 for every n from 2 to 39.
- The harmonic sum first passes 10 at $N = 12\,367$, so every valid answer must be at least this. Both 2^{18} (where the sum is about 13.05) and 2^{1024} are comfortably bigger.

Common slips

- Raising inequalities to powers without noting that both sides are positive (here guaranteed by $x > 1$).
- Using blocks $[2, 4]$, $[4, 16]$, $[16, 256]$, …, which share their endpoints, so some terms are counted twice.
- Writing $\geq$ instead of $>$ in part (ii) without checking that at least one term is strictly larger than $\frac{1}{n^2}$.

Beyond STEP

Part (ii) is a version of the Cauchy condensation idea. It proves that the harmonic series diverges, though astonishingly slowly: its partial sums grow like $\ln N + \gamma$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. Comparing a sum with an integral gives this growth rate and makes the smallest N easy to estimate ($e^{10-\gamma} \approx 12\,367$). Abbott's *Understanding Analysis* treats condensation and series convergence carefully.

WORKED SOLUTION

STEP I 1991, Question 10

Mechanics • Equilibrium of Hanging Chains

STEP I 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.14, *Parabola, Not Catenary*: https://beyondstep.keithmcnulty.org/ch11/bridge-parabola.html

The key idea

Every joint of the chain is in equilibrium under three forces: the tensions in the two neighbouring links and the weight w hung from it. Resolving horizontally shows that every link has the *same* horizontal component of tension. Resolving vertically then says that the vertical component drops by exactly w at each joint, and since the horizontal spacing is fixed, the slopes of the links form an arithmetic progression. Slopes in arithmetic progression are the discrete fingerprint of a parabola.

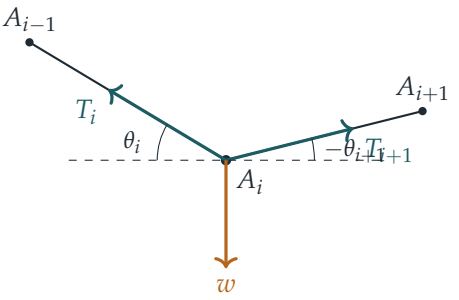
Solution

The equilibrium equations

Model the chain as light, flexible and inextensible, so the tension in each link $A_{i-1}A_i$ acts along it. Let θ_i be the angle that $A_{i-1}A_i$ makes with the horizontal, measured *downwards* as we move in the direction of increasing x, so that

$$\tan\theta_i = \frac{y_{i-1} - y_i}{h}.$$

(On the right-hand half of the bridge θ_i is negative; nothing below depends on its sign.) The joint A_i is pulled by T_i along A_iA_{i-1}, by T_{i+1} along A_iA_{i+1}, and down by the strut with force w.



Resolving horizontally and vertically at A_i, for $1 \le i \le n+1$ (indeed for every $1 \le i \le 2n+1$):

$$T_i \cos\theta_i = T_{i+1}\cos\theta_{i+1},$$
$$T_i \sin\theta_i - T_{i+1}\sin\theta_{i+1} = w.$$

These are the equations asked for.

The ratios

The first equation says that $T_i \cos\theta_i$ is the same for every link; call this common horizontal component H. It is positive, because a chain can only pull. Dividing the second equation by H and using $\tan\theta_i = (y_{i-1} - y_i)/h$,

$$\frac{H}{h}(y_{i-1} - y_i) - \frac{H}{h}(y_i - y_{i+1}) = w.$$

So if we write $V_i = H(y_{i-1} - y_i)/h$ for the vertical component of tension in link i, then

$$V_i - V_{i+1} = w \quad (1 \le i \le 2n+1), \qquad \text{so} \qquad V_i = V_1 - (i-1)w \quad (1 \le i \le 2n+2).$$

Ask yourself We have one unknown V_1 left. Which fact about the bridge fixes it?

The supports are at the same height. Adding up the drops along the whole chain,

$$0 = y_0 - y_{2n+2} = \sum_{i=1}^{2n+2}(y_{i-1} - y_i) = \frac{h}{H}\sum_{i=1}^{2n+2} V_i$$
$$= \frac{h}{H}\Big[(2n+2)V_1 - w\big(0 + 1 + \cdots + (2n+1)\big)\Big].$$

The bracket is $(2n+2)V_1 - w(2n+1)(n+1)$, so $V_1 = \frac{1}{2}(2n+1)w$. (This is also what overall vertical equilibrium suggests: the two supports share the total load $(2n+1)w$ equally.) Therefore

$$V_i = \tfrac{1}{2}(2n+3-2i)\,w, \qquad \text{that is,} \qquad y_{i-1} - y_i = \frac{(2n+3-2i)\,wh}{2H}.$$

Taking $i = n+1, n, \ldots, 1$ gives drops of $1, 3, \ldots, 2n+1$ times $wh/(2H)$:

$$\frac{h}{y_n - y_{n+1}} = \frac{3h}{y_{n-1} - y_n} = \cdots = \frac{(2n+1)h}{y_0 - y_1} = \frac{2H}{w},$$

which is the required chain of equalities. (The last numerator is printed as $(2n+1)y$ on the paper; it must be $(2n+1)h$, both from the pattern and for the ratio to be dimensionless like the others.)

The parabola

Let $k = wh/(2H)$. Summing the drops from A_0,

$$y_i = y_0 - k\sum_{j=1}^{i}(2n+3-2j) = y_0 - k\big[(2n+3)i - i(i+1)\big] = y_0 - k\,i\,(2n+2-i).$$

Writing $m = n+1$, we have $i(2m-i) = m^2 - (i-m)^2$, so

$$y_i = (y_0 - km^2) + k(i-m)^2 = y_{n+1} + k(i-n-1)^2,$$

since putting $i = m$ shows that $y_0 - km^2 = y_{n+1}$. Now $x_i = x_0 + ih$, so $(i-n-1)h = x_i - x_{n+1}$, and every point A_i, $0 \le i \le 2n+2$, satisfies

$$y = y_{n+1} + \frac{w}{2Hh}\,(x - x_{n+1})^2.$$

This is a single parabola with a vertical axis through the lowest point A_{n+1}, so the points $A_0, A_1, \ldots, A_{2n+2}$ all lie on it. The end points are included: $i = 0$ and $i = 2n+2$ both give y_0, as they should.

Checks

- With $n = 0$ there is one strut in the middle: the formula gives $V_1 = \frac{1}{2}w$, so each half of the chain carries half the load, as it obviously should.
- The result agrees with the continuous picture: a uniform load per unit horizontal length w/h hung from a cable with horizontal tension H gives $y'' = (w/h)/H$, and our parabola has $y'' = w/(Hh)$.
- In a numerical check we took 200 random bridges ($1 \le n \le 8$, random H, w, h), solved the joint equations directly as a linear system in exact fractions, and confirmed the ratios, the formula $y_i = y_0 - k\,i(2n+2-i)$, the parabola, and full force balance at every joint. For instance, with $n = 2$, $w = 2$, $H = 5$, $h = 1$ the drops are 1.0, 0.6, 0.2 on the way down, and each ratio equals $2H/w = 5$.

Common slips

- Using symmetry to say $y_n = y_{n+2}$ without comment. It is true, but it follows from the supports being level; the summing argument above proves it rather than assuming it.

- Forgetting that the horizontal component of tension is constant, and trying to find each T_i separately. The question never needs the T_i themselves (they are $\sqrt{H^2 + V_i^2}$).
- Mixing up the sign of θ_i on the rising half of the chain. Working with $\tan\theta_i = (y_{i-1} - y_i)/h$ throughout avoids this.

Beyond STEP

A chain hanging under its own weight forms a catenary, $y = c\cosh(x/c)$, but a chain carrying a load spread evenly along the *horizontal*, like a road deck, forms a parabola; this question is the discrete version of that fact. The second-difference equation $y_{i-1} - 2y_i + y_{i+1} = \text{const}$ is the finite-difference form of $y'' = \text{const}$, and the same idea underlies numerical methods for differential equations. Calculus of variations treats the catenary and its relatives. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see statics of strings and cables done carefully.

WORKED SOLUTION

STEP I 1991, Question 11

Mechanics • Pulleys and Impulses

STEP I 1991, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q11

The key idea

Each cycle has two stages. When the weight hits the pile it stops dead and the string goes slack, so the counterweight flies up and falls back freely under gravity. Then the string jerks taut and the two masses share momentum, and they move together with the constant acceleration from the first part until the weight lands again. Constant acceleration makes each stage symmetric in time. So every speed in one cycle is a fixed multiple of the speed in the cycle before, and a geometric progression follows straight away.

Solution

The constant acceleration

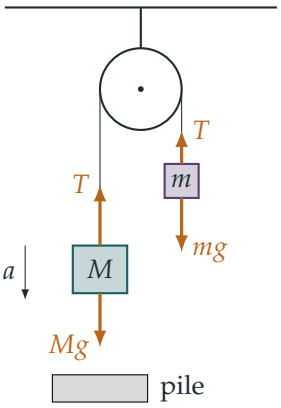
We model the weight and counterweight as particles, and the string as light and inextensible over a smooth light pulley. While the string is taut both particles move with the same speed v, the weight going down and the counterweight going up. Suppose that after moving a distance x from rest they have speed v. The weight has lost potential energy Mgx and the counterweight has gained mgx. The pulley is smooth and the string is light and inextensible, so no energy is lost and

$$\tfrac{1}{2}(M+m)v^2 = (M-m)gx.$$

Differentiating with respect to time, and using $\dot{x} = v$,

$$(M+m)v\dot{v} = (M-m)gv \quad \Longrightarrow \quad \dot{v} = g\,\frac{M-m}{M+m}$$

for $v \neq 0$. This is a constant. Call it a. The same result comes from Newton's second law: if T is the tension, then $Mg - T = Ma$ and $T - mg = ma$, and adding these gives $(M-m)g = (M+m)a$.



From one collision to the next

Just before a collision the weight is moving down with speed v, and the counterweight is moving up with the same speed v. Follow the motion in three stages.

The impact. The collision is completely inelastic and the pile is fixed, so the weight stops. The counterweight is still moving up at speed v, so the string goes slack. Tension then can't help the counterweight, and gravity alone pushes the weight onto the pile, so the weight stays put.

Free flight. With the string slack the counterweight is a projectile moving vertically. It rises and returns to the point where the string becomes taut again, which is where it was at the impact. Under constant acceleration g the motion is symmetric, so it arrives there moving *down* at speed v, after a time

$$\frac{2v}{g}.$$

The jerk. The string snaps taut with the counterweight moving down at v and the weight at rest. The masses must then move with a common speed u, the counterweight downwards and the weight upwards. By the given assumption, the momentum lost by the counterweight equals the momentum gained by the weight:

$$m(v-u) = Mu \quad \Longrightarrow \quad u = \frac{mv}{M+m}.$$

Ask yourself Why can't we just use conservation of momentum for the whole system here?

The two masses move in opposite directions and the pulley exerts an impulse, so there is no single direction in which total momentum is conserved. What is true is that the impulsive tension has the same size J on both masses, since the string is light and the pulley smooth. So the counterweight loses momentum J along the string and the weight gains J along the string. That's exactly the statement we were allowed to assume.

Together again. Now the string is taut and the weight is moving up at speed u. By the first part, the acceleration has the constant magnitude a and is directed so that the heavier mass M goes down. So the weight slows to rest, then falls back to the pile. Constant acceleration makes this symmetric too: the weight reaches the pile at speed u, after a time

$$\frac{2u}{a}.$$

So just before the next collision the counterweight is moving with speed

$$u = \frac{mv}{M+m},$$

as required.

The times form a geometric progression

Write $k = \dfrac{m}{M+m}$, so $0 < k < 1$. Let v_0 be the speed just before the first impact. By the previous part the speed just before the $(n+1)$th impact is $v_n = k^n v_0$. Adding the two stages, the time from the $(n+1)$th impact to the next one is

$$t_n = \frac{2v_n}{g} + \frac{2kv_n}{a} = \left(\frac{2}{g} + \frac{2k}{a}\right) k^n v_0.$$

This is a constant times k^n, so the times between successive collisions form a geometric progression with common ratio $k = m/(M+m)$.

The total time

Let T_0 be the time from release to the first impact. The weight starts from rest with acceleration a, so $v_0 = aT_0$. Since $0 < k < 1$, the geometric series converges, and the total time is

$$T_0 + \sum_{n=0}^{\infty} t_n = T_0 + \left(\frac{2}{g} + \frac{2k}{a}\right) \frac{v_0}{1-k}.$$

Now $1 - k = \dfrac{M}{M+m}$, and $\dfrac{k}{a} = \dfrac{m}{M+m} \cdot \dfrac{M+m}{g(M-m)} = \dfrac{m}{g(M-m)}$. Hence

$$\frac{2}{g} + \frac{2k}{a} = \frac{2}{g}\left(1 + \frac{m}{M-m}\right) = \frac{2M}{g(M-m)},$$

$$\left(\frac{2}{g} + \frac{2k}{a}\right) \frac{v_0}{1-k} = \frac{2M}{g(M-m)} \cdot \frac{M+m}{M} v_0 = \frac{2v_0}{a} = 2T_0.$$

So there are infinitely many collisions, but they take only a finite total time. The weight finally comes to rest after a time

$$T_0 + 2T_0 = 3T_0,$$

which is three times the time from the start to the first impact.

Checks

- Take the limit $m \to 0$. Then $k \to 0$ and $a \to g$, so the weight never lifts again after the first impact. The only remaining time is the counterweight's flight, $2v_0/g = 2gT_0/g = 2T_0$, and the total is $3T_0$, as the general formula predicts.
- We stepped the motion numerically for random M, m and drop heights (for example $M = 2.403$, $m = 1.571$, so $k = 0.3953$). The speed ratios and gap ratios both came out as 0.3953, and the total time, including the geometric tail, came out as $2.99998\,T_0$.

Common slips

- Forgetting the free-flight stage. The counterweight carries on upwards after the impact, and the cycle includes the time $2v/g$ it spends as a projectile.
- Using a instead of g for the slack phase. Once the string is slack the counterweight feels only gravity.
- Writing "total momentum is conserved" at the jerk with both velocities in the same direction. The pulley turns the string, so it is the momentum *along the string* that is shared.

Beyond STEP

Infinitely many events in a finite time, like this piledriver or a bouncing ball, are called Zeno behaviour. They turn up in the theory of hybrid systems, where smooth motion is interrupted by sudden jumps. The jerk in the string is a simple impulsive constraint. Lagrangian mechanics treats such constraints systematically in the first undergraduate year. For more on impulses, collisions and constrained motion, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1991, Question 12

Mechanics • Equilibrium with Friction

STEP I 1991, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q12

The key idea

The plank AB is light and is acted on only at its two ends, by smooth hinges. So the force it exerts on CD must act along AB. That reduces the problem to a single rigid body, CD, with three unknown forces. Taking moments about C gives the thrust in AB, and resolving then gives the friction and the normal reaction at C. The geometry (both planks meet at B) links x to α and β, and this is what makes l and x drop out of the answer.

Solution

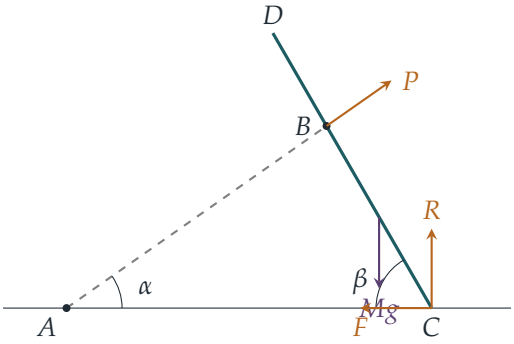
The model

We treat the planks as light rigid rods, the painter as a particle of mass M, and the hinges at A and B as smooth. The ground is rough at C, with coefficient μ.

? **Ask yourself** Why must the force between the planks at B act along AB?

Plank AB has no weight, and the only forces on it come from the smooth hinges at A and B. Taking moments about A, the force at B has no moment about A, so it acts along BA. The force at A must then be equal and opposite to it. By Newton's third law, AB pushes on CD at B with a force P along AB. We'll take P in the direction from A to B (a thrust), and the working confirms that $P > 0$.

So the forces on CD are: the thrust P at B, at angle α above the horizontal and pointing away from A; the weight Mg of the painter, at distance x from C along the plank; and at C a normal reaction R upwards and friction F towards A. The friction has to point towards A because the thrust pushes C away from A.



Geometry

The point B is at height $2l\sin\alpha$, since it is the top of AB. It is also at distance $2x$ from C along CD, so it is at height $2x\sin\beta$. Hence

$$x\sin\beta = l\sin\alpha, \qquad \text{that is,} \qquad x = \frac{l\sin\alpha}{\sin\beta}.$$

The horizontal distance AC is the sum of the horizontal projections of AB and BC:

$$AC = 2l\cos\alpha + 2x\cos\beta.$$

Moments and resolving for CD

Take moments about C. The line of action of P is the line AB, and its perpendicular distance from C is $AC\sin\alpha$. The weight acts at horizontal distance $x\cos\beta$ from C. The two moments are in opposite senses, so

$$P\cdot AC\sin\alpha = Mg\,x\cos\beta \quad\Longrightarrow\quad P = \frac{Mg\,x\cos\beta}{(2l\cos\alpha + 2x\cos\beta)\sin\alpha} > 0.$$

Resolving horizontally and vertically:

$$F = P\cos\alpha, \qquad R = Mg - P\sin\alpha.$$

Substituting for P,

$$R = Mg\left(1 - \frac{x\cos\beta}{2l\cos\alpha + 2x\cos\beta}\right) = Mg\,\frac{2l\cos\alpha + x\cos\beta}{2l\cos\alpha + 2x\cos\beta},$$
$$F = Mg\,\frac{x\cos\beta\cos\alpha}{(2l\cos\alpha + 2x\cos\beta)\sin\alpha}.$$

In particular $R > 0$, so the end C stays in contact with the ground. Dividing,

$$\frac{F}{R} = \frac{x\cos\alpha\cos\beta}{\sin\alpha\,(2l\cos\alpha + x\cos\beta)}.$$

Now we use the geometry, $x = l\sin\alpha/\sin\beta$. Then $x\cos\beta = l\sin\alpha\cot\beta$, and

$$\frac{F}{R} = \frac{l\sin\alpha\cot\beta\cos\alpha}{\sin\alpha\,(2l\cos\alpha + l\sin\alpha\cot\beta)} = \frac{\cos\alpha\cot\beta}{2\cos\alpha + \sin\alpha\cot\beta} = \frac{\cot\alpha\cot\beta}{2\cot\alpha + \cot\beta},$$

where in the last step we divided the top and bottom by $\sin\alpha$.

The plank CD is in equilibrium only if friction can supply this force, that is, $F \leq \mu R$. Since $R > 0$, this is

$$\mu \geq \frac{\cot\alpha \cot\beta}{2\cot\alpha + \cot\beta},$$

as required. (The plank AB is hinged at A, so it is held in place whatever force it needs.)

The final part

If B coincides with D, then $2x = 2l$, so $x = l$. The geometry gives $\sin\beta = \sin\alpha$, and both angles are acute, so $\beta = \alpha$. The planks and the ground form an isosceles triangle with $AC = 4l\cos\alpha$. The friction condition becomes

$$\mu \geq \frac{\cot^2\alpha}{3\cot\alpha} = \tfrac{1}{3}\cot\alpha, \qquad \text{that is,} \qquad \tan\alpha \geq \frac{1}{3\mu}.$$

So equilibrium is possible exactly when $\alpha \geq \alpha_0$, where $\tan\alpha_0 = 1/(3\mu)$.

Ask yourself Where is the painter, and which way does the distance change as α varies?

The painter stands at the midpoint of CD, at distance l from C. Taking A as the origin, C is at $(4l\cos\alpha, 0)$, and the painter is at

$$(4l\cos\alpha - l\cos\alpha,\ l\sin\alpha) = (3l\cos\alpha,\ l\sin\alpha).$$

So the square of the painter's distance from A is

$$d^2 = 9l^2\cos^2\alpha + l^2\sin^2\alpha = l^2\left(1 + 8\cos^2\alpha\right).$$

This decreases as α increases on $(0, \frac{\pi}{2})$. So the largest distance occurs at the smallest allowed angle, $\alpha = \alpha_0$. There $\tan\alpha_0 = 1/(3\mu)$, so

$$\cos^2\alpha_0 = \frac{1}{1+\tan^2\alpha_0} = \frac{9\mu^2}{1+9\mu^2},$$

and

$$d^2_{\max} = l^2\left(1 + \frac{72\mu^2}{1+9\mu^2}\right) = l^2\,\frac{1+81\mu^2}{1+9\mu^2}.$$

Therefore the maximum distance of the painter from A is

$$l\sqrt{\frac{1+81\mu^2}{1+9\mu^2}},$$

as required. This value is attained, with friction exactly limiting.

Checks

- Limiting cases of the final answer: as $\mu \to 0$ the distance tends to l, which corresponds to $\alpha \to \frac{\pi}{2}$, with both planks vertical. As $\mu \to \infty$ it tends to $3l$, which corresponds to the planks lying flat with the painter at $(3l, 0)$. Both are sensible.
- If $\beta \to \frac{\pi}{2}$, so that CD is vertical, then $\cot\beta \to 0$ and the required μ tends to 0. A vertical plank needs no friction, which agrees.
- Numerically, taking $\alpha = 60°$ and $\beta = 70°$ and solving the three equilibrium equations for CD directly gives $F/R = 0.13837$, which matches the formula. A scan over α with $\mu = 0.5$ gives a maximum distance of $2.55704\,l$, and the formula gives $l\sqrt{21.25/3.25} = 2.55704\,l$.

Common slips

- Assuming that the force at B is horizontal or vertical. Because AB is light and smoothly hinged at both ends, the force must act along AB.
- Forgetting the vertical component of the thrust in R. It reduces the normal reaction, which makes friction harder to supply.
- In the final part, maximising over all α instead of only over the angles for which equilibrium is possible.

Beyond STEP

A light rod with smooth pins at its ends is a *two-force member*. Recognising these is the first step in analysing pin-jointed frameworks (trusses) in engineering statics, where each such member carries a pure tension or thrust. The final part is a small constrained optimisation: the objective is monotonic, so the optimum sits on the boundary of the feasible set. This is a theme that recurs throughout linear programming and optimisation. For more on equilibrium of rigid bodies, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1991, Question 13

Mechanics • Conservation of Momentum

STEP I 1991, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q13

The key idea

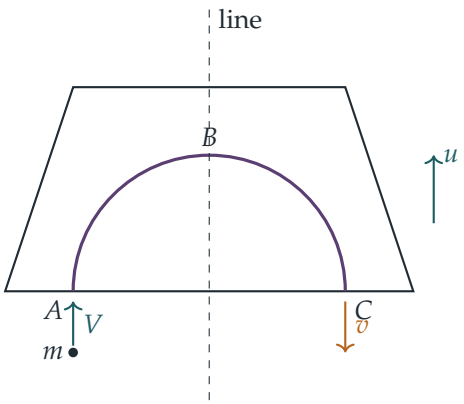
The table is smooth and the groove is smooth, so nothing outside the system pushes along the line, and no force does any work. That gives us two conservation laws: momentum along the line and kinetic energy. The groove's tangent at the exit is parallel to the line, so at the exit *both* velocities lie along the line, and the two laws then work just like a one-dimensional elastic collision. For the fixed lamina, the average force is simply the total impulse divided by the time taken.

A note on the wording: the question says the tangents at A and B are parallel to the line, but the diagram shows that it is the tangents at the two ends, A and C, that are parallel to it (at B the groove is at right angles to the line). We use A and C, which is clearly what is intended.

Solution

The moving lamina

Model the particle as a point mass, the lamina as a rigid body that can only translate along the line, and all contacts as smooth. Take the line as an axis, with the positive direction being the particle's direction of motion when it enters at A.



Let the particle leave C with velocity v along the line (positive in the direction of V), and let the lamina then have velocity u along the line.

Ask yourself Why must the particle's velocity at C be along the line?

Relative to the lamina, the particle moves along the groove, so its relative velocity at C is along the tangent there, which is parallel to the line. The lamina itself only moves along the line. Adding the two, the particle's actual velocity at C is along the line as well. So at the exit we only need one number for each body.

During the motion, the forces on the system (particle plus lamina) from outside are the table's normal reaction, which is vertical, and whatever keeps the lamina on its line, which is perpendicular to the line (the line is smooth). The groove's force on the particle and the particle's force on the groove are internal and cancel in pairs. So momentum along the line is conserved:

$$mV = mv + Mu. \tag{1}$$

Every contact is smooth, and the table is horizontal, so no work is done and kinetic energy is conserved. (The normal reaction of the groove does work on the particle and on the lamina separately, but these two amounts cancel, because the reaction is perpendicular to the *relative* velocity.) So

$$\tfrac{1}{2}mV^2 = \tfrac{1}{2}mv^2 + \tfrac{1}{2}Mu^2. \tag{2}$$

Rewrite (1) and (2) as

$$m(V - v) = Mu, \qquad m(V - v)(V + v) = Mu^2.$$

One solution is $v = V$, $u = 0$: this is the situation at A, before anything has happened. At C the particle is moving *backwards* relative to the lamina, so $v \neq V$ and $u \neq 0$, and we may divide the second equation by the first to get

$$V + v = u.$$

(This says the relative speed of the particle with respect to the lamina is V at both ends, just as in an elastic collision.) Substituting $u = V + v$ into (1):

$$mV = mv + MV + Mv \quad \Longrightarrow \quad v = -\frac{M - m}{M + m}V, \qquad u = V + v = \frac{2m}{M + m}V.$$

Since $m < M$, v is negative: the particle comes out of C moving back the way it came. So when the particle leaves the groove,

$$\text{speed of particle} = \frac{M-m}{M+m}\,V, \qquad \text{speed of lamina} = \frac{2m}{M+m}\,V,$$

with the lamina moving in the original direction of the particle and the particle moving the opposite way.

The particle does get round the groove: at any point where it has stopped relative to the lamina, it would share the lamina's velocity, and then the kinetic energy would be $\frac{1}{2}(mV)^2/(M+m) < \frac{1}{2}mV^2$, which energy conservation forbids. So its relative speed never drops to zero.

The lamina held fixed

Now the lamina cannot move. The groove's reaction on the particle is perpendicular to its velocity, so it does no work, and the particle goes round the semicircle at constant speed V. The semicircle has length πr, so the time in the groove is

$$T = \frac{\pi r}{V}.$$

The particle's momentum changes from mV forwards to mV backwards, so the total impulse the lamina gives the particle is $2mV$, backwards along the line. Its momentum perpendicular to the line is zero at both ends, so the perpendicular part of the impulse is zero.

By Newton's third law the particle gives the lamina an impulse $2mV$ forwards along the line. The lamina stays at rest, so the peg must supply the opposite impulse, and the lamina pushes on the peg with total impulse $2mV$ in the direction of the particle's original motion. (Any sideways push from the line averages to zero, by the previous paragraph.) Dividing by T, the average force on the peg is

$$\frac{2mV}{\pi r/V} = \frac{2mV^2}{\pi r},$$

in the direction in which the particle entered the groove. (The question's "average force per unit time" means exactly this: impulse per unit time.)

We can also see this directly. The particle needs a force mV^2/r towards the centre, so it pushes on the lamina with mV^2/r outwards. When the radius to the particle makes angle θ with CA (where θ runs from 0 to π at a steady rate), the component of this push along the line is $(mV^2/r)\sin\theta$, and the average of $\sin\theta$ over $[0, \pi]$ is $2/\pi$.

Checks

- If $M \to \infty$ the lamina barely moves: $u \to 0$ and the particle's speed tends to V, as for a fixed groove.
- If $m = M$ the particle comes out at rest and the lamina takes all its speed V, like an elastic collision between equal masses.

- With $M = 3$, $m = 1$, $V = 2$: $v = -1$ and $u = 1$. Momentum: $1 \cdot (-1) + 3 \cdot 1 = 2 = mV$. Energy: $\frac{1}{2} + \frac{3}{2} = 2 = \frac{1}{2} \cdot 1 \cdot 2^2$. A numerical simulation of the equations of motion (for a semicircular groove and for a half-elliptical one) gave the same values to six decimal places, and numerical integration of $(mV^2/r)\sin\theta$ for the fixed lamina gave average force $2mV^2/(\pi r)$, e.g. 1.5117 for $m = 0.7$, $r = 1.3$, $V = 2.1$.

Common slips

- Keeping the root $v = V$, $u = 0$: it satisfies both equations but describes the moment of entry, not exit.
- Using the maximum force mV^2/r instead of the average along the line; the sideways components cancel, and the along-line component reaches mV^2/r only at B.

Beyond STEP

The first part is a hidden collision: whatever the shape of the groove, only the conserved quantities and the exit direction matter, which is the idea behind treating scattering problems by conservation laws without knowing the forces. In Lagrangian mechanics, the coordinate along the line does not appear in the Lagrangian, so its conjugate momentum is conserved automatically; this is the start of the link between symmetries and conservation laws (Noether's theorem). Kleppner and Kolenkow's *An Introduction to Mechanics* treats momentum, impulse and collisions in this spirit.

WORKED SOLUTION

STEP I 1991, Question 14

Probability and Statistics • Counting Triangles

STEP I 1991, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q14

The key idea

Three lengths $a > b > c$ make a non-degenerate triangle exactly when the longest is *strictly* shorter than the sum of the other two, $b + c > a$; the other two triangle inequalities hold automatically. So with a fixed we count, for each b, the values of c that squeeze between $a - b$ and b. These counts go up in steps of 2, which is where the sums of odd (or even) numbers come from. Adding over every possible a then needs only the standard formula for $\sum k^2$.

Solution

When does a triangle exist?

Suppose $a > b > c > 0$. The three triangle inequalities for a non-degenerate triangle are

$$b + c > a, \qquad a + c > b, \qquad a + b > c.$$

The last two hold automatically, because a is already bigger than b and than c. So the three rods form a non-degenerate triangle if and only if $b + c > a$. (If $b + c = a$ the "triangle" is flat, with zero area, and is excluded.)

The largest rod has even length

Fix a. We need integers b, c with

$$a - b < c < b, \qquad b < a.$$

For a given b, the integers c with $a - b + 1 \leq c \leq b - 1$ number

$$(b - 1) - (a - b + 1) + 1 = 2b - a - 1,$$

provided this is positive; otherwise there are none. (Note that $c \geq a - b + 1 \geq 2$ here, so c is automatically a genuine rod length.)

Now let a be even. Then $2b-a-1$ is odd, and it is at least 1 exactly when $b \geq \frac{a}{2}+1$. So b runs from $\frac{a}{2}+1$ up to $a-1$, giving

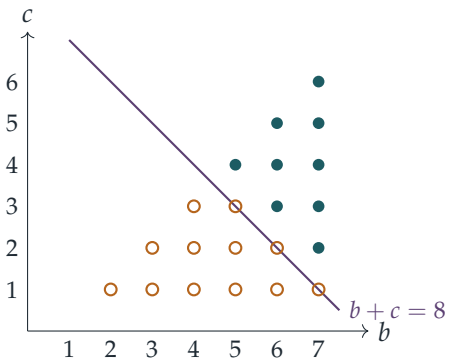
$$\begin{aligned} b = \tfrac{a}{2}+1: &\quad 2b-a-1 = 1, \\ b = \tfrac{a}{2}+2: &\quad 2b-a-1 = 3, \\ &\vdots \\ b = a-1: &\quad 2b-a-1 = a-3. \end{aligned}$$

Each increase of b by 1 adds 2 to the count, so the number of triangles with this largest rod is

$$1+3+5+\cdots+(a-3),$$

as required. (For $a=2$ the sum is empty, which is right: no triangle has longest side 2.) The sum of the first k odd numbers is k^2, and here there are $\frac{a-2}{2}$ terms, so this count is $\left(\frac{a-2}{2}\right)^2$.

The picture below shows the case $a=8$: each dot is a pair (b,c) with $8>b>c$, and the filled dots are those above the line $b+c=8$. The columns contain 1, 3 and 5 filled dots.



The largest rod has odd length

Now let a be odd. Then $2b-a-1$ is even, and it is positive exactly when $b \geq \frac{a+3}{2}$. As b runs from $\frac{a+3}{2}$ to $a-1$ the counts are $2,4,\ldots,a-3$, so the number of triangles is

$$2+4+6+\cdots+(a-3) = 2\left(1+2+\cdots+\tfrac{a-3}{2}\right) = \frac{a-3}{2}\cdot\frac{a-1}{2} = \frac{(a-1)(a-3)}{4}.$$

(Again $a=3$ correctly gives 0.)

The total number of triangles

The largest rod can be any of $3,4,\ldots,2N+1$ (smaller values give zero anyway).

Even $a=2m$, $m=1,\ldots,N$: the count is $(m-1)^2$, so these contribute

$$\sum_{m=1}^{N}(m-1)^2 = \sum_{k=1}^{N-1}k^2 = \frac{(N-1)N(2N-1)}{6}.$$

Odd $a = 2m+1$, $m = 1, \ldots, N$: the count is $\frac{(2m)(2m-2)}{4} = m(m-1)$, so these contribute

$$\sum_{m=1}^{N} (m^2 - m) = \frac{N(N+1)(2N+1)}{6} - \frac{N(N+1)}{2}$$
$$= \frac{N(N+1)(2N-2)}{6} = \frac{(N-1)N(N+1)}{3}.$$

Adding, the number of selections that form a non-degenerate triangle is

$$T_N = \frac{N(N-1)}{6}\big[(2N-1) + 2(N+1)\big] = \frac{N(N-1)(4N+1)}{6}.$$

Ask yourself Why split the sum by parity of a rather than summing a single formula?

The count for a given a is $\frac{(a-2)^2}{4}$ when a is even but $\frac{(a-1)(a-3)}{4} = \frac{(a-2)^2-1}{4}$ when a is odd. These are different polynomials in a, so writing $a = 2m$ or $a = 2m+1$ turns each into a simple polynomial in m that we can sum with the standard formulae.

Verifying for $N = 3$

The formula gives $T_3 = \frac{3\cdot 2\cdot 13}{6} = 13$. Listing directly with rods $1, \ldots, 7$, writing triangles as (a, b, c):

- $a = 4$ (one): $(4,3,2)$;
- $a = 5$ (two): $(5,4,2)$, $(5,4,3)$;
- $a = 6$ (four): $(6,4,3)$, $(6,5,2)$, $(6,5,3)$, $(6,5,4)$;
- $a = 7$ (six): $(7,5,3)$, $(7,5,4)$, $(7,6,2)$, $(7,6,3)$, $(7,6,4)$, $(7,6,5)$.

That is $1 + 2 + 4 + 6 = 13$ triangles, in agreement with the formula (and with $1 = 1^2$, $4 = 2^2$ for even a and $2 = 2\cdot 1$, $6 = 3\cdot 2$ for odd a).

The probability

Drawing three rods at random without replacement, every one of the $\binom{2N+1}{3}$ unordered selections is equally likely. Since

$$\binom{2N+1}{3} = \frac{(2N+1)(2N)(2N-1)}{6},$$

the probability of a non-degenerate triangle is

$$\frac{T_N}{\binom{2N+1}{3}} = \frac{N(N-1)(4N+1)}{(2N+1)\,2N\,(2N-1)} = \frac{(N-1)(4N+1)}{2(4N^2-1)},$$

cancelling the factor N and using $(2N+1)(2N-1) = 4N^2 - 1$. This is the required result.

Checks

- For $N = 3$ the probability is $\frac{2 \cdot 13}{2 \cdot 35} = \frac{13}{35}$, matching 13 triangles out of $\binom{7}{3} = 35$ selections.
- As $N \to \infty$ the probability tends to $\frac{4N^2}{8N^2} = \frac{1}{2}$. This matches the continuous version: for three independent uniform lengths on $[0, 1]$, the chance that the largest is less than the sum of the other two is $\frac{1}{2}$.
- A brute-force count over all triples, for every N from 2 to 29, agreed exactly with T_N and with the probability formula.

Common slips

- Using $b + c \geq a$: equality gives a flat triangle, which the question excludes, and every count comes out too big.
- Checking all three triangle inequalities separately, or forgetting that only $b + c > a$ matters once a is the largest.
- Getting the range of b off by one; test it on a small case such as $a = 6$ (triangles $(6, 4, 3)$, $(6, 5, 2)$, $(6, 5, 3)$, $(6, 5, 4)$, i.e. $1 + 3$).

Beyond STEP

Counting lattice points in a region, as in the picture, is a discrete version of computing an area, and as N grows the count divided by $\binom{2N+1}{3}$ approaches a volume ratio in the unit cube. This passage from discrete to continuous uniform distributions, and "geometric probability" questions like the broken-stick problem, are classic first-year topics. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read more.

WORKED SOLUTION

STEP I 1991, Question 15

Probability and Statistics • Normal Approximation

STEP I 1991, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q15

The key idea

Each X_j has mean 0 and variance 1, and the throws are independent, so S_n has mean 0 and variance n; for large n the central limit theorem lets us treat S_n as approximately normal. The final part looks similar but has a trap: every Y_k contains the *same* first throw X_1, so the Y_k are not independent, and that one shared throw ends up dominating the whole total.

Solution

Each X_j takes the values ± 1 with probability $\frac{1}{2}$ each, so

$$\mathrm{E}(X_j) = 0, \qquad \mathrm{E}(X_j^2) = 1, \qquad \mathrm{Var}(X_j) = 1.$$

The throws are independent, so $\mathrm{E}(S_n) = 0$ and $\mathrm{Var}(S_n) = n$. By the central limit theorem, for large n,

$$S_n \approx \mathrm{N}(0,\, n).$$

(Equivalently, if H is the number of heads then $H \sim \mathrm{B}(n, \frac{1}{2})$ and $S_n = H - (n - H) = 2H - n$.)

Part (i)

With $n = 10\,000$ the standard deviation of S_n is $\sqrt{10\,000} = 100$, so

$$\mathrm{P}(S_n > 100) \approx \mathrm{P}\left(Z > \frac{100}{100}\right) = 1 - \Phi(1) \approx 0.16.$$

If you want to be a little more careful, use a continuity correction on H: $S_n > 100$ means $2H - 10\,000 > 100$, that is $H \geq 5051$. Since H has mean 5000 and standard deviation 50,

$$\mathrm{P}(H \geq 5051) \approx \mathrm{P}\left(Z > \frac{5050.5 - 5000}{50}\right) = 1 - \Phi(1.01) \approx 0.156.$$

Either way the answer is about 0.16.

Part (ii)

Standardising,

$$\mathrm{P}(S_n > 0.01n) \approx \mathrm{P}\left(Z > \frac{0.01n}{\sqrt{n}}\right) = 1 - \Phi\left(0.01\sqrt{n}\right).$$

This decreases as n increases, and it is less than 0.01 exactly when $\Phi(0.01\sqrt{n}) > 0.99$. From tables, $\Phi(2.326) = 0.99$, so we need

$$0.01\sqrt{n} > 2.326 \quad \Longleftrightarrow \quad \sqrt{n} > 232.6 \quad \Longleftrightarrow \quad n > 54\,100 \text{ (approximately).}$$

So the least such n is roughly 54 000. (Using 2.33 instead gives about 54 300; at this level of approximation any answer close to 5.4×10^4 is the right one.)

The new scoring

For $2 \le k \le n$ there are four cases for the first throw and the kth throw:

first throw	kth throw	score Y_k	$X_1 + X_k$
H	H	2	$1+1=2$
T	T	-2	$-1-1=-2$
H	T	0	$1-1=0$
T	H	0	$-1+1=0$

The last two columns agree in every case, so $Y_k = X_1 + X_k$, as required.

Mean and variance. By linearity, $\mathrm{E}(Y_k) = \mathrm{E}(X_1) + \mathrm{E}(X_k) = 0$. Since $k \ge 2$, the throws X_1 and X_k are different throws and so independent, giving

$$\mathrm{Var}(Y_k) = \mathrm{Var}(X_1) + \mathrm{Var}(X_k) = 2.$$

(Directly: Y_k is 2, -2, 0 with probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$, so $\mathrm{E}(Y_k^2) = 2$.)

The limit. Write $m = n - 1$ and $T = Y_2 + Y_3 + \cdots + Y_n$. Then

$$T = \sum_{k=2}^{n} (X_1 + X_k) = mX_1 + S', \qquad \text{where } S' = X_2 + X_3 + \cdots + X_n.$$

Ask yourself Each Y_k has mean 0 and variance 2, just like a scaled X_j. Why can't we simply copy the argument of part (ii)?

Because the argument of part (ii) needed the terms to be independent, and the Y_k are not: they all share X_1. Indeed for $j \ne k$, $\mathrm{Cov}(Y_j, Y_k) = \mathrm{Var}(X_1) = 1$, and $\mathrm{Var}(T) = m^2 + m$, not $2m$. So we look at the two values of X_1 separately. Note that S' is a sum of m terms each ± 1, so $-m \le S' \le m$.

- If the first throw is a tail, $X_1 = -1$ and $T = -m + S' \leq 0$. So $T > 0.01m$ is impossible.
- If the first throw is a head, $X_1 = 1$ and $T = m + S'$, so $T > 0.01m$ exactly when $S' > -0.99m$. By the same normal approximation as in part (ii), with $S' \approx \mathrm{N}(0, m)$,

$$\mathrm{P}(S' \leq -0.99m) \approx \Phi\left(-0.99\sqrt{m}\right) \to 0 \quad \text{as } m \to \infty.$$

  (No approximation is needed: by Chebyshev's inequality this probability is at most $m/(0.99m)^2 \to 0$.)

Since S' is independent of X_1,

$$\mathrm{P}(T > 0.01m) = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}\,\mathrm{P}(S' > -0.99m) \;\to\; \tfrac{1}{2} \quad \text{as } n \to \infty.$$

So the statement is **false**: the probability tends to $\frac{1}{2}$, not 0. In words, for large n the total is positive essentially when the first throw is a head, and otherwise it is not.

Checks

- Part (i): the exact binomial probability $\mathrm{P}(H \geq 5051)$ for $H \sim \mathrm{B}(10\,000, \frac{1}{2})$ is 0.15625, matching the continuity-corrected value 0.156.
- Part (ii): the exact binomial value of $\mathrm{P}(S_n > 0.01n)$ is 0.0125 at $n = 50\,000$, about 0.0100 near $n = 54\,000$, and 0.0094 at $n = 55\,000$, so the threshold is indeed close to 54 000.
- Final part: enumerating all 2^n outcomes for $n \leq 14$ agrees exactly with $\frac{1}{2}\,\mathrm{P}(S' > -0.99m)$; at $n = 14$ the probability is already 0.49994, and a simulation with $n = 100$ gave 0.497.

Common slips

- Treating $Y_2, \ldots, Y_n$ as independent and concluding $\mathrm{Var}(T) = 2(n-1)$, which makes the probability appear to tend to 0.
- Using $\mathrm{Var}(S_n) = n/4$ (the variance of the number of heads) instead of n for the score.

Beyond STEP

Part (ii) is a quantitative form of the weak law of large numbers: S_n/n is very unlikely to be far from 0 once n is large. The final part shows how that law fails when the terms are strongly dependent: here every term shares the first throw, so the average $T/m = X_1 + S'/m$ tends to X_1, a random quantity, rather than to the mean 0. Laws of large numbers for dependent sequences (ergodic theorems, exchangeable sequences) are a rich topic in undergraduate probability; Grimmett and Welsh, *Probability: An Introduction*, covers the weak law and Chebyshev's inequality.

WORKED SOLUTION

STEP I 1991, Question 16

Probability and Statistics • Conditional Probability

STEP I 1991, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q16

The key idea

Organise everything by the inspection times $n = 0, 1, 2, \ldots$ minutes after each reptile first looks at the road. The toad's waiting time is geometric, so she crosses for certain. The frog is different: his rule forces a decision by $n = 3$ at the latest, so his story has only four stages, and we can list the probability that he gets across safely at each one. Once we have that list, the comparison with the toad and the conditional probability are just a few sums.

Solution

Throughout, we assume the state of the road at different inspections is independent, each safe with probability 0.1, and that the toad and the frog act independently. That's reasonable because the frog crosses "later on", at a different time. Since crossing takes negligible time, a reptile who crosses at inspection n arrives at time n minutes.

The toad

The toad only ever crosses when it is safe, so she can never be run over. She is still waiting after inspections $0, 1, \ldots, k$ with probability 0.9^{k+1}, which tends to 0. So she eventually crosses, safely, with probability

$$\sum_{k=0}^{\infty} 0.9^k(0.1) = \frac{0.1}{1 - 0.9} = 1.$$

Write T for the time at which she crosses. Then $P(T = k) = 0.9^k(0.1)$ and, for later use, $P(T > k) = 0.9^{k+1}$.

The frog

At inspection n, if he is still waiting:

- with probability 0.1 the road is safe and he crosses safely;
- with probability $0.9 \cdot \frac{n}{3}$ it is unsafe but he goes anyway, arriving safely with probability 0.2 and being run over with probability 0.8;
- with probability $0.9\left(1 - \frac{n}{3}\right)$ he waits for the next inspection.

At $n = 0$ he never takes a risk, and at $n = 3$ the chance of going when it's unsafe is 1, so he certainly sets off at $n = 3$ if he hasn't already. In particular the case $n > 3$ never arises.

The probability that he is still waiting at inspection n is

$$w_0 = 1, \quad w_1 = 0.9, \quad w_2 = 0.9 \cdot 0.9 \cdot \tfrac{2}{3} = 0.54, \quad w_3 = 0.54 \cdot 0.9 \cdot \tfrac{1}{3} = 0.162.$$

At inspection n he arrives safely with probability $w_n\left(0.1 + 0.9 \cdot \frac{n}{3} \cdot 0.2\right) = w_n(0.1 + 0.06n)$. Writing s_n for this probability,

$$s_0 = 1 \times 0.1 = 0.1, \qquad s_1 = 0.9 \times 0.16 = 0.144,$$
$$s_2 = 0.54 \times 0.22 = 0.1188, \qquad s_3 = 0.162 \times 0.28 = 0.04536.$$

Hence

$$P(\text{frog crosses safely}) = 0.1 + 0.144 + 0.1188 + 0.04536 = \boxed{0.40816 = \tfrac{2551}{6250}}.$$

He is run over with the remaining probability $1 - 0.40816 = 0.59184$.

Both safe, frog quicker

Let F be the time at which the frog arrives safely. We need both to arrive safely with $F < T$. The toad is certainly safe, and by independence

$$P(\text{frog safe at } n \text{ and } T > n) = s_n\, P(T > n) = s_n (0.9)^{n+1}.$$

Ask yourself Why is it $T > n$ and not $T \geq n$?

If both cross at the same inspection time n, they take the same time, and the frog isn't strictly quicker. So the toad must still be waiting after her inspection at time n, which has probability 0.9^{n+1}. Summing over the frog's four possible times,

$$\begin{aligned} P &= 0.1(0.9) + 0.144(0.81) + 0.1188(0.729) + 0.04536(0.6561) \\ &= 0.09 + 0.11664 + 0.0866052 + 0.029760696 \\ &= 0.323005896 \approx \boxed{0.323}. \end{aligned}$$

As an exact fraction this is $\frac{40375737}{125000000}$.

The final part

Let A be the event that the frog has not reached the other side 2 minutes after his first inspection, and R the event that he is run over. Because crossing is instantaneous, a

frog who gets across safely at the inspection at $n = 2$ *has* arrived by 2 minutes. So A is the complement of "safe at $n = 0$, 1 or 2":

$$P(A) = 1 - (s_0 + s_1 + s_2) = 1 - 0.3628 = 0.6372.$$

A frog who is run over never arrives, at any time, so $R \subseteq A$ and $P(R \cap A) = P(R) = 0.59184$. Therefore

$$P(R \mid A) = \frac{0.59184}{0.6372} = \frac{59184}{63720} = \boxed{\frac{274}{295} \approx 0.929}.$$

This includes frogs who have already been run over (at $n = 1$ or $n = 2$) and frogs still waiting, who are then run over at $n = 3$, as "at some stage" requires.

(Some readers take the condition to mean that he hasn't arrived after the inspections at $n = 0$ and $n = 1$, that is, *before* the $n = 2$ look. That reading gives $0.59184/(1 - 0.244) = \frac{137}{175} \approx 0.783$. We prefer the reading above, since an arrival at exactly 2 minutes is an arrival "2 minutes after" he began.)

Checks

- The frog's outcomes must account for everything: the run-over probabilities at $n = 1, 2, 3$ are $0.9(0.24)$, $0.54(0.48)$ and $0.162(0.72)$, i.e. $0.216 + 0.2592 + 0.11664 = 0.59184$, which is exactly $1 - 0.40816$.
- The frog's safe probability is well below the toad's certain 1, as it should be for the impatient one.
- An exact enumeration with fractions and a Monte Carlo simulation of 10^6 pairs of reptiles gave 0.4087, 0.3234 and 0.9286, matching 0.40816, 0.32301 and 0.92881.

Common slips

- Letting the frog take a risk at $n = 0$: the rule gives $n/3 = 0$ there.
- Forgetting that the frog's process stops at $n = 3$, and summing a geometric series for him as for the toad.
- In the final part, dividing by $P(\text{not safe at } n \leq 2)$ but putting only the run-overs *up to* $n = 2$ on top. The question asks about being run over at any stage, including $n = 3$.

Beyond STEP

The frog's journey is a small *Markov chain* with absorbing states ("safe" and "run over"), and the toad's waiting time is the geometric distribution, the discrete version of the memoryless exponential. Absorption probabilities and hitting times of Markov chains are a core topic in a first probability course. Grimmett and Welsh, *Probability: An Introduction*, develops both.

WORKED SOLUTION

STEP II 1991, Question 1

Pure • Quadratics and Discriminants

STEP II 1991, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q1

The key idea

A quadratic is a constant multiple of a perfect square exactly when it has a repeated root, that is, when its discriminant is zero. Every later part is that one condition applied to a quadratic whose coefficients depend on λ. Asking for the λ that make the discriminant vanish gives a quadratic *in* λ, and the decomposition then comes from solving two linear equations for f and g.

Solution

The condition for the form $(*)$

Expanding,

$$a(x+k)^2 = ax^2 + 2akx + ak^2.$$

If $\mathrm{h}(x)$ has this form then, comparing coefficients, $b = 2ak$ and $c = ak^2$, so

$$b^2 = 4a^2k^2 = 4a \cdot ak^2 = 4ac.$$

Conversely, if $b^2 = 4ac$, take $k = \dfrac{b}{2a}$ (allowed since $a \neq 0$). Then $2ak = b$ and $ak^2 = \dfrac{b^2}{4a} = c$, so $\mathrm{h}(x) = a(x+k)^2$. The condition is therefore

$$b^2 = 4ac,$$

which says that the discriminant is zero.

The two values of λ

We have

$$\mathrm{f}(x) + \lambda \mathrm{g}(x) = (3+\lambda)x^2 + 4x - 2\lambda.$$

If $\lambda = -3$ this is $4x + 6$. That is not of the form $(*)$, which needs a nonzero coefficient of x^2. So $3 + \lambda \neq 0$, and by the condition above we need

$$16 = 4(3+\lambda)(-2\lambda) \iff \lambda^2 + 3\lambda + 2 = 0 \iff (\lambda+1)(\lambda+2) = 0.$$

So $\lambda = -1$ or $\lambda = -2$, and neither is -3. Explicitly,

$$f(x) - g(x) = 2x^2 + 4x + 2 = 2(x+1)^2,$$
$$f(x) - 2g(x) = x^2 + 4x + 4 = (x+2)^2.$$

The decomposition

Ask yourself How do two facts about $f - g$ and $f - 2g$ tell us f and g separately?

Treat the two displayed identities as simultaneous linear equations in the unknowns $f(x)$ and $g(x)$. Subtracting the second from the first gives

$$g(x) = 2(x+1)^2 - (x+2)^2,$$

and then $f(x) = (f(x) - 2g(x)) + 2g(x) = (x+2)^2 + 4(x+1)^2 - 2(x+2)^2$, that is,

$$f(x) = 4(x+1)^2 - (x+2)^2.$$

Checking by expansion: $4(x^2 + 2x + 1) - (x^2 + 4x + 4) = 3x^2 + 4x$, and $2(x^2 + 2x + 1) - (x^2 + 4x + 4) = x^2 - 2$. So one choice is

$$A = 4, \quad B = -1, \quad C = 2, \quad D = -1, \quad m = 1, \quad n = 2.$$

Swapping the roles of m and n gives the equivalent answer $m = 2$, $n = 1$, $A = -1$, $B = 4$, $C = -1$, $D = 2$.

The final part

Now

$$f(x) + \lambda g(x) = (3 + \lambda)x^2 + 4x + \lambda\alpha.$$

As before, $\lambda = -3$ gives the linear expression $4x - 3\alpha$, which is never of the form $(*)$. For $\lambda \neq -3$ the condition $b^2 = 4ac$ reads

$$16 = 4(3 + \lambda)\lambda\alpha \iff \alpha\lambda^2 + 3\alpha\lambda - 4 = 0. \qquad (\dagger)$$

We want exactly one value of λ satisfying $(\dagger)$ with $\lambda \neq -3$.

Ask yourself Could $(\dagger)$ have two roots, one of which is the forbidden value $\lambda = -3$?

No. Putting $\lambda = -3$ into the left-hand side of $(\dagger)$ gives $9\alpha - 9\alpha - 4 = -4 \neq 0$, so $\lambda = -3$ is never a root. The values of λ that work are therefore exactly the roots of $(\dagger)$.

- If $\alpha = 0$, $(\dagger)$ becomes $-4 = 0$, which has no solutions. So $\alpha \neq 0$.
- If $\alpha \neq 0$, $(\dagger)$ is a genuine quadratic in λ. It has exactly one root precisely when its discriminant is zero:

  $$9\alpha^2 + 16\alpha = 0 \iff \alpha(9\alpha + 16) = 0 \iff \alpha = -\tfrac{16}{9},$$

  using $\alpha \neq 0$.

Hence

$$\alpha = -\frac{16}{9}.$$

The repeated root is $\lambda = -\frac{3\alpha}{2\alpha} = -\frac{3}{2}$, and indeed

$$f(x) - \tfrac{3}{2}\left(x^2 - \tfrac{16}{9}\right) = \tfrac{3}{2}x^2 + 4x + \tfrac{8}{3} = \tfrac{3}{2}\left(x + \tfrac{4}{3}\right)^2.$$

Checks

Expand the decomposition at a test value: at $x = 1$, $4(2)^2 - (3)^2 = 7 = f(1)$ and $2(2)^2 - 3^2 = -1 = g(1)$. In the final part, $\frac{3}{2}(x + \frac{4}{3})^2$ at $x = 0$ is $\frac{3}{2} \cdot \frac{16}{9} = \frac{8}{3}$, matching $\lambda\alpha = (-\frac{3}{2})(-\frac{16}{9})$. A computer testing every rational $\lambda = p/q$ with $|p| \leq 60$, $q \leq 12$ found exactly $\lambda = -1, -2$ in the middle part and only $\lambda = -\frac{3}{2}$ when $\alpha = -\frac{16}{9}$.

Common slips

- Forgetting the requirement $a \neq 0$ in $(*)$: at $\lambda = -3$ the expression is linear, and you should say why that value is excluded.
- In the final part, stopping at "discriminant of $(\dagger)$ is zero" without dealing with $\alpha = 0$, where $(\dagger)$ is not a quadratic at all and has no roots.

Beyond STEP

Writing f and g as combinations of two squares of linear forms is simultaneous diagonalisation of a pair of quadratic forms. The values of λ that make $f + \lambda g$ degenerate are the roots of $\det(F + \lambda G) = 0$ for the corresponding symmetric matrices, a "generalised eigenvalue" problem. The final part, where the two values of λ coincide, is exactly the case in which such a diagonalisation can fail. Strang's *Introduction to Linear Algebra* treats quadratic forms and eigenvalues in this spirit.

WORKED SOLUTION

STEP II 1991, Question 2

Pure • Asymptotes of a Hyperbola

STEP II 1991, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q2

The key idea

The quadratic part $3y^2 - 10xy + 3x^2$ factorises as $(3y - x)(y - 3x)$, and this is the whole story: once you complete the factorisation to include the linear terms, the curve becomes "product of two linear expressions $= 1$". Far out along the curve one factor must blow up, so the other must tend to zero, and that factor, set equal to zero, is an asymptote. This is exactly the "relationship between s and t in the limit" that the hint asks for.

Solution

The tangent

Differentiating $(\dagger)$ implicitly with respect to x,

$$6y\frac{\mathrm{d}y}{\mathrm{d}x} - 10y - 10x\frac{\mathrm{d}y}{\mathrm{d}x} + 6x + 16\frac{\mathrm{d}y}{\mathrm{d}x} - 16 = 0,$$

so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{10y - 6x + 16}{6y - 10x + 16} = \frac{5y - 3x + 8}{3y - 5x + 8}.$$

At (s, t) the gradient is $m = \dfrac{5t - 3s + 8}{3t - 5s + 8}$, and the tangent is $y = mx + (t - ms)$. Putting the intercept over the common denominator,

$$t - ms = \frac{t(3t - 5s + 8) - s(5t - 3s + 8)}{3t - 5s + 8} = \frac{3t^2 - 10st + 3s^2 + 8t - 8s}{3t - 5s + 8}.$$

Since (s, t) lies on the curve, $3t^2 - 10st + 3s^2 = -16t + 16s - 15$, so the numerator is $-8t + 8s - 15 = -(8t - 8s + 15)$. Hence the tangent is

$$y = \left(\frac{5t - 3s + 8}{3t - 5s + 8}\right)x - \left(\frac{8t - 8s + 15}{3t - 5s + 8}\right),$$

as required. (Where $3t - 5s + 8 = 0$ the tangent is vertical, and the formula is to be read in the cleared-denominator form.)

The asymptotes

Try to write (†) as $(3y - x + a)(y - 3x + b) + k = 0$. Expanding,

$$(3y - x + a)(y - 3x + b) = 3y^2 - 10xy + 3x^2 + (3b + a)y - (b + 3a)x + ab,$$

so we need $3b + a = 16$ and $b + 3a = 16$, giving $a = b = 4$ and $ab = 16$. Therefore (†) is

$$(3y - x + 4)(y - 3x + 4) = 1.$$

For the point (s, t) write

$$u = 3t - s + 4, \qquad v = t - 3s + 4, \qquad \text{so that} \qquad uv = 1.$$

Every expression in the tangent is a combination of u, v and constants. Matching coefficients of t and s gives

$$\begin{aligned} 5t - 3s + 8 &= \tfrac{1}{2}(3u + v), \\ 3t - 5s + 8 &= \tfrac{1}{2}(u + 3v), \\ 8t - 8s + 15 &= 2(u + v) - 1 = 2(u + v) - uv, \end{aligned}$$

using $uv = 1$ in the last one. So the gradient and intercept of the tangent are

$$m = \frac{3u + v}{u + 3v}, \qquad c = \frac{8t - 8s + 15}{3t - 5s + 8} = \frac{2(2u + 2v - uv)}{u + 3v}.$$

Ask yourself As $s \to \infty$, what happens to u and v?

Eliminating t, $u - 3v = 8s - 8$, so $u - 3v \to \infty$. Since $uv = 1 > 0$, u and v have the same sign.

- If both are positive, $u - 3v \to \infty$ forces $u \to \infty$, and then $v = 1/u \to 0$.
- If both are negative, $u - 3v = 3|v| - |u| \to \infty$ forces $|v| \to \infty$, and then $u = 1/v \to 0$.

This is the relationship the hint wants: on one branch $t - 3s + 4 \to 0$, on the other $3t - s + 4 \to 0$.

Case $v \to 0$, $u \to \infty$. Dividing top and bottom by u,

$$m = \frac{3 + v/u}{1 + 3v/u} \to 3, \qquad c = \frac{2(2 + 2v/u - v)}{1 + 3v/u} \to 4,$$

so the limiting tangent is $y = 3x - 4$.

Case $u \to 0$, $v \to \infty$. Dividing top and bottom by v,

$$m = \frac{3u/v + 1}{u/v + 3} \to \tfrac{1}{3}, \qquad c = \frac{2(2u/v + 2 - u)}{u/v + 3} \to \tfrac{4}{3},$$

so the limiting tangent is $y = \frac{1}{3}x - \frac{4}{3}$, that is, $3y = x - 4$.

So the asymptotes are $y = 3x - 4$ and $3y = x - 4$. (Letting $s \to -\infty$ instead gives the same two lines with the roles of the cases swapped.) Notice that these are exactly the lines $v = 0$ and $u = 0$: the factorised form shows them straight away.

Angles and axes

The line $y = 3x - 4$ has gradient 3, so it makes an angle θ with the x-axis where $\tan\theta = 3$. Its angle with the y-axis is then $\frac{\pi}{2} - \theta$, and $\tan(\frac{\pi}{2} - \theta) = \cot\theta = \frac{1}{3}$. The line $3y = x - 4$ has gradient $\frac{1}{3}$, so its angle with the x-axis is $\arctan\frac{1}{3}$, the same as the angle between the first asymptote and the y-axis.

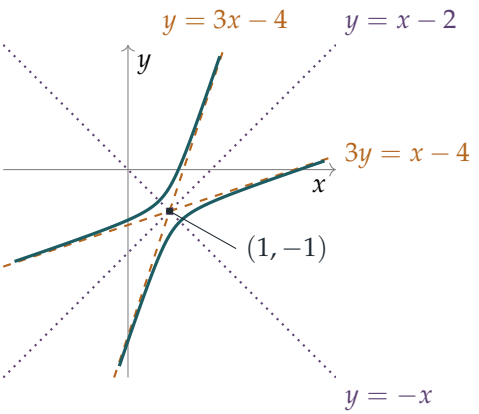
? **Ask yourself** Why does this tell us the bisectors?

Reflection in the line $y = x$ swaps the x- and y-axes, and by what we've just shown it swaps the directions of the two asymptotes: direction $(1,3)$ goes to $(3,1)$. The angle between the asymptotes is therefore symmetric about the direction of $y = x$, which bisects it: the asymptote directions are at $\arctan 3$ and $\arctan\frac{1}{3}$ from the x-axis, and these are $\frac{\pi}{4} \pm (\arctan 3 - \frac{\pi}{4})$. The other bisector is perpendicular to the first. So the bisectors have gradients 1 and -1.

The axes of a hyperbola are the angle bisectors of its asymptotes through its centre, the point where the asymptotes meet. Solving $y = 3x - 4$ and $3y = x - 4$: $9x - 12 = x - 4$, so $x = 1$, $y = -1$. The centre is $(1,-1)$, and the axes are

$$y + 1 = x - 1 \iff y = x - 2, \qquad y + 1 = -(x - 1) \iff y = -x.$$

To see which is which: on $y = -x$ we get $u = v = 4 - 4x$, so $uv = 1$ gives $x = 1 \pm \frac{1}{4}$, and the vertices are $(\frac{5}{4}, -\frac{5}{4})$ and $(\frac{3}{4}, -\frac{3}{4})$. So $y = -x$ is the transverse axis. On $y = x - 2$ we get $u = 2x - 2 = -v$, so $uv = -(2x-2)^2 \leq 0$ and that line (the conjugate axis) never meets the curve.



Checks

- The point $(\frac{5}{4}, -\frac{5}{4})$ satisfies (†): $3 \cdot \frac{25}{16} + 10 \cdot \frac{25}{16} + 3 \cdot \frac{25}{16} - 20 - 20 + 15 = 25 - 25 = 0$.
- Numerically, taking $u = 10^8$ (so $s \approx 1.25 \times 10^7$), the tangent formula gives gradient 3.0000000 and intercept term 3.99999998; taking $u = -10^{-8}$ gives 0.33333333 and 1.33333334.

- The centre $(1,-1)$ gives $u = v = 0$, so it is not on the curve, which is why the tangent formula never has $0/0$.

Common slips

- Finding only the gradients 3 and $\frac{1}{3}$ from $3k^2 - 10k + 3 = 0$ (with $t \approx ks$) and forgetting that the intercept needs the next term: $t = 3s - 4 + o(1)$, not just $t \approx 3s$.
- Taking the bisectors through the origin rather than through the centre $(1,-1)$.

Beyond STEP

Every conic $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ can be written as $\mathbf{x}^{\mathsf{T}} A\mathbf{x} + \ldots$ with a symmetric matrix $A = \begin{pmatrix} a & h \\ h & b \end{pmatrix}$, and its axes point along the eigenvectors of A. Here $A = \begin{pmatrix} 3 & -5 \\ -5 & 3 \end{pmatrix}$ has eigenvectors $(1,1)$ and $(1,-1)$, with eigenvalues -2 and 8 of opposite sign, which is what makes the curve a hyperbola. Diagonalising quadratic forms like this is a central topic of first-year linear algebra; see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP II 1991, Question 3

Pure • Symmetric Simultaneous Equations

STEP II 1991, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q3

The key idea

Three equal expressions give two equations, and with three unknowns we can't hope to find x, y, z themselves. What we can do is exploit the cyclic symmetry. Subtracting pairs of expressions turns each difference $x - y$ into a multiple of the next difference $y - z$, and multiplying the three relations together makes almost everything cancel. For the second result, giving the common value a name k and clearing fractions produces three cyclic linear relations that pin down xyz in three different ways.

Solution

Showing that $x^2y^2z^2 = 1$

Equating the first two expressions,

$$x - y = \frac{1}{z} - \frac{1}{y} = \frac{y - z}{yz}.$$

In the same way, equating the second and third, and then the third and first,

$$y - z = \frac{1}{x} - \frac{1}{z} = \frac{z - x}{zx}, \qquad z - x = \frac{1}{y} - \frac{1}{x} = \frac{x - y}{xy}.$$

Multiplying these three equations together gives

$$(x - y)(y - z)(z - x) = \frac{(y - z)(z - x)(x - y)}{x^2y^2z^2}.$$

Because x, y and z are distinct, the product $(x - y)(y - z)(z - x)$ is not zero, so we may divide both sides by it. This leaves $1 = 1/(x^2y^2z^2)$, that is,

$$x^2y^2z^2 = 1,$$

as required.

Ask yourself Where exactly did we use that x, y, z are distinct and non-zero?

Non-zero is needed for the fractions $1/x$, $1/y$, $1/z$ to make sense at all. Distinctness is what lets us cancel $(x-y)(y-z)(z-x)$. Without it the result fails: $x=y=z=2$ satisfies the equations, with $x^2y^2z^2=64$.

The common value is ± 1

Let k be the common value, so

$$x+\frac{1}{y}=k, \qquad y+\frac{1}{z}=k, \qquad z+\frac{1}{x}=k.$$

Multiplying through by y, z and x respectively (all non-zero),

$$xy=ky-1, \qquad yz=kz-1, \qquad zx=kx-1. \tag{$*$}$$

Now multiply the first of these by z and use the second to replace yz:

$$\begin{aligned} xyz &= k\,yz - z \\ &= k(kz-1)-z \\ &= (k^2-1)z-k. \end{aligned}$$

The relations $(*)$ are unchanged when we replace $x \to y \to z \to x$, so the same calculation, started from the second and third relations, gives

$$xyz=(k^2-1)x-k, \qquad xyz=(k^2-1)y-k.$$

Subtracting the first two expressions for xyz,

$$(k^2-1)(z-x)=0.$$

Since $z \neq x$, we must have $k^2=1$, so the value of $x+\frac{1}{y}$ is either $+1$ or -1, as required.

As a bonus, putting $k^2=1$ back in gives $xyz=-k$. This is a second route to the first result, since then $x^2y^2z^2=k^2=1$.

Checks

Both values really occur. With $k=1$, start from any $x \neq 0,1$ and set $y=\frac{1}{1-x}$ and $z=\frac{1}{1-y}=\frac{x-1}{x}$. Then $z+\frac{1}{x}=\frac{x-1}{x}+\frac{1}{x}=1$, so all three expressions equal 1. For instance $x=2$ gives $y=-1$, $z=\frac{1}{2}$: indeed $2-1=-1+2=\frac{1}{2}+\frac{1}{2}=1$, and $xyz=-1=-k$. Replacing (x,y,z) by $(-x,-y,-z)$ gives solutions with $k=-1$, such as $(-2,1,-\frac{1}{2})$. The three numbers are always distinct here, because $x=\frac{1}{1-x}$ would need $x^2-x+1=0$, which has no real roots. We checked about 4000 such triples with exact fractions, confirming each time that $x^2y^2z^2=1$ and $xyz=-k$.

Common slips

- Dividing by $(x-y)(y-z)(z-x)$ without saying why it is non-zero. The distinctness condition is there precisely for this.
- Concluding $xyz = 1$ instead of $xyz = \pm 1$: in fact $xyz = -k$, so $xyz = -1$ when the common value is 1.

Beyond STEP

The map $t \mapsto \dfrac{1}{k-t}$ takes x to y, y to z and z back to x. It is a Möbius transformation, with matrix $\begin{pmatrix} 0 & 1 \\ -1 & k \end{pmatrix}$, and composing such maps corresponds to multiplying their matrices. The question is really asking when this map has a 3-cycle, and the answer, $k^2 = 1$, is exactly when the cube of the matrix is a multiple of the identity. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations and their classification by the trace of the matrix.

WORKED SOLUTION

STEP II 1991, Question 4

Pure • Sums of Cosines

STEP II 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 6.2, *Gauss's 17-gon*: https://beyondstep.keithmcnulty.org/ch06/gauss-17gon.html

The key idea

Because $5\phi = 2\pi$ and $17\theta = 2\pi$, cosines of large multiples of the angle fold back onto small ones: $\cos(n-k)\phi = \cos k\phi$ when $n\phi = 2\pi$. So when we square or multiply sums of cosines with the product formulae, everything collapses back into the same few cosines. In the second half we split the eight cosines $\cos\theta, \ldots, \cos 8\theta$ into two groups of four, find their sum and their product, and so get a quadratic of which z is a root.

Solution

The pentagon: $\phi = 2\pi/5$

We use the product formula $2\cos A\cos B = \cos(A+B) + \cos(A-B)$, together with $\cos^2 A = \frac{1}{2}(1+\cos 2A)$. Since $5\phi = 2\pi$,

$$\cos 3\phi = \cos(2\pi - 2\phi) = \cos 2\phi, \qquad \cos 4\phi = \cos(2\pi - \phi) = \cos\phi.$$

Hence

$$\begin{aligned} y^2 &= \cos^2\phi + 2\cos\phi\cos 2\phi + \cos^2 2\phi \\ &= \tfrac{1}{2}(1+\cos 2\phi) + (\cos 3\phi + \cos\phi) + \tfrac{1}{2}(1+\cos 4\phi) \\ &= \tfrac{1}{2}(1+\cos 2\phi) + (\cos 2\phi + \cos\phi) + \tfrac{1}{2}(1+\cos\phi) \\ &= 1 + \tfrac{3}{2}(\cos\phi + \cos 2\phi) = 1 + \tfrac{3}{2}y. \end{aligned}$$

Multiplying by 2 gives $2y^2 = 3y + 2$, as required.

This factorises as $(2y+1)(y-2) = 0$, so $y = -\frac{1}{2}$ or $y = 2$. But $2\phi = \frac{4\pi}{5}$ lies between $\frac{\pi}{2}$ and π, so $\cos 2\phi < 0$, while $\cos\phi \le 1$. Thus $y < 1$, which rules out $y = 2$, and so

$$y = \cos\tfrac{2\pi}{5} + \cos\tfrac{4\pi}{5} = -\tfrac{1}{2}.$$

The sum of the cosines for $\theta = 2\pi/17$

Let $\omega = \cos\theta + \mathrm{i}\sin\theta$. By de Moivre's theorem $\omega^k = \cos k\theta + \mathrm{i}\sin k\theta$, and $\omega^{17} = \cos 2\pi + \mathrm{i}\sin 2\pi = 1$, while $\omega \neq 1$. Summing the geometric series,

$$\sum_{k=0}^{16} \omega^k = \frac{\omega^{17} - 1}{\omega - 1} = 0.$$

Taking real parts gives $\displaystyle\sum_{k=0}^{16} \cos k\theta = 0$, as required.

The value of z

Write $c_k = \cos k\theta$. Since $17\theta = 2\pi$ and cosine is even,

$$c_{17-k} = c_k \quad \text{and} \quad c_{-k} = c_k,$$

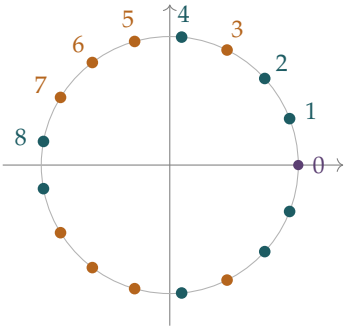
so every c_k equals one of $c_1, c_2, \ldots, c_8$. In particular the sum just proved, with $c_0 = 1$ removed, reads $2(c_1 + c_2 + \cdots + c_8) = -1$, so

$$c_1 + c_2 + \cdots + c_8 = -\tfrac{1}{2}. \qquad (*)$$

The angles in z are $1, 2, 4, 8$ (each double the last, reduced as above: $16 \to 1$). The other four, $3, 5, 6, 7$, give the companion sum

$$w = c_3 + c_5 + c_6 + c_7.$$

The figure shows the seventeen points ω^k on the unit circle: the teal ones (and their mirror images) carry the cosines in z, the amber ones those in w.



The sum. By $(*)$, $z + w = c_1 + c_2 + \cdots + c_8 = -\tfrac{1}{2}$.

The product. Expanding zw gives sixteen products $c_a c_b$ with $a \in \{1, 2, 4, 8\}$ and $b \in \{3, 5, 6, 7\}$, and each is $\frac{1}{2}(c_{a+b} + c_{a-b})$. Reducing each index to the range 1 to 8 (using $c_{-k} = c_k$ and $c_{17-k} = c_k$), the pairs of indices are:

	$b=3$	$b=5$	$b=6$	$b=7$
$a=1$	4, 2	6, 4	7, 5	8, 6
$a=2$	5, 1	7, 3	8, 4	8, 5
$a=4$	7, 1	8, 1	7, 2	6, 3
$a=8$	6, 5	4, 3	3, 2	2, 1

(For example $a=8$, $b=7$ gives c_{15} and c_1, and $c_{15}=c_2$.)

Ask yourself Do we need to know which cosines appear, or only how often?

Only how often. Counting the table, each of the indices $1,2,\dots,8$ appears exactly four times among the 32 entries. So

$$zw=\tfrac{1}{2}\cdot 4\,(c_1+c_2+\cdots+c_8)=2\cdot\left(-\tfrac{1}{2}\right)=-1,$$

using $(*)$ again.

The quadratic. Since $z+w=-\frac{1}{2}$ and $zw=-1$, the numbers z and w are the roots of

$$t^2+\tfrac{1}{2}t-1=0,\qquad\text{that is,}\qquad 2t^2+t-2=0,$$

so $\{z,w\}=\left\{\dfrac{-1+\sqrt{17}}{4},\ \dfrac{-1-\sqrt{17}}{4}\right\}$. The first root is positive and the second negative.

Choosing the root. We show $z>0$. Since $\frac{4}{17}<\frac{1}{4}$, both θ and 2θ lie in $(0,\frac{\pi}{4})$, so c_1 and c_2 both exceed $\frac{1}{\sqrt{2}}$ and $c_1+c_2>\sqrt{2}>1$. Also $4\theta=\frac{8\pi}{17}<\frac{\pi}{2}$, so $c_4>0$, and $c_8\geq -1$. Therefore

$$z=(c_1+c_2)+c_4+c_8>1+0-1=0,$$

and z must be the positive root:

$$z=\frac{-1+\sqrt{17}}{4}=-\frac{1-\sqrt{17}}{4}.$$

Checks

Numerically, $\cos 72^\circ+\cos 144^\circ=0.30902-0.80902=-0.5$. With $\theta=2\pi/17$, a calculator gives $z=0.932472+0.739009+0.092268-0.982973=0.780776$, and $(\sqrt{17}-1)/4=0.780776$; also $w=-1.280776=(-1-\sqrt{17})/4$, so $z+w=-0.5$ and $zw=-1$, as found.

Common slips

- Forgetting to rule out $y=2$ in the first part, or w's root in the second: a quadratic gives two candidates, and a separate sign argument is needed to pick one.

- Including $k = 0$ in $(*)$: the full sum is 0, but $c_0 = 1$, so $c_1 + \cdots + c_{16} = -1$, and halving gives $-\frac{1}{2}$.
- Mis-reducing indices in the product table, such as treating c_9 as new rather than as c_8.

Beyond STEP

This is Gauss's famous calculation showing that the regular 17-gon can be constructed with ruler and compasses. Splitting the cosines by the powers of 3 (a generator modulo 17) into nested groups of 8, 4, 2 and 1 gives a chain of quadratics, so $\cos\frac{2\pi}{17}$ can be written using only square roots; z and w are the first stage of that chain, called Gaussian periods. The full story belongs to Galois theory: see Allenby, *Rings, Fields and Groups*, and, for the arithmetic of roots of unity, Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP II 1991, Question 5

Pure • Integration Reduction Formulae

STEP II 1991, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q5

The key idea

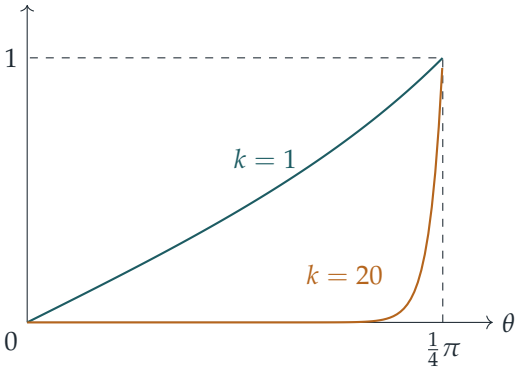
Write $I_k = \int_0^{\pi/4} \tan^k \theta \, \mathrm{d}\theta$. Odd and even powers of tan don't integrate directly, but two of them added together do: $\tan^k \theta + \tan^{k+2} \theta = \tan^k \theta \sec^2 \theta$, which is the derivative of $\tan^{k+1} \theta / (k+1)$. That reduction formula, unwound by induction, gives the formula; the sketch for large k tells you the integral shrinks to zero, and that is what turns the formula into the sum of an infinite series.

Solution

The sketches

On $[0, \frac{1}{4}\pi]$ we have $0 \leq \tan\theta \leq 1$, with $\tan 0 = 0$ and $\tan \frac{1}{4}\pi = 1$. For $k = 1$ the graph rises from $(0,0)$ to $(\frac{1}{4}\pi, 1)$, gently curving upwards (it starts with gradient 1 and ends with gradient $\sec^2 \frac{1}{4}\pi = 2$).

For $k \gg 1$, every value of $\tan\theta$ below 1 is crushed towards 0 by raising it to a high power, while the end values 0 and 1 stay put. So $\tan^k \theta$ hugs the axis for most of the interval and then shoots up to 1 very close to $\theta = \frac{1}{4}\pi$. In fact, writing $\theta = \frac{1}{4}\pi - u$ with u small, $\tan\theta \approx 1 - 2u$, so $\tan^k \theta \approx (1-2u)^k \approx \mathrm{e}^{-2ku}$: the spike has width about $1/(2k)$, and the area under it is about $1/(2k)$.



The reduction formula

For $k \geq 0$,

$$I_k + I_{k+2} = \int_0^{\pi/4} \tan^k\theta\,(1+\tan^2\theta)\,\mathrm{d}\theta = \int_0^{\pi/4} \tan^k\theta\sec^2\theta\,\mathrm{d}\theta = \Big[\frac{\tan^{k+1}\theta}{k+1}\Big]_0^{\pi/4} = \frac{1}{k+1}.$$

We also need the two starting values:

$$I_0 = \tfrac{1}{4}\pi, \qquad I_1 = \int_0^{\pi/4} \frac{\sin\theta}{\cos\theta}\,\mathrm{d}\theta = \Big[-\ln\cos\theta\Big]_0^{\pi/4} = -\ln\tfrac{1}{\sqrt{2}} = \tfrac{1}{2}\ln 2.$$

The odd powers

Let $J_n = I_{2n+1}$. Taking $k = 2n-1$ in the reduction formula gives

$$J_n = \frac{1}{2n} - J_{n-1} \qquad (n \geq 1).$$

We prove the required formula by induction. For $n = 1$: $J_1 = \frac{1}{2} - J_0 = \frac{1}{2} - \frac{1}{2}\ln 2 = (-1)^1\left(\frac{1}{2}\ln 2 - \frac{1}{2}\right)$, which is the formula with $n = 1$.

Suppose it holds for $n - 1 \geq 1$. Then

$$\begin{aligned} J_n &= \frac{1}{2n} - (-1)^{n-1}\Big(\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n-1}\frac{(-1)^m}{2m}\Big) \\ &= (-1)^n\Big(\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n-1}\frac{(-1)^m}{2m}\Big) + (-1)^n\,\frac{(-1)^n}{2n} \\ &= (-1)^n\Big(\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n}\frac{(-1)^m}{2m}\Big), \end{aligned}$$

using $(-1)^n(-1)^n = 1$ to write $\frac{1}{2n}$ as the $m = n$ term inside the bracket. By induction the formula holds for every positive integer n, as required.

The first series

Ask yourself How do we know $J_n \to 0$, rather than just "it looks small on the sketch"?

We need an honest bound. On $[0, \frac{1}{4}\pi]$ we have $\tan\theta \geq 0$ and $\sec^2\theta \geq 1$, so

$$0 \leq J_n = \int_0^{\pi/4} \tan^{2n+1}\theta \,\mathrm{d}\theta \leq \int_0^{\pi/4} \tan^{2n+1}\theta \sec^2\theta \,\mathrm{d}\theta = \frac{1}{2n+2}.$$

Hence $J_n \to 0$ as $n \to \infty$, which is exactly what the large-k sketch suggested (and the bound $1/(2n+2)$ matches the spike's area of about $1/(2k)$). Since $|J_n|$ is the absolute value of the bracket in the formula,

$$\left|\tfrac{1}{2}\ln 2 + \sum_{m=1}^{n} \frac{(-1)^m}{2m}\right| = J_n \leq \frac{1}{2n+2} \to 0,$$

so the partial sums $\sum_{m=1}^{n} \frac{(-1)^m}{2m}$ converge to $-\frac{1}{2}\ln 2$. Multiplying by -1,

$$\sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{2m} = \tfrac{1}{2}\ln 2.$$

The second series

Now do the same with even powers. Let $K_n = I_{2n}$. Taking $k = 2n-2$ in the reduction formula gives

$$K_n = \frac{1}{2n-1} - K_{n-1} \qquad (n \geq 1), \qquad K_0 = \tfrac{1}{4}\pi.$$

Exactly the same induction (now starting from $K_1 = 1 - \frac{1}{4}\pi$) gives

$$K_n = \int_0^{\pi/4} \tan^{2n}\theta \,\mathrm{d}\theta = (-1)^n\left(\tfrac{1}{4}\pi + \sum_{m=1}^{n} \frac{(-1)^m}{2m-1}\right):$$

the inductive step is $K_n = \frac{1}{2n-1} - (-1)^{n-1}(\cdots) = (-1)^n(\cdots) + (-1)^n\frac{(-1)^n}{2n-1}$, just as before. The same bound gives

$$0 \leq K_n \leq \int_0^{\pi/4} \tan^{2n}\theta \sec^2\theta \,\mathrm{d}\theta = \frac{1}{2n+1} \to 0,$$

so $\sum_{m=1}^{n} \frac{(-1)^m}{2m-1} \to -\frac{1}{4}\pi$, that is,

$$\sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{2m-1} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots = \frac{\pi}{4}.$$

Checks

- With $n = 1$: $\int_0^{\pi/4} \tan^3\theta \,\mathrm{d}\theta = \frac{1}{2} - \frac{1}{2}\ln 2 \approx 0.15343$, and Simpson's rule on the integral gives the same to ten decimal places. We checked the formulas for I_{2n+1} and I_{2n} this way for $n = 1, \ldots, 11$.

- For large k, numerically $I_{100} \approx 0.0049995$ and $I_{1000} \approx 0.00049999996$, very close to $1/(2k)$, as the sketch predicts.
- A million terms of the two series give $0.346573\ldots$ and $0.785398\ldots$, against $\frac{1}{2}\ln 2 = 0.346574$ and $\frac{1}{4}\pi = 0.785398$. The error is about the size of the next term, as the bounds above suggest.

Common slips

- Saying "$\tan^{2n+1}\theta \to 0$, so the integral tends to 0" without a bound. The integrand does *not* tend to 0 at $\theta = \frac{1}{4}\pi$, so a reason such as the $\sec^2\theta \geq 1$ bound is needed.
- Sign errors in the induction: keep $(-1)^n$ outside the bracket and check the case $n = 1$ against a direct calculation.

Beyond STEP

The first series is half of the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2$, and the second is the Leibniz series for π. Both converge only *conditionally*: rearranging their terms can change the sum, a striking fact (Riemann's rearrangement theorem) met early in a first analysis course. The argument here, an exact error term that is squeezed to zero, is the same idea as Taylor's theorem with remainder. Abbott's *Understanding Analysis* treats both carefully.

WORKED SOLUTION

STEP II 1991, Question 6

Pure • Bounding Integrals

STEP II 1991, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q6

The key idea

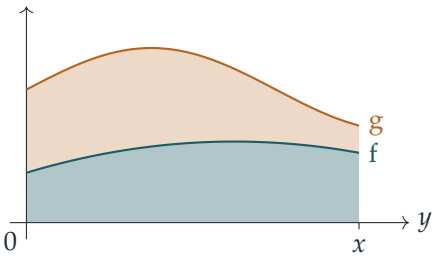
An integral from 0 to x is an area, and a smaller non-negative function encloses a smaller area. That one fact, used over and over, turns a known inequality into a new one. Integrating $0 \leq \cos y \leq 1$ gives a bound on $\sin x$; integrating that gives a bound on $\cos x$; integrating again gives a sharper bound on $\sin x$. Each integral bound then follows by trapping the integrand between two functions we can integrate exactly.

Solution

Comparing integrals

Since $\mathrm{f}(y) \geq 0$ on $[0, x]$, the graph of f lies on or above the y-axis, so $\int_0^x \mathrm{f}(y)\,\mathrm{d}y$ is the area between the graph and the axis, which is non-negative. Since $\mathrm{f}(y) \leq \mathrm{g}(y)$, the region under f sits inside the region under g, so its area is no larger:

$$0 \leq \int_0^x \mathrm{f}(y)\,\mathrm{d}y \leq \int_0^x \mathrm{g}(y)\,\mathrm{d}y.$$



(Analytically: $\mathrm{g} - \mathrm{f} \geq 0$, so $\int_0^x (\mathrm{g} - \mathrm{f})\,\mathrm{d}y \geq 0$, and the integral is linear.)

The bounds on $\sin x$ and $\cos x$

Take $0 \le x \le \frac{1}{2}\pi$. For $0 \le y \le x$ we have $0 \le \cos y \le 1$, so by the result above with $\mathrm{f} = \cos$, $\mathrm{g} = 1$,

$$0 \le \int_0^x \cos y \,\mathrm{d}y \le \int_0^x 1 \,\mathrm{d}y, \qquad \text{that is} \qquad 0 \le \sin x \le x.$$

This holds for every x in $[0, \frac{1}{2}\pi]$, so in particular $0 \le \sin y \le y$ for $0 \le y \le x$. Applying the result again with $\mathrm{f} = \sin$, $\mathrm{g}(y) = y$:

$$\int_0^x \sin y \,\mathrm{d}y \le \int_0^x y \,\mathrm{d}y \quad \Longrightarrow \quad 1 - \cos x \le \tfrac{1}{2}x^2 \quad \Longrightarrow \quad \cos x \ge 1 - \tfrac{1}{2}x^2.$$

The first integral

On $[0, \frac{1}{10}]$ (well inside $[0, \frac{1}{2}\pi]$) we now have $1 - \frac{1}{2}x^2 \le \cos x \le 1$, so

$$3 - \tfrac{1}{2}x^2 \le 2 + \cos x \le 3.$$

Ask yourself Can we just square and take reciprocals?

Yes, because every quantity here is positive: $3 - \frac{1}{2}x^2 \ge 3 - \frac{1}{200} > 0$. For positive numbers, squaring keeps the order and taking reciprocals reverses it. Multiplying by $x \ge 0$ then gives

$$\frac{x}{9} \le \frac{x}{(2+\cos x)^2} \le \frac{x}{(3 - \frac{1}{2}x^2)^2} \qquad (0 \le x \le \tfrac{1}{10}).$$

All three functions are non-negative, so we may integrate from 0 to $\frac{1}{10}$. The left gives

$$\int_0^{1/10} \frac{x}{9} \,\mathrm{d}x = \frac{1}{18} \cdot \frac{1}{100} = \frac{1}{1800}.$$

For the right, the numerator x is, up to sign, the derivative of $3 - \frac{1}{2}x^2$, and $\frac{\mathrm{d}}{\mathrm{d}x}\left(3 - \frac{1}{2}x^2\right)^{-1} = x\left(3 - \frac{1}{2}x^2\right)^{-2}$. So

$$\int_0^{1/10} \frac{x}{(3 - \frac{1}{2}x^2)^2} \,\mathrm{d}x = \left[\frac{1}{3 - \frac{1}{2}x^2}\right]_0^{1/10} = \frac{200}{599} - \frac{1}{3} = \frac{600 - 599}{1797} = \frac{1}{1797}.$$

Hence

$$\frac{1}{1800} \le \int_0^{1/10} \frac{x}{(2+\cos x)^2} \,\mathrm{d}x \le \frac{1}{1797}.$$

A sharper bound on $\sin x$

For $0 \le y \le x \le \frac{1}{2}\pi$ we have $\cos y \ge 1 - \frac{1}{2}y^2$, i.e. $0 \le 1 - \cos y \le \frac{1}{2}y^2$ (the left inequality because $\cos y \le 1$). Integrating from 0 to x:

$$\int_0^x (1 - \cos y) \,\mathrm{d}y \le \int_0^x \tfrac{1}{2}y^2 \,\mathrm{d}y \quad \Longrightarrow \quad x - \sin x \le \tfrac{1}{6}x^3 \quad \Longrightarrow \quad \sin x \ge x - \tfrac{1}{6}x^3.$$

The second integral

On $[0, \frac{1}{10}]$ we have $x - \frac{1}{6}x^3 \le \sin x \le x$, so

$$1 - \tfrac{1}{6}x^3 \le 1 - x + \sin x \le 1.$$

Ask yourself Why is this the right combination to look at?

The $-x$ in the denominator is there to cancel the leading term of $\sin x$, leaving 1 plus something of order x^3. The lower bound $1 - \frac{1}{6}x^3 \ge 1 - \frac{1}{6000}$ is positive, so as before we square, take reciprocals (reversing the order) and multiply by $x^2 \ge 0$:

$$x^2 \le \frac{x^2}{(1 - x + \sin x)^2} \le \frac{x^2}{(1 - \frac{1}{6}x^3)^2}.$$

Integrating, the left gives $\int_0^{1/10} x^2\,\mathrm{d}x = \frac{1}{3}\cdot\frac{1}{1000} = \frac{1}{3000}$. For the right, $\frac{\mathrm{d}}{\mathrm{d}x}\left(1 - \frac{1}{6}x^3\right)^{-1} = \frac{1}{2}x^2\left(1 - \frac{1}{6}x^3\right)^{-2}$, so

$$\int_0^{1/10} \frac{x^2}{(1 - \frac{1}{6}x^3)^2}\,\mathrm{d}x = 2\left[\frac{1}{1 - \frac{1}{6}x^3}\right]_0^{1/10} = 2\left(\frac{6000}{5999} - 1\right) = \frac{2}{5999}.$$

Hence

$$\frac{1}{3000} \le \int_0^{1/10} \frac{x^2}{(1 - x + \sin x)^2}\,\mathrm{d}x \le \frac{2}{5999}.$$

Checks

- Simpson's rule gives $\int_0^{1/10} x(2 + \cos x)^{-2}\,\mathrm{d}x \approx 0.00055648$, between $1/1800 \approx 0.00055556$ and $1/1797 \approx 0.00055648$ (the upper bound is extremely tight: it agrees to about 5×10^{-10}).
- Similarly $\int_0^{1/10} x^2(1 - x + \sin x)^{-2}\,\mathrm{d}x \approx 0.000333389$, between $1/3000 \approx 0.000333333$ and $2/5999 \approx 0.000333389$.
- The three pointwise inequalities held at 10^5 random points of $[0, \frac{1}{2}\pi]$, with equality at $x = 0$.

Common slips

- Taking reciprocals without checking signs: $a \le b$ gives $1/a \ge 1/b$ only when a and b have the same sign.
- Forgetting that the comparison result needs f ≥ 0 for the left-hand inequality; here $\sin y \ge 0$ and $1 - \cos y \ge 0$ supply it.
- Using the bound $\cos x \le 1$ for the upper bound of the integral: $\cos x \le 1$ makes the denominator *smaller than* 9, which gives the lower bound $1/1800$, not the upper one.

Beyond STEP

Repeatedly integrating $0 \leq \cos y \leq 1$ produces the whole chain of alternating Taylor bounds $x - \frac{1}{6}x^3 \leq \sin x \leq x - \frac{1}{6}x^3 + \frac{1}{120}x^5 \leq \cdots$, which is one way to prove that the Taylor series of sin and cos converge to them. Monotonicity of the integral and the comparison of integrands are basic tools of real analysis, used to bound remainders, errors and limits. Abbott's *Understanding Analysis* develops the integral and Taylor's theorem with remainder carefully.

WORKED SOLUTION

STEP II 1991, Question 7

Pure • Functional Equations

STEP II 1991, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q7

The key idea

A functional equation holds for *all* x and y, so we may differentiate it with respect to one variable while holding the other fixed. That turns it into a differential equation linking g' at two different points. Choosing one of the variables cleverly then pins g' down everywhere, and integrating finishes the job. Throughout we assume, as the question does, that g and f are differentiable.

Solution

The derivative relation

Hold $y > 0$ fixed and differentiate $\mathrm{g}(x) + \mathrm{g}(y) = \mathrm{g}(z)$ with respect to x. Since $\mathrm{g}(y)$ is now a constant, the chain rule gives

$$\mathrm{g}'(x) = \mathrm{g}'(z)\,\frac{\partial z}{\partial x}.$$

By the quotient rule,

$$\frac{\partial z}{\partial x} = \frac{y(x+y+1) - xy}{(x+y+1)^2} = \frac{y^2+y}{(x+y+1)^2},$$

which is the first form. For the second, notice that $z+1$ factorises neatly:

$$z+1 = \frac{xy+x+y+1}{x+y+1} = \frac{(x+1)(y+1)}{x+y+1}, \qquad \text{so} \qquad z(z+1) = \frac{xy(x+1)(y+1)}{(x+y+1)^2}.$$

Dividing by $x(x+1)$ gives exactly $\dfrac{y(y+1)}{(x+y+1)^2}$, so

$$\mathrm{g}'(x) = \frac{y^2+y}{(x+y+1)^2}\,\mathrm{g}'(z) = \frac{z(z+1)}{x(x+1)}\,\mathrm{g}'(z),$$

as required. It is worth writing this in the symmetric form

$$x(x+1)\,\mathrm{g}'(x) = z(z+1)\,\mathrm{g}'(z). \qquad (\dagger)$$

Deducing $2g'(1) = (u^2+u)g'(u)$

Ask yourself We want a statement about every u in $(0,1)$. Which variable should we fix, and at what value?

Put $x = 1$ in $(\dagger)$; the left side becomes $2g'(1)$ and $z = \dfrac{y}{y+2}$. As y runs through $(0,\infty)$, $\dfrac{y}{y+2} = 1 - \dfrac{2}{y+2}$ increases continuously from 0 towards 1, so it takes every value u with $0 < u < 1$ (explicitly, $y = \frac{2u}{1-u}$). Hence

$$2g'(1) = (u^2+u)\,g'(u) \qquad \text{for all } 0 < u < 1.$$

Integrating

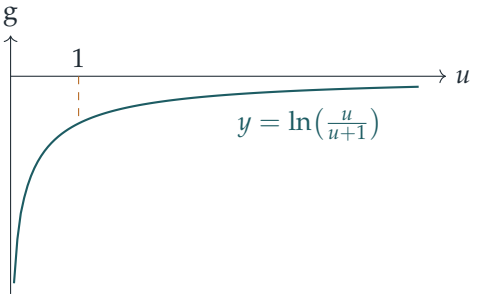
Write $A = 2g'(1)$, a constant. Then for $0 < u < 1$, using partial fractions,

$$g'(u) = \frac{A}{u(u+1)} = A\left(\frac{1}{u} - \frac{1}{u+1}\right),$$

and integrating (everything is positive, so no modulus signs are needed),

$$g(u) = A\big(\ln u - \ln(u+1)\big) + B = A\ln\left(\frac{u}{u+1}\right) + B.$$

In fact the formula holds for *every* $u > 0$, not just $u < 1$. For fixed x, $z = \dfrac{xy}{x+y+1}$ increases from 0 towards x as y goes from 0 to ∞, so we can choose y to make $z < 1$. Then $(\dagger)$ and the result above give $x(x+1)g'(x) = z(z+1)g'(z) = A$. So the same integration works on all of $(0,\infty)$, with one constant B, since $(0,\infty)$ is an interval.



(The curve shown is the case $A = 1$, $B = 0$: it is negative, increasing, and tends to 0 as $u \to \infty$.)

Checking $(*)$ and finding B

Substitute $g(u) = A\ln\frac{u}{u+1} + B$. The left-hand side of $(*)$ is

$$g(x) + g(y) = A\ln\frac{xy}{(x+1)(y+1)} + 2B.$$

For the right-hand side, the factorisation of $z+1$ found earlier gives

$$\frac{z}{z+1} = \frac{xy}{x+y+1}\cdot\frac{x+y+1}{(x+1)(y+1)} = \frac{xy}{(x+1)(y+1)},$$

so

$$\mathrm{g}(z) = A\ln\frac{xy}{(x+1)(y+1)} + B.$$

The two sides agree for all $x, y > 0$ if and only if $2B = B$, that is, $B = 0$. So $\mathrm{g}(u) = A\ln\dfrac{u}{u+1}$ satisfies $(*)$.

Can A be determined?

No. The verification above works for *every* value of A, including $A = 0$ (the zero function). Nor does the definition $A = 2\mathrm{g}'(1)$ help: for our solution, $\mathrm{g}'(1) = A\left(1-\frac{1}{2}\right) = \frac{A}{2}$, so $A = 2\mathrm{g}'(1)$ is automatically true and says nothing new. The equation $(*)$ is linear in g, so any multiple of a solution is again a solution, and A is just that scale factor.

The function f

The same method works. Fix $y > 0$ and differentiate $\mathrm{f}(x) + \mathrm{f}(y) = \mathrm{f}(xy)$ with respect to x:

$$\mathrm{f}'(x) = y\,\mathrm{f}'(xy).$$

Ask yourself What is the analogue here of putting $x = 1$?

Putting $x = 1$ gives $\mathrm{f}'(1) = y\,\mathrm{f}'(y)$ for every $y > 0$, so with $C = \mathrm{f}'(1)$,

$$\mathrm{f}'(y) = \frac{C}{y} \qquad\Longrightarrow\qquad \mathrm{f}(y) = C\ln y + D \quad (y > 0).$$

Substituting back, $\mathrm{f}(x) + \mathrm{f}(y) = C\ln(xy) + 2D$ while $\mathrm{f}(xy) = C\ln(xy) + D$, so $D = 0$. (Equivalently, $x = y = 1$ in the equation gives $2\mathrm{f}(1) = \mathrm{f}(1)$, so $\mathrm{f}(1) = 0$.) Hence $\mathrm{f}(x) = C\ln x$, as required, and once again C is arbitrary.

Checks

- Take $x = 1$, $y = 2$: then $z = \frac{1}{2}$ and the two derivative factors are $\frac{6}{16}$ and $\frac{\frac{1}{2}\cdot\frac{3}{2}}{2} = \frac{3}{8}$, which agree.
- With $A = 1$: $\mathrm{g}(1) + \mathrm{g}(2) = \ln\frac{1}{2} + \ln\frac{2}{3} = \ln\frac{1}{3}$, and $\mathrm{g}\left(\frac{1}{2}\right) = \ln\frac{1/2}{3/2} = \ln\frac{1}{3}$. ✓
- A computer confirmed the derivative identity by finite differences at 1000 random points, and that $\mathrm{g}(x) + \mathrm{g}(y) - \mathrm{g}(z)$ equals B exactly (so vanishes only for $B = 0$) for several values of A.

Common slips

- Forgetting that $\mathrm{g}(y)$ is a constant when differentiating with respect to x, or forgetting the chain-rule factor $\partial z/\partial x$.
- Putting $x = 1$ but not checking that z then really covers all of $(0, 1)$; this is what turns a single equation into a statement for every u.
- Assuming B must be 0 "because constants of integration don't matter": it is the equation $(*)$ that forces $B = 0$, and that needs to be shown.

Beyond STEP

The second equation is Cauchy's logarithmic functional equation: putting $x = \mathrm{e}^s$ turns it into $h(s + t) = h(s) + h(t)$. Here we assumed differentiability; in a first analysis course you see that continuity at a single point is already enough to force $h(s) = cs$, whereas without any regularity there are wild non-linear solutions (built using a Hamel basis). The first equation is the same thing in disguise: $w = \ln \frac{u}{u+1}$ transforms $(*)$ into the additive equation. Abbott's *Understanding Analysis* is a good place to read about continuity arguments of this kind.

WORKED SOLUTION

STEP II 1991, Question 8

Pure • Nonlinear First-Order ODEs

STEP II 1991, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q8

The key idea

Both differential equations are quadratics in $\frac{dy}{dx}$, so we can solve for the derivative algebraically first and only then integrate. Each quadratic has two roots, so we get two candidate first-order equations, and the conditions at $x = 0$ decide which branch is allowed. Hyperbolic identities, in particular $\cosh^2 x - \sinh^2 x = 1$ and the half-angle formulae, keep the roots tidy.

Solution

The quadratic

By the quadratic formula,

$$u = \frac{-2\sinh x \pm \sqrt{4\sinh^2 x + 4}}{2} = -\sinh x \pm \sqrt{\sinh^2 x + 1} = -\sinh x \pm \cosh x,$$

using $\cosh^2 x - \sinh^2 x = 1$ and $\cosh x > 0$. Since $\cosh x - \sinh x = \mathrm{e}^{-x}$ and $\cosh x + \sinh x = \mathrm{e}^{x}$, the two roots are

$$u = \mathrm{e}^{-x} \qquad \text{or} \qquad u = -\mathrm{e}^{x}.$$

(Check: their product is -1 and their sum is $\mathrm{e}^{-x} - \mathrm{e}^{x} = -2\sinh x$, as they should be.)

The first differential equation

With $u = \frac{dy}{dx}$ this is exactly the quadratic above, so at each x either $y' = \mathrm{e}^{-x}$ or $y' = -\mathrm{e}^{x}$.

Ask yourself Could y' follow one root for some values of x and the other root elsewhere?

No. The first root is always positive and the second always negative, so they never meet. A derivative has the intermediate value property (Darboux's theorem; and if

you assume y' is continuous, the ordinary intermediate value theorem), so y' can never jump from a positive value to a negative one without passing through 0, which neither root takes. Hence y' stays on one root throughout.

At $x = 0$ the roots are 1 and -1, and we need $y' > 0$, so $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{-x}$. Integrating, $y = -\mathrm{e}^{-x} + c$, and $y = 0$ at $x = 0$ gives $c = 1$:

$$\boxed{y = 1 - \mathrm{e}^{-x}}$$

The second differential equation

Write $p = \frac{\mathrm{d}y}{\mathrm{d}x}$, so $p^2 \sinh x + 2p - \sinh x = 0$.

At $x = 0$, $\sinh 0 = 0$ and the equation reduces to $2p = 0$, so we must have $y'(0) = 0$.

For $x \neq 0$, $\sinh x \neq 0$ and the quadratic formula gives

$$p = \frac{-2 \pm \sqrt{4 + 4\sinh^2 x}}{2\sinh x} = \frac{-1 \pm \cosh x}{\sinh x}.$$

Using $\cosh x = 1 + 2\sinh^2 \frac{x}{2} = 2\cosh^2 \frac{x}{2} - 1$ and $\sinh x = 2\sinh \frac{x}{2} \cosh \frac{x}{2}$, the two roots simplify to

$$\frac{\cosh x - 1}{\sinh x} = \frac{2\sinh^2 \frac{x}{2}}{2\sinh \frac{x}{2}\cosh \frac{x}{2}} = \tanh \tfrac{x}{2},$$
$$\frac{-1 - \cosh x}{\sinh x} = -\frac{2\cosh^2 \frac{x}{2}}{2\sinh \frac{x}{2}\cosh \frac{x}{2}} = -\coth \tfrac{x}{2}.$$

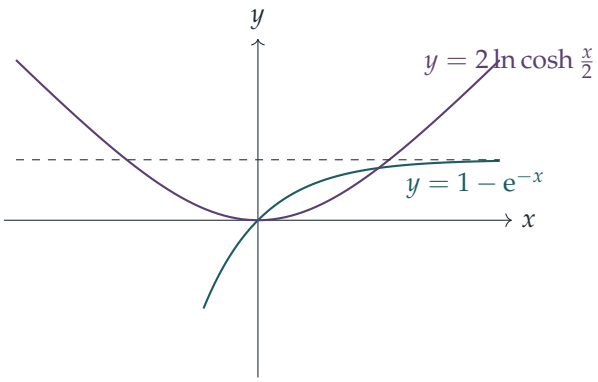
Choosing the branch. The second root $-\coth \frac{x}{2}$ is unbounded as $x \to 0$, and integrating it gives $y = -2\ln\left|\sinh \frac{x}{2}\right| + c$, which tends to $+\infty$ as $x \to 0$ from either side. So on neither side of the origin can this branch give a solution that is continuous at $x = 0$ with $y(0) = 0$. Near $x = 0$ (on each side) we must therefore take the first root, which agrees with $y'(0) = 0$ since $\tanh 0 = 0$. The two roots never coincide (one has the sign of x, the other the opposite sign, and neither vanishes for $x \neq 0$), so by the same intermediate value argument as before the solution stays on this branch for all x:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tanh \tfrac{x}{2} \quad \text{for all } x.$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}\cosh \frac{x}{2} = \frac{1}{2}\sinh \frac{x}{2}$, we have $\tanh \frac{x}{2} = 2 \cdot \frac{\frac{1}{2}\sinh \frac{x}{2}}{\cosh \frac{x}{2}}$, which is of the form $2f'/f$, so integrating gives $y = 2\ln\cosh \frac{x}{2} + c$ (no modulus is needed as $\cosh > 0$). The condition $y(0) = 0$ gives $c = 0$, so

$$\boxed{y = 2\ln\cosh \tfrac{x}{2} = \ln\left(\frac{1 + \cosh x}{2}\right)}$$

the second form following from $2\cosh^2 \frac{x}{2} = 1 + \cosh x$.



Checks

- For $y = 1 - \mathrm{e}^{-x}$: $y'^2 + 2y' \sinh x = \mathrm{e}^{-2x} + \mathrm{e}^{-x}(\mathrm{e}^{x} - \mathrm{e}^{-x}) = 1$, as required, and $y'(0) = 1 > 0$.
- For $y = 2 \ln \cosh \frac{x}{2}$, write $t = \tanh \frac{x}{2}$, so $\sinh x = \frac{2t}{1-t^2}$; then $\sinh x\,(t^2 - 1) + 2t = -2t + 2t = 0$. The solution is even, as it should be, because the equation is unchanged by $x \mapsto -x$, $y \mapsto y$.
- Numerically, at $x = 1$ the solutions are $1 - \mathrm{e}^{-1} \approx 0.6321$ and $2 \ln \cosh \frac{1}{2} \approx 0.2402$; substituting finite-difference derivatives at many random points satisfies both equations to within 10^{-6}.

Common slips

- Dividing the second equation by $\sinh x$ without noticing that it vanishes at $x = 0$, which is exactly where the initial condition lives; the original equation forces $y'(0) = 0$.
- Keeping both branches in the second part: the $-\coth \frac{x}{2}$ branch cannot satisfy $y(0) = 0$, because its integral blows up at the origin.

Beyond STEP

Equations that are not linear in y', such as $F(x, y, y') = 0$, can have several solutions through one point, or "singular" solutions that no single branch describes, so existence and uniqueness theorems (Picard–Lindelöf) are stated for equations already solved in the form $y' = f(x, y)$. Here the roots were separated, so each branch behaved well; when the discriminant can vanish, branches can be glued together and uniqueness fails. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence, uniqueness and these nonlinear first-order equations.

WORKED SOLUTION

STEP II 1991, Question 9

Pure • Matrix Groups

STEP II 1991, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q9

The key idea

Everything rests on one fact: the integers modulo 5 form a field. Every non-zero residue has a multiplicative inverse, and a product of non-zero residues is never zero. That is what makes the diagonal entries behave, so closure and inverses come almost for free. For the last part, find the elements of order 1 or 2 explicitly by solving $M^2 = I$, then test whether multiplying two of them keeps you inside the set.

Solution

Throughout, all arithmetic is modulo 5, and we write -1 for 4 when it is convenient. We write a general element as

$$M(a,b,c) = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, \qquad a,c \in \{1,2,3,4\},\ b \in \{0,1,2,3,4\}.$$

G is a group

Closure. Multiplying two elements gives

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}\begin{pmatrix} d & e \\ 0 & f \end{pmatrix} = \begin{pmatrix} ad & ae+bf \\ 0 & cf \end{pmatrix}.$$

This is upper triangular with entries modulo 5. Its diagonal entries ad and cf are non-zero: since 5 is prime, if $5 \mid ad$ then $5 \mid a$ or $5 \mid d$, and neither holds. So the product lies in G.

Associativity is given, since matrix multiplication is associative.

Identity. $I = M(1,0,1)$ is in G, and $IM = MI = M$ for every M.

Inverses. Every non-zero residue modulo 5 has an inverse ($1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$,

$4^{-1} = 4$). So, given $M(a,b,c) \in G$, we can form

$$N = \begin{pmatrix} a^{-1} & -a^{-1}bc^{-1} \\ 0 & c^{-1} \end{pmatrix},$$

which lies in G because $a^{-1}, c^{-1} \neq 0$. Using the product formula,

$$MN = \begin{pmatrix} aa^{-1} & -aa^{-1}bc^{-1} + bc^{-1} \\ 0 & cc^{-1} \end{pmatrix} = I, \qquad NM = \begin{pmatrix} a^{-1}a & a^{-1}b - a^{-1}bc^{-1}c \\ 0 & c^{-1}c \end{pmatrix} = I.$$

So every element has an inverse in G, and G is a group.

The order of G

There are 4 choices for a, 5 for b and 4 for c, and different choices give different matrices. So

$$|G| = 4 \times 5 \times 4 = 80.$$

Is G commutative?

No. Take $A = M(1,1,1)$ and $B = M(2,0,1)$. Then

$$AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \qquad BA = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix},$$

so $AB \neq BA$ and G is not commutative.

Elements of order 1 or 2

An element M has order 1 or 2 exactly when $M^2 = I$. By the product formula,

$$M(a,b,c)^2 = \begin{pmatrix} a^2 & b(a+c) \\ 0 & c^2 \end{pmatrix},$$

so we need $a^2 = 1$, $c^2 = 1$ and $b(a+c) = 0$.

The squares of $1, 2, 3, 4$ modulo 5 are $1, 4, 4, 1$, so $a^2 = 1$ forces $a = \pm 1$, and likewise $c = \pm 1$.

- If $a = c$, then $a + c = \pm 2 \neq 0$. Since $\mathbb{Z}_5$ has no zero divisors, $b(a+c) = 0$ forces $b = 0$. This gives I and $-I$.
- If $c = -a$, then $a + c = 0$, so b can be anything. This gives the 10 matrices $M(1, b, -1)$ and $M(-1, b, 1)$, for $b = 0, 1, \ldots, 4$.

So the set H of elements of order 1 or 2 has exactly 12 members.

Ask yourself Is a set of self-inverse elements automatically closed?

No. It contains I, and each element is its own inverse, so the identity and inverse axioms hold automatically. The only thing that can fail is closure, and in a non-commutative

group it usually does: if $X^2 = Y^2 = I$, then $(XY)^2 = XYXY$, and this need not collapse to I unless X and Y commute.

Here it fails. Take $X = M(1,1,-1)$ and $Y = M(1,0,-1)$, both in H. Then

$$XY = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

This has $a = c = 1$ but $b = -1 \neq 0$, so by the classification above it is not in H. (In fact $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}^k = \begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix}$, so it has order 5.) Hence H is not closed, and it is *not* a subgroup of G.

If you know Lagrange's theorem, there is a quicker argument: the order of a subgroup must divide the order of the group, and $12 \nmid 80$.

Checks

A short computer enumeration of all 80 matrices confirms closure, the inverse formula and $AB \neq BA$ for the pair above. Squaring every element finds exactly the 12 predicted solutions of $M^2 = I$, and $XY = M(1,4,1)$ has order 5. By hand, check that $M(-1,3,1)^2 = I$: the off-diagonal entry is $3(-1+1) = 0$.

Common slips

- Forgetting to show that the inverse lies in G. Its diagonal entries a^{-1}, c^{-1} must be non-zero.
- Missing the elements $M(1,b,-1)$ and $M(-1,b,1)$ with $b \neq 0$ by assuming that $M^2 = I$ forces M to be diagonal.
- Checking only the identity and inverse axioms for H. Closure is the axiom that fails.

Beyond STEP

G is the group of invertible upper triangular 2×2 matrices over the finite field $\mathbb{F}_5$, a *Borel subgroup* of $\mathrm{GL}_2(\mathbb{F}_5)$. The matrices with $a = c = 1$ form a normal subgroup of order 5, and G is built from it and the diagonal matrices as a semidirect product. Groups of matrices over finite fields are a rich source of examples in a first course on group theory. For more, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 1991, Question 10

Pure • Lines and Planes in 3D

STEP II 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q10

The key idea

Despite the database tag, this is a question about lines and planes in three dimensions. The tip of the shadow is where the ray of sunlight that grazes the top of the stick meets the ground. So we write the ray as a line through the Sun and the tip of the stick, write the ground as a plane through the base perpendicular to the stick, and intersect them. Everything is easy once we write all positions in terms of the unit vector along the stick.

Solution

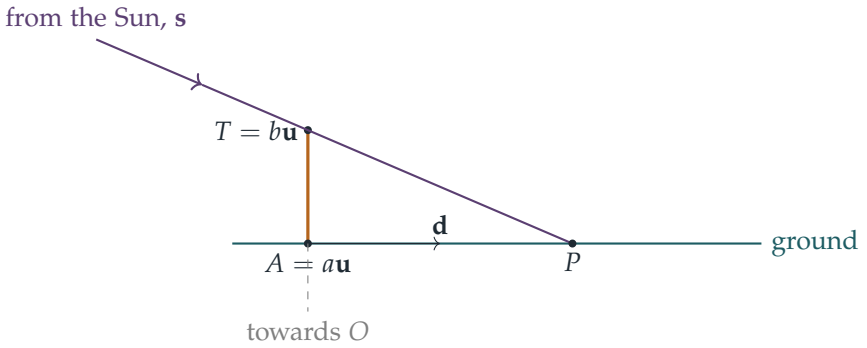
Let O be the centre of the Earth, and write

$$\mathbf{u} = \begin{pmatrix} 0 \\ \cos\phi \\ \sin\phi \end{pmatrix}, \qquad \mathbf{s} = R\begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix}.$$

Then $\mathbf{u}$ is a unit vector along the stick, the base is $A = a\mathbf{u}$, the tip is $T = b\mathbf{u}$, and the Sun is at $\mathbf{s}$. The one dot product we need is

$$\mathbf{s}\cdot\mathbf{u} = R\sin\theta\cos\phi.$$

We write $c = R\cos\phi\sin\theta$ for short.



The ray through the tip

Light reaching the tip of the shadow, P, has just passed the tip of the stick. So P lies on the line through $\mathbf{s}$ and $b\mathbf{u}$:

$$\mathbf{p} = \mathbf{s} + t\,(b\mathbf{u} - \mathbf{s}) = (1-t)\,\mathbf{s} + tb\,\mathbf{u}$$

for some real t.

The ground

The ground is horizontal, meaning perpendicular to the stick, and it passes through the base $a\mathbf{u}$. Since $\mathbf{u}\cdot\mathbf{u} = 1$, it is the plane $\mathbf{r}\cdot\mathbf{u} = a$.

Intersecting

Substituting the line into the plane, and using $\mathbf{u}\cdot\mathbf{u} = 1$ and $\mathbf{s}\cdot\mathbf{u} = c$,

$$(1-t)c + tb = a \qquad \Longrightarrow \qquad t = \frac{c-a}{c-b}.$$

We can divide by $c-b$ because the Sun is shining on the stick. The Sun is above the horizon at the tip exactly when it is on the far side of the tangent plane there, i.e. $\mathbf{s}\cdot\mathbf{u} > b$, so $c > b$.

> **Ask yourself** Is this P really on the ground beyond the stick, and not somewhere behind the Sun?

Since $c > b > a$, we have $c-a > c-b > 0$, so $t > 1$. The point $t = 0$ is the Sun and $t = 1$ is the tip of the stick, so $t > 1$ is past the tip, travelling away from the Sun, as it should be.

From the value of t,

$$1 - t = \frac{(c-b)-(c-a)}{c-b} = \frac{a-b}{c-b} = -\frac{h}{c-b},$$
$$tb - a = \frac{b(c-a)-a(c-b)}{c-b} = \frac{c(b-a)}{c-b} = \frac{ch}{c-b}.$$

So the displacement from the base to the tip of the shadow is

$$\begin{aligned}
\mathbf{d} = \mathbf{p} - a\mathbf{u} &= (1-t)\,\mathbf{s} + (tb-a)\,\mathbf{u} \\
&= \frac{h}{c-b}\left(c\,\mathbf{u} - \mathbf{s}\right) \\
&= \frac{Rh}{c-b}\left[\cos\phi\sin\theta\begin{pmatrix}0\\ \cos\phi\\ \sin\phi\end{pmatrix} - \begin{pmatrix}\cos\theta\\ \sin\theta\\ 0\end{pmatrix}\right] \\
&= \frac{Rh}{c-b}\begin{pmatrix}-\cos\theta\\ -(1-\cos^2\phi)\sin\theta\\ \cos\phi\sin\phi\sin\theta\end{pmatrix}.
\end{aligned}$$

Since $1 - \cos^2\phi = \sin^2\phi$ and $c - b = R\cos\phi\sin\theta - b$, this is

$$\mathbf{d} = Rh\,(R\cos\phi\sin\theta - b)^{-1}\begin{pmatrix} -\cos\theta \\ -\sin^2\phi\sin\theta \\ \cos\phi\sin\phi\sin\theta \end{pmatrix},$$

as required.

The middle line, $\mathbf{d} = \frac{h}{c-b}(c\,\mathbf{u} - \mathbf{s})$, is worth noticing. The vector $c\,\mathbf{u}$ is the component of $\mathbf{s}$ along the stick, so $c\,\mathbf{u} - \mathbf{s}$ is minus the horizontal part of the Sun's position vector. The shadow points directly away from the Sun's horizontal direction, as experience says it should.

Checks

- $\mathbf{d}\cdot\mathbf{u} = \frac{Rh}{c-b}\left(-\sin^2\phi\sin\theta\cos\phi + \cos\phi\sin^2\phi\sin\theta\right) = 0$, so the shadow lies along the ground.
- Take $\phi = 0$, so the stick points along the y-axis. Then $\mathbf{d} = \frac{Rh}{R\sin\theta - b}(-\cos\theta, 0, 0)$. When $\theta = \frac{\pi}{2}$ the Sun is directly overhead and the shadow vanishes. When R is huge, the length tends to $h\cot\theta$. That is the familiar flat-Earth answer for a Sun at elevation θ.
- Numerically, we located the shadow tip directly, by intersecting the ray with the ground plane, for thousands of random choices of R, a, h, θ, ϕ with the Sun above the horizon. It always agreed with the formula, with a largest relative error of about 10^{-13}.

Common slips

- Taking the ground to be the plane $y = a\cos\phi$, or some other coordinate plane. "Horizontal" means perpendicular to $\mathbf{u}$, which is $\mathbf{r}\cdot\mathbf{u} = a$.
- Expanding everything in components from the start. Keeping $\mathbf{u}$ and $\mathbf{s}$ as vectors until the last line saves a lot of algebra.

Beyond STEP

Here sunlight is a family of lines through one point, the Sun, so the shadow is a *central projection* of the stick onto the ground plane. When the Sun is very far away, the rays are almost parallel and the projection becomes a parallel projection. That is the kind of map linear algebra studies, and the vector $c\,\mathbf{u} - \mathbf{s}$ is the projection of $\mathbf{s}$ onto the stick minus $\mathbf{s}$ itself. Strang's *Introduction to Linear Algebra* treats projections onto lines and planes in detail.

WORKED SOLUTION

STEP II 1991, Question 11

Mechanics • Projectile Distance from Launch

STEP II 1991, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q11

The key idea

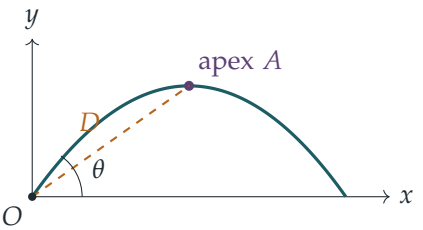
Work with the *square* of the distance from the gun: it avoids square roots and turns both parts into polynomial problems. In the first part the apex distance squared is a quadratic in $\sin^2\theta$, so completing the square (or differentiating) finds the best angle. In the second part, the shell gets nearer exactly when the distance squared is decreasing, so we look at the sign of its time derivative along the whole flight.

Solution

The model

We treat the shell as a particle moving freely under gravity, with no air resistance, fired from the origin O (where the gunners stand) with speed u at elevation θ, where $0 < \theta < \frac{\pi}{2}$. Take x horizontal and y vertically up. At time t,

$$x = ut\cos\theta, \qquad y = ut\sin\theta - \tfrac{1}{2}gt^2.$$



The angle of elevation

The apex is reached when $\dot{y} = u\sin\theta - gt = 0$, that is at $t = u\sin\theta/g$. There

$$x_A = \frac{u^2\sin\theta\cos\theta}{g}, \qquad y_A = \frac{u^2\sin^2\theta}{2g}.$$

Write D for the distance OA and put $s = \sin^2\theta$, so that $\cos^2\theta = 1 - s$. Then

$$\begin{aligned} D^2 = x_A^2 + y_A^2 &= \frac{u^4}{g^2}\left(\sin^2\theta\cos^2\theta + \tfrac{1}{4}\sin^4\theta\right) \\ &= \frac{u^4}{g^2}\left(s(1-s) + \tfrac{1}{4}s^2\right) = \frac{u^4}{g^2}\left(s - \tfrac{3}{4}s^2\right). \end{aligned}$$

Since the damage decreases with distance, the gunners want D as large as possible, which is the same as making D^2 as large as possible. Completing the square,

$$s - \tfrac{3}{4}s^2 = \tfrac{1}{3} - \tfrac{3}{4}\left(s - \tfrac{2}{3}\right)^2,$$

which is largest, with value $\frac{1}{3}$, exactly when $s = \frac{2}{3}$. This value is allowed, since $s = \sin^2\theta$ takes every value in $(0, 1)$. So $\sin^2\theta = \frac{2}{3}$ and $\cos^2\theta = \frac{1}{3}$, giving

$$\boxed{\tan\theta = \sqrt{2}, \qquad \theta = \arctan\sqrt{2} \approx 54.7^\circ}$$

and then the apex is at distance $D = \dfrac{u^2}{\sqrt{3}\,g}$ from the gun.

Does the shell come nearer during its descent?

Let $r^2 = x^2 + y^2$ be the squared distance from the gun at time t. The shell gets nearer exactly when r^2 is decreasing, so we need the sign of

$$\tfrac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}(r^2) = x\dot{x} + y\dot{y}.$$

Ask yourself Why not just compare the distance at the apex with the distance at a few later points?

A few points can't rule out a dip in between. The derivative tells us about *every* instant of the flight at once. With $\dot{x} = u\cos\theta$ and $\dot{y} = u\sin\theta - gt$,

$$\begin{aligned} x\dot{x} + y\dot{y} &= u^2 t\cos^2\theta + \left(ut\sin\theta - \tfrac{1}{2}gt^2\right)(u\sin\theta - gt) \\ &= u^2 t - \tfrac{3}{2}ugt^2\sin\theta + \tfrac{1}{2}g^2t^3 \\ &= \tfrac{1}{2}t\left[(gt)^2 - 3u\sin\theta\,(gt) + 2u^2\right], \end{aligned}$$

using $\cos^2\theta + \sin^2\theta = 1$. For $t > 0$ the sign is that of the bracket, a quadratic in gt. Completing the square,

$$(gt)^2 - 3u\sin\theta\,(gt) + 2u^2 = \left(gt - \tfrac{3}{2}u\sin\theta\right)^2 + u^2\left(2 - \tfrac{9}{4}\sin^2\theta\right).$$

At the chosen angle $\sin^2\theta = \frac{2}{3}$, so the constant term is $u^2\left(2 - \frac{3}{2}\right) = \frac{1}{2}u^2 > 0$. The bracket is therefore strictly positive for every t, so r^2 is strictly increasing throughout the flight.

So **no**: the shell moves steadily further from the gunners all the way down, and the nearest point of the descending path is the apex itself. This shows that the gunners' shortcut of considering only an explosion at the maximum height was justified: for this angle it really does give the nearest point of the descent, at distance $u^2/(\sqrt{3}\,g)$.

Checks

- For $\theta = \frac{\pi}{4}$ the apex is at $(u^2/2g,\, u^2/4g)$, so $D^2 = \frac{5}{16}u^4/g^2$, a little less than our maximum $\frac{1}{3}u^4/g^2$, as it should be.
- The general calculation shows that the shell first comes nearer only when $\frac{9}{4}\sin^2\theta \geq 2$, i.e. $\sin\theta \geq \frac{2\sqrt{2}}{3}$, an angle of about $70.5°$, comfortably above $54.7°$. (The limit $\theta \to \frac{\pi}{2}$ makes sense: a shell fired straight up falls straight back towards the gun.)
- Numerically, with $u = g = 1$, a search over θ gives the largest apex distance $0.57735 \approx 1/\sqrt{3}$ at $\theta \approx 54.736°$, and stepping along the trajectory at that angle shows the distance from O never decreasing; at $70.5° \pm 0.6°$ it switches from monotonic to not.

Common slips

- Maximising the horizontal distance or the height of the apex instead of its straight-line distance from the gun.
- Assuming the distance from the gun obviously increases after the apex because the shell keeps moving away horizontally. It is falling at the same time, and for steep enough shots it really does come back nearer.

Beyond STEP

The trick of differentiating r^2 rather than r, and reading off the sign of $\mathbf{r}\cdot\dot{\mathbf{r}}$, is the standard way to decide whether a moving point is approaching or receding, and it appears throughout vector mechanics and in the study of orbits. The threshold angle $\arcsin(2\sqrt{2}/3)$ is a classic result: below it, a projectile's distance from the launch point always increases. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the vector approach to motion that makes this kind of argument natural.

WORKED SOLUTION

STEP II 1991, Question 12

Mechanics • Forced Elastic Oscillations

STEP II 1991, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 11.1, *Resonance*: https://beyondstep.keithmcnulty.org/ch11/resonance.html

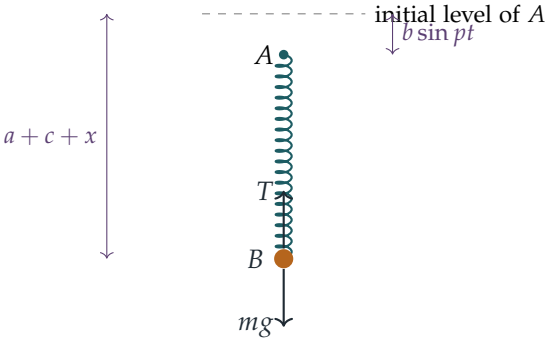
The key idea

The tension depends only on the *extension*, which is the particle's position minus the position of the moving end, minus the natural length. Write that down in terms of x and $b\sin pt$, use the initial equilibrium to replace the modulus by something involving c, and the equation of motion falls out. What follows is a forced simple harmonic oscillator. Solve it by adding a particular solution to the complementary function, and remember that "taut" means "extension never negative".

Solution

The equation of motion

Model the particle as a point of mass m on a light elastic string of modulus λ obeying Hooke's law. Measure all positions downwards from the initial position of A.



In the initial equilibrium the extension is c and the tension balances the weight:

$$\frac{\lambda c}{a} = mg, \qquad \text{so} \qquad \frac{\lambda}{ma} = \frac{g}{c} = n^2.$$

At time t the particle is at depth $a + c + x$ and the end A is at depth $b \sin pt$. While the string is taut its length is $a + c + x - b \sin pt$, so its extension is

$$e = c + x - b \sin pt,$$

and the tension is $T = \lambda e / a$. Resolving downwards for the particle,

$$\begin{aligned} m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} &= mg - \frac{\lambda}{a}(c + x - b \sin pt) \\ &= mg - \frac{\lambda c}{a} - \frac{\lambda}{a}(x - b \sin pt) \\ &= -\frac{\lambda}{a}(x - b \sin pt), \end{aligned}$$

using $\lambda c / a = mg$. Dividing by m and using $\lambda/(ma) = n^2$,

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -n^2(x - b \sin pt),$$

as required. This needs $e \geq 0$: if the string went slack, the tension would be zero rather than $\lambda e / a$.

Solving for $x(t)$

Rewrite the equation as $\ddot{x} + n^2 x = n^2 b \sin pt$. The complementary function is $C \cos nt + D \sin nt$. For a particular integral try $x = K \sin pt$:

$$-p^2 K + n^2 K = n^2 b \quad \Longrightarrow \quad K = \frac{n^2 b}{n^2 - p^2},$$

which is valid because $p < n$ makes $n^2 - p^2 \neq 0$. So

$$x = \frac{n^2 b}{n^2 - p^2} \sin pt + C \cos nt + D \sin nt.$$

At $t = 0$ the particle is at its equilibrium position and at rest, so $x(0) = 0$ and $\dot{x}(0) = 0$. The first gives $C = 0$. The second gives

$$\frac{n^2 bp}{n^2 - p^2} + nD = 0 \quad \Longrightarrow \quad D = -\frac{nbp}{n^2 - p^2}.$$

Hence

$$x(t) = \frac{bn}{n^2 - p^2}(n \sin pt - p \sin nt),$$

as required.

When the string stays taut

Ask yourself What exactly has to stay positive: x, the tension, or something else?

It is the extension $e = c + x - b\sin pt$. The formula for $x(t)$ describes the motion only while $e \geq 0$, so the string stays taut throughout exactly when this holds for every $t \geq 0$. Substituting for x,

$$\begin{aligned} x - b\sin pt &= b\sin pt\left(\frac{n^2}{n^2-p^2} - 1\right) - \frac{bnp}{n^2-p^2}\sin nt \\ &= \frac{bp}{n^2-p^2}\,(p\sin pt - n\sin nt). \end{aligned}$$

So the necessary and sufficient condition is

$$c + \frac{bp}{n^2-p^2}\,(p\sin pt - n\sin nt) \geq 0 \quad \text{for all } t \geq 0,$$

or equivalently, since $n^2 - p^2 > 0$,

$$bp\,(n\sin nt - p\sin pt) \leq c\,(n^2-p^2) \quad \text{for all } t \geq 0.$$

(If e only touches 0 at isolated instants the tension is momentarily zero, but the equation of motion still holds and the string never actually goes slack. That is why the inequality is not strict.)

Now suppose $pb < (n-p)c$. For every t, since $\sin nt \leq 1$ and $-\sin pt \leq 1$, and $n, p > 0$,

$$n\sin nt - p\sin pt \leq n + p.$$

Taking $b > 0$ (the amplitude), we have $bp > 0$, so

$$bp\,(n\sin nt - p\sin pt) \leq bp\,(n+p) < (n-p)c\,(n+p) = c\,(n^2-p^2).$$

The condition holds, in fact with strict inequality, so the string remains taut throughout. (If b were allowed to be negative, the same argument works with $|b|$ in place of b, since $|n\sin nt - p\sin pt| \leq n + p$.)

Checks

- With $t = 0$: $x(0) = 0$ and $\dot{x}(0) = \frac{bn}{n^2-p^2}(np - pn) = 0$, as they should be.
- As $p \to 0$ the end A barely moves, and $x(t) \approx \frac{bn}{n^2}\,(npt - pnt) \to 0$. Nothing happens, as expected.
- The minimum extension can't go below $c - bp(n+p)/(n^2-p^2) = c - bp/(n-p)$, which is positive exactly when $pb < (n-p)c$.
- Numerically, a step-by-step (RK4) integration of the full Hooke's-law equation, with random m, λ, a, c, b, p, matched the formula for $x(t)$ to within 4×10^{-10}. With $n = 1$, $p = 1/\sqrt{2}$, $c = 1$ and b just 5% above $(n-p)c/p$, the extension does become negative (about -0.05) by $t = 20\,000$.

Common slips

- Using x itself as the extension. The top end moves, so the extension is $c + x - b\sin pt$.
- Forgetting that $\dot{x}(0) = 0$ forces a $\sin nt$ term in the solution. Without it the particle would start with velocity $n^2bp/(n^2 - p^2)$.
- Writing the taut condition as $x \geq -c$. That would be right only if A were fixed.

Beyond STEP

This is a forced, undamped oscillator. The response is a mix of the driving frequency p and the natural frequency n, and the factor $1/(n^2 - p^2)$ blows up as $p \to n$, which is resonance. When n/p is irrational the values of (nt, pt) modulo 2π fill out densely, so $n\sin nt - p\sin pt$ gets arbitrarily close to $n + p$. The sufficient condition $pb < (n - p)c$ is then essentially also necessary (equality is still allowed). Forced oscillations, resonance and damping are treated thoroughly in Kleppner and Kolenkow, *An Introduction to Mechanics*, and the differential-equation side in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 1991, Question 13

Mechanics • Rigid Body Rotation

STEP II 1991, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q13

The key idea

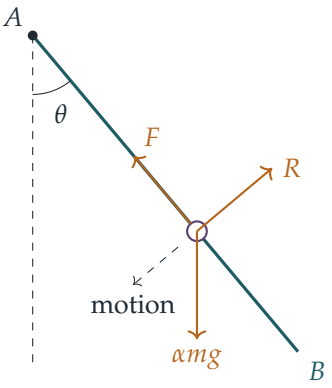
Until the ring slips, the rod and ring move together as a single rigid body pivoted at A. So we can find the angular speed (from energy) and the angular acceleration (from moments) of the combined body. We then use them to work out what forces the rod must exert on the ring to keep it moving with the rod. The ring stays put only while the friction it needs is at most μ times the normal reaction, and slipping starts at the first angle where that fails.

Solution

The model

We treat the ring as a particle at distance x from A, the pivot as smooth, and the rod as a rigid body. Let θ be the angle between the rod and the downward vertical, so $\theta = \frac{\pi}{2}$ at release, and θ decreases as the rod swings down. Write $\omega = |\dot{\theta}|$ for the angular speed and $\beta = |\ddot{\theta}|$ for the size of the angular acceleration.

The rod exerts two forces on the ring: a normal reaction R perpendicular to the rod, taken positive in the direction the ring is moving, and a friction force F along the rod, taken positive towards A. Because the ring is threaded on the rod, R may come out with either sign.



In the diagram, R is drawn opposing the motion because that is how it turns out when $k^2 > xd$, but we keep the signs general in the algebra.

Motion of the rigid body

With the ring fixed on the rod, the moment of inertia about A is $mk^2 + \alpha m x^2 = m(k^2 + \alpha x^2)$. From release, the centre of mass of the rod has fallen $d\cos\theta$ and the ring has fallen $x\cos\theta$, so conservation of energy gives

$$\tfrac{1}{2}m(k^2 + \alpha x^2)\,\omega^2 = mg(d + \alpha x)\cos\theta \quad \Longrightarrow \quad \omega^2 = \frac{2g(d + \alpha x)\cos\theta}{k^2 + \alpha x^2}.$$

Taking moments about A (the pivot force has no moment), the weights have total moment $mg(d + \alpha x)\sin\theta$, so

$$\beta = \frac{g(d + \alpha x)\sin\theta}{k^2 + \alpha x^2}.$$

Forces on the ring

The ring moves on a circle of radius x. Its acceleration is $x\omega^2$ towards A along the rod and $x\beta$ perpendicular to the rod in the direction of motion. The outward direction along the rod makes angle θ with the downward vertical, so the weight has component $\alpha mg\cos\theta$ outward along the rod and $\alpha mg\sin\theta$ in the direction of motion.

Along the rod, towards A: $F - \alpha mg\cos\theta = \alpha m x\omega^2$, so

$$\begin{aligned} F &= \alpha mg\cos\theta\left(1 + \frac{2x(d + \alpha x)}{k^2 + \alpha x^2}\right) \\ &= \frac{\alpha mg\cos\theta\,(k^2 + 3\alpha x^2 + 2xd)}{k^2 + \alpha x^2}. \end{aligned}$$

Perpendicular to the rod, in the direction of motion:

$$R + \alpha mg\sin\theta = \alpha m x\beta \quad \Longrightarrow \quad R = \alpha mg\sin\theta\left(\frac{x(d + \alpha x)}{k^2 + \alpha x^2} - 1\right) = \frac{\alpha mg\sin\theta\,(xd - k^2)}{k^2 + \alpha x^2}.$$

Notice that the αx^2 terms cancel in R, which is why only $k^2 - xd$ appears.

Ask yourself Which way does the ring try to slide, and what does the sign of R mean?

For $0 < \theta < \frac{\pi}{2}$ we have $F > 0$: the rod must pull the ring *towards* A, so without enough friction the ring slides outwards, towards B. If $k^2 > xd$ then $R < 0$, so the rod is holding the ring back. On its own the ring would fall faster than this point of the rod does, and the rod has to push against it.

When does it slip?

The ring stays fixed on the rod as long as $|F| \le \mu|R|$. With $k^2 > xd$, both F and $|R|$ have the same positive factor $\alpha mg/(k^2 + \alpha x^2)$, so the condition is

$$\cos\theta\,(k^2 + 3\alpha x^2 + 2xd) \le \mu \sin\theta\,(k^2 - xd), \quad \text{that is,} \quad \mu\tan\theta \ge \frac{3\alpha x^2 + k^2 + 2xd}{k^2 - xd}.$$

At release, $\theta = \frac{\pi}{2}$ and $F = 0$, so the ring does not slip immediately. As θ decreases from $\frac{\pi}{2}$ to 0, $\mu\tan\theta$ decreases steadily from $+\infty$ to 0, while the right-hand side is a fixed positive number. So the inequality first fails, and the ring starts to slide outwards, at the unique angle where

$$\boxed{\mu\tan\theta = \frac{3\alpha x^2 + k^2 + 2xd}{k^2 - xd}}$$

as required. The ring always slips before the rod reaches the vertical.

Part (i): $k^2 = xd$

Now $R = 0$ at every angle, so the maximum available friction $\mu|R|$ is zero. But $F > 0$ as soon as $\theta < \frac{\pi}{2}$. So the ring starts sliding outwards along the rod *as soon as the rod is released*, whatever the value of μ. Physically, at release the point of the rod at distance $x = k^2/d$ from A starts to fall with acceleration exactly g (since $x\beta = g$ at $\theta = \frac{\pi}{2}$). So the ring presses on neither side of the rod and there is nothing for friction to act through.

Part (ii): $k^2 < xd$

Now $R > 0$: the rod pushes the ring along in the direction of motion, because this point of the rod would fall faster than g at release. The ring again tends to slide outwards, and the condition $F \le \mu R$ becomes

$$\mu\tan\theta \ge \frac{3\alpha x^2 + k^2 + 2xd}{xd - k^2}.$$

By the same argument as before, the ring holds at first and starts to slide outwards when

$$\mu\tan\theta = \frac{3\alpha x^2 + k^2 + 2xd}{xd - k^2},$$

which is the given result with $k^2 - xd$ replaced by $|k^2 - xd|$. The only difference is that the normal reaction now acts on the other side of the ring.

Checks

- With $k^2 > xd$, as $\mu \to \infty$ the slipping angle $\theta \to 0$: very rough rods hold the ring almost to the bottom. As $k^2 - xd \to 0$, the angle tends to $\frac{\pi}{2}$, which matches part (i).
- A uniform rod of length $2a$ has $d = a$ and $k^2 = \frac{4}{3}a^2$, so the critical position $x = k^2/d = \frac{4}{3}a$ is the centre of percussion, the point that starts off in free fall when the rod is released.
- Numerically, we integrated the rigid motion, worked out the ring's acceleration by finite differences, and recovered F and R to within 10^{-5} for random parameter values. For example, with $k^2 - xd \approx 2.87$, the ring first needed more than $\mu|R|$ at $\theta \approx 1.3270$, in agreement with the formula. Cases with $k^2 < xd$ matched the formula in (ii) in the same way.

Common slips

- Forgetting the ring's own contribution to the moment of inertia and to the moment of the weight. The ring is part of the rigid body until it slips.
- Using the friction law with the weight's normal component instead of the actual normal reaction R, which here also depends on the angular acceleration.
- Missing the centripetal term $x\omega^2$ in the equation along the rod, which is where the factor 3 in $3ax^2$ comes from.

Beyond STEP

The point at distance k^2/d from the pivot is the *centre of percussion*. A blow struck there produces no impulsive reaction at the pivot, which is why a cricket bat or tennis racket has a "sweet spot". It is also the length of the simple pendulum with the same period as the compound pendulum. The method here, of finding the motion of a rigid body and then asking what internal forces are needed to sustain it, is the basis of the analysis of constraint forces in Lagrangian mechanics. For more on rigid-body rotation and the compound pendulum, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1991, Question 14

Mechanics • Relative Velocity Pursuit

STEP II 1991, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

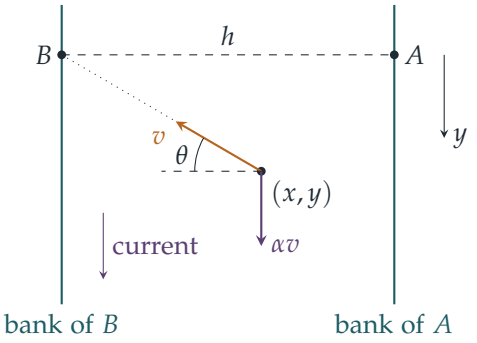
step.maths.org/questions/91-s2-q14

The key idea

The boat's ground velocity is its velocity through the water plus the velocity of the water. "Always pointing at B" is a geometric condition, $y = x\tan\theta$, and differentiating it gives a second expression for the same velocity. Matching the two produces a separable equation linking x and θ. The substitution $\tan\theta = \sinh(\alpha s)$ is chosen because it makes $\sec\theta + \tan\theta$ an exponential, which turns everything into simple exponentials of s.

Solution

We model the boat as a particle. Take B as the origin, with x measured from B's bank towards A and y measured downstream, so that A is at $(h, 0)$ and the boat starts there.



The velocity in terms of θ

The boat points towards B's bank, at angle θ to the direction AB, angled upstream (to counter the drift). Its velocity relative to the water is therefore v along the unit vector $-(\cos\theta, \sin\theta)$ in our (x, y) components. Adding the current, which is αv in the

y-direction,

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-v\cos\theta\\ \alpha v - v\sin\theta\end{pmatrix}.$$

Pointing at B

The direction from the boat at (x, y) to B is along $(-x, -y)$, and this makes angle θ with AB exactly when $\tan\theta = y/x$. So, throughout the motion,

$$y = x\tan\theta.$$

Differentiating with respect to t by the product and chain rules,

$$\dot{y} = \tan\theta\,\frac{\mathrm{d}x}{\mathrm{d}t} + x\sec^2\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}t},$$

so the velocity vector $(\dot{x}, \dot{y})$ has the required form.

The differential equation for x and θ

Now we match the two expressions. The x-components give $\dot{x} = -v\cos\theta$. Putting this into the y-component,

$$\begin{aligned}\tan\theta\,(-v\cos\theta) + x\sec^2\theta\,\dot{\theta} &= \alpha v - v\sin\theta\\ x\sec^2\theta\,\dot{\theta} &= \alpha v, \qquad (*)\end{aligned}$$

since $\tan\theta\cos\theta = \sin\theta$ and the $-v\sin\theta$ terms cancel. In particular $\dot{\theta} > 0$, so θ increases steadily from 0 and we may treat x as a function of θ. Dividing,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\dot{x}}{\dot{\theta}} = \frac{-v\cos\theta\cdot x\sec^2\theta}{\alpha v} = -\frac{x\sec\theta}{\alpha}, \qquad \text{so} \qquad \alpha\frac{\mathrm{d}x}{\mathrm{d}\theta} = -x\sec\theta,$$

as required.

Solving for x

The equation separates (we have $x > 0$ until the boat arrives):

$$\alpha\int\frac{\mathrm{d}x}{x} = -\int\sec\theta\,\mathrm{d}\theta \quad\Longrightarrow\quad \alpha\ln x = -\ln(\sec\theta + \tan\theta) + C,$$

using the standard integral $\int\sec\theta\,\mathrm{d}\theta = \ln(\sec\theta + \tan\theta)$, valid for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. At the start the boat is at A, where $x = h$ and $\theta = 0$, so $C = \alpha\ln h$. Hence

$$\alpha\ln\frac{x}{h} = -\ln(\sec\theta + \tan\theta), \qquad \text{that is,} \qquad \left(\frac{x}{h}\right)^{\alpha} = (\sec\theta + \tan\theta)^{-1}.$$

The substitution $\tan\theta = \sinh(\alpha s)$

Ask yourself What is $\sec\theta$ in terms of s?

Since $\sec^2\theta = 1 + \tan^2\theta = 1 + \sinh^2(\alpha s) = \cosh^2(\alpha s)$, and both $\sec\theta$ and $\cosh(\alpha s)$ are positive (as $0 \le \theta < \frac{\pi}{2}$), we get $\sec\theta = \cosh(\alpha s)$. Then

$$\sec\theta + \tan\theta = \cosh(\alpha s) + \sinh(\alpha s) = \mathrm{e}^{\alpha s},$$

so $(x/h)^\alpha = \mathrm{e}^{-\alpha s}$, and taking αth roots, $x = h\mathrm{e}^{-s}$.

For the rate equation, differentiate $\tan\theta = \sinh(\alpha s)$ with respect to t:

$$\sec^2\theta\,\dot\theta = \alpha\cosh(\alpha s)\,\dot s.$$

Substituting this and $x = h\mathrm{e}^{-s}$ into $(*)$ gives $h\mathrm{e}^{-s}\cdot\alpha\cosh(\alpha s)\,\dot s = \alpha v$, and dividing by α,

$$h\mathrm{e}^{-s}\cosh(\alpha s)\frac{\mathrm{d}s}{\mathrm{d}t} = v,$$

as required.

The time of crossing

At A, $\theta = 0$ so $s = 0$. Since $\dot s > 0$, s increases, and the boat reaches B when $x = h\mathrm{e}^{-s} \to 0$, that is, as $s \to \infty$. (It really does arrive at B: $y = x\tan\theta = h\mathrm{e}^{-s}\sinh(\alpha s) \to 0$ because $\alpha < 1$.) Separating the variables, the crossing time is

$$\begin{aligned} T &= \frac{h}{v}\int_0^\infty \mathrm{e}^{-s}\cosh(\alpha s)\,\mathrm{d}s = \frac{h}{2v}\int_0^\infty \left(\mathrm{e}^{-(1-\alpha)s} + \mathrm{e}^{-(1+\alpha)s}\right)\mathrm{d}s \\ &= \frac{h}{2v}\left(\frac{1}{1-\alpha} + \frac{1}{1+\alpha}\right) = \frac{h}{2v}\cdot\frac{2}{1-\alpha^2} = \frac{h}{v(1-\alpha^2)}, \end{aligned}$$

where both integrals converge because $1 - \alpha > 0$. So the time of crossing is $hv^{-1}(1-\alpha^2)^{-1}$.

Checks

- With no current ($\alpha = 0$) the boat heads straight across and $T = h/v$, as it should. As $\alpha \to 1$ the time blows up: a boat no faster than the current can never make headway against it near B, where it is heading straight upstream.
- We stepped the motion numerically (fourth-order Runge–Kutta, steering at B at every step). With $h = v = 1$ and $\alpha = \frac{1}{2}$ the boat arrives at $t \approx 1.33333$, matching $1/(1-\frac{1}{4}) = \frac{4}{3}$. Along the path, $(x/h)^\alpha(\sec\theta + \tan\theta)$ stayed equal to 1 to about 10^{-13}.

Common slips

- Getting the sign of the heading's y-component wrong. The boat drifts downstream, so pointing at B means pointing partly upstream; with $\theta > 0$ that component is $-v\sin\theta$.

- Forgetting to justify $\sec\theta = +\cosh(\alpha s)$: it needs θ to stay below $\frac{\pi}{2}$, which holds because θ is the angle of the line from B to a point with $x > 0$.

Beyond STEP

This is a classical *pursuit curve* problem: one object steers continually towards a target, and the path is found from a differential equation for the direction of motion. The same ideas describe a dog chasing a moving owner, or a missile homing on a target, and they lead naturally to curves such as the tractrix. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to build fluency with relative velocity and setting up equations of motion in moving frames.

WORKED SOLUTION

STEP II 1991, Question 15

Probability and Statistics • Linearity of Expectation

STEP II 1991, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q15

The key idea

Think of the pair (n_1, n_2) as a cell in an $m \times m$ grid, all m^2 cells equally likely. The cells with $|n_1 - n_2| = k$ lie on two diagonals parallel to the main one, so they are easy to count. After that, everything follows from *linearity of expectation*: a sum of differences has expected value equal to the sum of the expected values, even though neighbouring terms share a variable and are not independent. The invigilator's walk is the same kind of sum, except that the first leg starts from a fixed desk rather than a random one.

Solution

The distribution of $|n_1 - n_2|$

Since n_1 and n_2 are independent and each is uniform on $\{1, \dots, m\}$, each of the m^2 ordered pairs (n_1, n_2) has probability $1/m^2$.

Fix k with $1 \le k \le m-1$. The pairs with $n_1 - n_2 = k$ are $(k+1, 1), (k+2, 2), \dots, (m, m-k)$, which is $m - k$ pairs. By symmetry there are another $m - k$ pairs with $n_2 - n_1 = k$, and the two sets don't overlap because $k \neq 0$. Hence

$$\mathrm{P}(|n_1 - n_2| = k) = \frac{2(m-k)}{m^2},$$

as required. (For $k = 0$ there are m pairs, so $\mathrm{P}(|n_1 - n_2| = 0) = 1/m$. As a check, $\frac{1}{m} + \sum_{k=1}^{m-1} \frac{2(m-k)}{m^2} = \frac{1}{m} + \frac{m(m-1)}{m^2} = 1$.)

The expectation of $|n_1 - n_2|$

The value $k = 0$ adds nothing to the expectation, so

$$\begin{aligned}\mathrm{E}|n_1 - n_2| &= \sum_{k=1}^{m-1} k \cdot \frac{2(m-k)}{m^2} = \frac{2}{m^2}\left(m\sum_{k=1}^{m-1} k - \sum_{k=1}^{m-1} k^2\right)\\ &= \frac{2}{m^2}\left(\frac{m\cdot m(m-1)}{2} - \frac{(m-1)m(2m-1)}{6}\right)\\ &= \frac{2}{m^2}\cdot\frac{m(m-1)\big(3m-(2m-1)\big)}{6} = \frac{(m-1)(m+1)}{3m} = \frac{m^2-1}{3m},\end{aligned}$$

using the standard sums $\sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$ and $\sum_{k=1}^{n} k^2 = \frac{1}{6}n(n+1)(2n+1)$ with $n = m-1$.

The case $m = 2$ with three integers

When $m = 2$ there are $2^3 = 8$ equally likely triples (n_1, n_2, n_3). Writing $S = |n_1 - n_2| + |n_2 - n_3|$:

(n_1, n_2, n_3)	111	112	121	122	211	212	221	222
S	0	1	2	1	1	2	1	0

The total is 8, so $\mathrm{E}(S) = 8/8 = 1$. The claimed value is $2(m^2-1)/(3m) = 2\cdot 3/6 = 1$, which agrees.

> **Ask yourself** The terms $|n_1 - n_2|$ and $|n_2 - n_3|$ both involve n_2. Can we still just add their expectations?

Yes. Linearity of expectation, $\mathrm{E}(X+Y) = \mathrm{E}(X) + \mathrm{E}(Y)$, needs no independence at all. Each pair (n_i, n_{i+1}) on its own consists of two independent uniform integers, so each term has expectation $(m^2-1)/(3m)$. There are $r-1$ terms, so

$$\mathrm{E}\big(|n_1 - n_2| + |n_2 - n_3| + \cdots + |n_{r-1} - n_r|\big) = \frac{(r-1)(m^2-1)}{3m}.$$

The case $r = 3$ is the result we just checked for $m = 2$.

The invigilator

Number the desks $1, 2, \ldots, m$ along the line, so desks i and j are a distance $d|i-j|$ apart. Let c_j be the number of the desk making the jth call, for $j = 1, \ldots, N+1$. We model the callers as independent, each uniform on $\{1, \ldots, m\}$. This is how we read "equally likely to be called by any of the m candidates". Because candidates never call at the same time or during another call, the invigilator deals with the calls one after another and is standing at desk c_j when call $j+1$ comes in. (If the same candidate calls twice running, that leg has length 0, and the formula allows for this.)

The total distance walked is therefore

$$D = d\,|c_1 - 1| + d\sum_{j=1}^{N} |c_{j+1} - c_j|.$$

The first leg. The invigilator starts at desk 1, and c_1 is uniform, so

$$\mathrm{E}|c_1 - 1| = \frac{1}{m}\sum_{i=1}^{m}(i-1) = \frac{m-1}{2}.$$

Notice that this is larger than $(m^2-1)/(3m)$. Starting at the end of the line makes the first walk longer than average.

The other N legs. Here $c_1, \ldots, c_{N+1}$ play the part of $n_1, \ldots, n_r$ with $r = N+1$, so the result above gives

$$\mathrm{E}\sum_{j=1}^{N} |c_{j+1} - c_j| = \frac{N(m^2-1)}{3m}.$$

Combining. By linearity again,

$$\begin{aligned}\mathrm{E}(D) &= d\left(\frac{m-1}{2} + \frac{N(m-1)(m+1)}{3m}\right)\\ &= \frac{d(m-1)}{6m}\big(3m + 2N(m+1)\big) = \frac{d(m-1)}{6m}\big[2N(m+1) + 3m\big],\end{aligned}$$

which is the required result.

Checks

- With $m = 1$ there is only one desk, so the invigilator never moves. The formula gives 0, as it should.
- With $N = 0$ there is only the first call, and the formula reduces to $d(m-1)\cdot 3m/(6m) = \frac{1}{2}d(m-1)$, the expected first leg.
- Exact enumeration of every possible sequence of callers confirms the formula for all $m \le 4$ and $N \le 3$. A simulation with $m = 10$, $N = 20$, $d = 1$ and 2×10^5 runs gave a mean of 70.50, against the formula's $\frac{9}{60}(440 + 30) = 70.5$.

Common slips

- Treating the first leg like the others. The invigilator starts at desk 1, not at a random desk, so that leg has expectation $\frac{1}{2}(m-1)$, not $(m^2-1)/(3m)$.
- Counting only $n_1 - n_2 = k$ and forgetting the factor 2 for $n_2 - n_1 = k$, or wrongly doubling the $k = 0$ case.
- Thinking that the terms of the sum must be independent before their expectations can be added.

Beyond STEP

The invigilator's position is a *Markov chain*: where they go next depends only on where they are now. Here the chain is so simple that each step "forgets" the past completely. In richer models (calls more likely from nearby desks, say) we would need the chain's stationary distribution to find the long-run average walk per call. The continuous analogue of the first result is that two independent uniform points on $[0,1]$ are on average $\frac{1}{3}$ apart, which is the limit of $(m^2-1)/(3m)$ divided by m. Grimmett and Welsh, *Probability: An Introduction*, covers both linearity of expectation and Markov chains.

WORKED SOLUTION

STEP II 1991, Question 16

Probability and Statistics • Binomial Normal Approximation

STEP II 1991, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q16

The key idea

Track the *deficit*: how far the water level is below the top of the dam. Outflow raises it steadily, and each shower lowers it by exactly V; overflow means the deficit goes negative. In the two-day problem only one pattern of rain can possibly cause overflow, and then just one uniform time matters. In the summer the outflow is switched off, so everything comes down to counting showers, and that count is binomial.

Solution

Part (i): the two-day holiday

Measure time in days from the midnight the holiday starts, and let $D(t)$ be the deficit at time t, so $D(0) = V$. Between showers D increases at rate u, and each shower decreases it by V. The dam overflows if D ever becomes negative. (We assume, as the question intends, that the reservoir holds enough water for the outflow to continue throughout the two days.)

Let R_1, R_2 be the events that it rains on day 1 and on day 2, and let the rain times (as fractions of a day) be T_1 and T_2, each uniform on $[0,1]$. The days are independent, as is implicit in "a probability p on any given day".

- **No rain:** the deficit only grows; no overflow.
- **Rain on one day only**, at time s: just after the shower $D = V + us - V = us \geq 0$, so no overflow (apart from the probability-zero event $s = 0$).
- **Rain on both days:** after the first shower $D = uT_1 \geq 0$, which is safe. Just after the second, at time $1 + T_2$,

$$D = V + u(1 + T_2) - 2V = u(1 + T_2) - V.$$

  Notice that T_1 has dropped out: the outflow runs at the same rate whenever the first shower falls.

So overflow happens exactly when both days are wet and $u(1+T_2) < V$, that is, $T_2 < \frac{V}{u} - 1$. If $\frac{1}{2}V \le u \le V$, the number $\frac{V}{u} - 1$ lies in $[0,1]$, and since T_2 is uniform,

$$\mathrm{P}(\text{overflow}) = p^2\left(\frac{V}{u} - 1\right).$$

Setting this equal to Q:

$$\frac{V}{u} = 1 + \frac{Q}{p^2} \quad \Longrightarrow \quad u = \frac{p^2 V}{p^2 + Q}.$$

The condition $Q < p^2$ is what makes this consistent: $0 < Q/p^2 < 1$ gives $\frac{1}{2}V < u < V$, as assumed. (Without it there's no solution. The overflow probability can never exceed p^2, because two wet days are needed.)

Part (i): the summer holiday

Now $u = 0$, so the deficit never increases. It starts at kV and drops by V with each shower, so if N is the number of wet days among the 18,

$$\text{overflow} \iff NV > kV \iff N > k.$$

The days are independent, each wet with probability $\frac{1}{3}$, so $N \sim \mathrm{B}(18, \frac{1}{3})$, with mean 6 and variance $18 \cdot \frac{1}{3} \cdot \frac{2}{3} = 4$.

> **Ask yourself** Which approximation is "suitable" here, and why do we need a continuity correction?

With $np = 6$ and $n(1-p) = 12$, the normal approximation $N \approx \mathrm{N}(6,4)$ is reasonable. (A Poisson approximation isn't, because p isn't small.) N takes whole-number values, so for an integer k the event $N > k$ is the event $N \ge k+1$, which we approximate by $X > k + \frac{1}{2}$ with $X \sim \mathrm{N}(6,4)$. We need

$$\mathrm{P}\left(Z > \frac{k + \frac{1}{2} - 6}{2}\right) < \frac{1}{10} \iff \frac{k - 5.5}{2} > 1.2816 \iff k > 8.06.$$

Concretely, $k = 8$ gives $\mathrm{P}(Z > 1.25) \approx 0.106$, which is too big, and $k = 9$ gives $\mathrm{P}(Z > 1.75) \approx 0.040$, which is fine. Since overflow depends only on whether $N > k$, a non-integer k between 8 and 9 is no better than $k = 9$. So the engineer should drain the reservoir to

$$k = 9,$$

that is, $9V$ below the top.

Part (ii)

Now there are 36 half-days, independent, each wet with probability $\frac{1}{6}$. The outflow is still zero, so overflow again means $N > k$, where now $N \sim \mathrm{B}(36, \frac{1}{6})$. The mean is still 6, but the variance is $36 \cdot \frac{1}{6} \cdot \frac{5}{6} = 5$, which is larger than 4.

With the same mean, a larger variance means a heavier upper tail: for each k above the mean, $\mathrm{P}(N > k)$ is larger than before. So the required k is *higher*. Approximating by $\mathrm{N}(6, 5)$ with the continuity correction, we need

$$\frac{k - 5.5}{\sqrt{5}} > 1.2816 \iff k > 8.37,$$

compared with $k > 8.06$ before. (The normal approximation gives $\mathrm{P}(N \geq 9) \approx 0.132$ here, against 0.106 in part (i).) So the threshold rises. If k is rounded up to a whole number, both cases still give $k = 9$. But the tail at $k = 9$ is now about 0.06 rather than 0.04, so the margin of safety is thinner.

Checks

- A Monte Carlo simulation of the two-day holiday (400,000 runs) with $p = 0.5$, $Q = 0.1$ and $u = \frac{0.25}{0.35}V \approx 0.714V$ gave an overflow frequency of 0.0996. Other choices of p and Q agreed just as well.
- Limiting cases: as $Q \to p^2$ we get $u \to \frac{1}{2}V$, the slowest outflow at which two wet days can still cause overflow. As $Q \to 0$ we get $u \to V$, at which a second shower can never overflow.
- The exact binomial tails confirm the normal approximation. For $\mathrm{B}(18, \frac{1}{3})$, $\mathrm{P}(N \geq 9) = 0.1076$ and $\mathrm{P}(N \geq 10) = 0.0433$. For $\mathrm{B}(36, \frac{1}{6})$, $\mathrm{P}(N \geq 9) = 0.1332$ and $\mathrm{P}(N \geq 10) = 0.0652$.

Common slips

- Thinking the first rain time T_1 matters in the two-day problem. It cancels, because the outflow is constant.
- Forgetting the continuity correction. Without it the inequality becomes $k > 8.56$, which hides the fact that N is a whole number.
- Saying the answer to (ii) is "the same" because the means are equal. It's the spread that controls the tail.

Beyond STEP

Rain arriving at random instants, and water draining away at a steady rate, is a classic model in applied probability: storage theory, developed by Moran for dams. It is closely related to queues and insurance ruin. In part (ii), as the day is cut into more and smaller pieces with the mean held fixed, $\mathrm{B}(n, 6/n)$ tends to the Poisson distribution with mean 6 and variance 6, and the arrivals become a Poisson process. For more, see Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 1991, Question 1

Pure • Series and Expansions

STEP III 1991, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q1

The key idea

All three parts reward a moment's thought before any grinding. In (i), partial fractions turn the sum into pieces that telescope. In (ii), the cubic $1+x+x^2+x^3$ is a geometric series, so it is a quotient of two very simple expressions, and the logarithm splits into two standard series. In (iii), write the exponent as a series first; since it starts at x^2, only a couple of terms of the exponential series can reach x^4.

Solution

Part (i)

We look for constants with

$$\frac{6}{r(r+1)(r+3)} = \frac{A}{r} + \frac{B}{r+1} + \frac{C}{r+3}.$$

By the cover-up rule, $A = \dfrac{6}{1\cdot 3} = 2$, $B = \dfrac{6}{(-1)(2)} = -3$ and $C = \dfrac{6}{(-3)(-2)} = 1$. (As a quick check, $A+B+C=0$, as it must be because the left-hand side behaves like $6/r^3$ for large r, with no $1/r$ term.)

The coefficients $2, -3, 1$ sum to zero, which suggests splitting them as $2-2$ and $-1+1$:

$$\frac{6}{r(r+1)(r+3)} = 2\left(\frac{1}{r} - \frac{1}{r+1}\right) - \left(\frac{1}{r+1} - \frac{1}{r+3}\right).$$

Each bracket now telescopes. The first gives

$$\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+1}\right) = 1 - \frac{1}{n+1}.$$

In the second, the terms are two steps apart, so two terms survive at each end:

$$\sum_{r=1}^{n}\left(\frac{1}{r+1} - \frac{1}{r+3}\right) = \frac{1}{2} + \frac{1}{3} - \frac{1}{n+2} - \frac{1}{n+3}.$$

Putting these together,

$$\sum_{r=1}^{n} \frac{6}{r(r+1)(r+3)} = 2 - \frac{2}{n+1} - \frac{5}{6} + \frac{1}{n+2} + \frac{1}{n+3}$$
$$= \frac{7}{6} - \frac{2}{n+1} + \frac{1}{n+2} + \frac{1}{n+3}.$$

Part (ii)

? **Ask yourself** Is there a neater way to write $1 + x + x^2 + x^3$?

It is a geometric series with ratio x, so for $x \neq 1$

$$1 + x + x^2 + x^3 = \frac{1 - x^4}{1 - x}.$$

For $|x| < 1$ both $1 - x^4$ and $1 - x$ are positive, so we may split the logarithm:

$$\ln(1 + x + x^2 + x^3) = \ln(1 - x^4) - \ln(1 - x).$$

Using $\ln(1 - y) = -\sum_{k \geq 1} y^k / k$, valid for $|y| < 1$ (and $|x^4| < 1$ here too),

$$-\ln(1 - x) = \sum_{k=1}^{\infty} \frac{x^k}{k}, \qquad \ln(1 - x^4) = -\sum_{m=1}^{\infty} \frac{x^{4m}}{m}.$$

The second series only touches powers x^k with k a multiple of 4. For such $k = 4m$ the combined coefficient is

$$\frac{1}{4m} - \frac{1}{m} = -\frac{3}{4m} = -\frac{3}{k}.$$

So

$$\ln(1 + x + x^2 + x^3) = x + \frac{x^2}{2} + \frac{x^3}{3} - \frac{3x^4}{4} + \frac{x^5}{5} + \cdots,$$

and the general term is

$$\frac{x^k}{k} \text{ if } 4 \nmid k, \qquad -\frac{3x^k}{k} \text{ if } 4 \mid k.$$

Every coefficient is non-zero, so the five terms displayed are the first five non-zero terms.

Part (iii)

For $-1 < x \leq 1$ we have $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so

$$u = x \ln(1 + x) = x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \cdots.$$

? **Ask yourself** How many terms of $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \cdots$ do we need?

Because u starts at x^2, the term u^k starts at x^{2k}. So u^3 and beyond contribute nothing up to x^4, and from u^2 we need only its leading term x^4. Hence

$$\begin{aligned} \mathrm{e}^{x\ln(1+x)} &= 1 + \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3}\right) + \frac{1}{2}x^4 + \cdots \\ &= 1 + x^2 - \frac{x^3}{2} + \frac{5x^4}{6} + \cdots . \end{aligned}$$

(Of course $\mathrm{e}^{x\ln(1+x)}$ is just $(1+x)^x$.)

Checks

- In (i), $n = 1$ gives $\frac{7}{6} - 1 + \frac{1}{3} + \frac{1}{4} = \frac{3}{4}$, which matches the single term $\frac{6}{1\cdot 2\cdot 4}$. Letting $n \to \infty$ gives the sum to infinity $\frac{7}{6}$.
- In (ii), expanding directly to second order, $\ln(1+y) \approx y - \frac{y^2}{2}$ with $y = x + x^2 + \cdots$, gives $x + x^2 - \frac{x^2}{2} = x + \frac{x^2}{2}$, agreeing with our first two terms. Numerically, at $x = 0.5$ the series summed to many terms agrees with $\ln 1.875 \approx 0.6286$.
- In (iii), at $x = 0.01$ the expansion differs from $(1.01)^{0.01}$ by about 7.4×10^{-11}, which is of size x^5, as it should be (the next coefficient turns out to be $-\frac{3}{4}$).

Common slips

- In (i), forgetting that a telescoping sum with a gap of two leaves two terms at each end, not one.
- In (ii), giving the coefficient of x^4 as $\frac{1}{4} - 1$ but then writing the general term as $(-1)^{k+1}x^k/k$ or similar; the pattern depends on k modulo 4.
- In (iii), dropping the $\frac{1}{2}u^2$ term: it contributes $\frac{1}{2}x^4$.

Beyond STEP

Part (iii) is a small example of composing power series, and the bookkeeping trick of tracking the lowest power in each term is exactly how asymptotic expansions are handled at university. The questions of when a power series converges, and why one can add, multiply and compose them freely inside the radius of convergence, are the business of a first course in analysis; Abbott's *Understanding Analysis* treats them clearly.

WORKED SOLUTION

STEP III 1991, Question 2

Pure • Complex Numbers in Geometry

STEP III 1991, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q2

The key idea

A single complex number can carry both a ratio of lengths and an angle. The quotient $(z_1 - z_2)/(z_1 - z_3)$ records the ratio of the two sides of the triangle at P_1 (its modulus) and the turn from one side to the other (its argument), and those two pieces of data are exactly what the SAS test for similarity needs. Once the condition becomes a determinant, each part reduces to factorising a determinant, or to choosing a clever triangle $Q_1Q_2Q_3$ to compare with.

Solution

A misprint first

As printed, the left-hand denominator is $z_2 - z_3$. That can't be right: take $w_i = z_i$, so that the two triangles are identical. The right-hand side is then $(z_1 - z_2)/(z_1 - z_3)$, which differs from the left-hand side because $z_2 \neq z_1$. The intended condition, and the one that matches the determinant, is

$$\frac{z_1 - z_2}{z_1 - z_3} = \frac{w_1 - w_2}{w_1 - w_3}. \qquad (*)$$

This is what we prove.

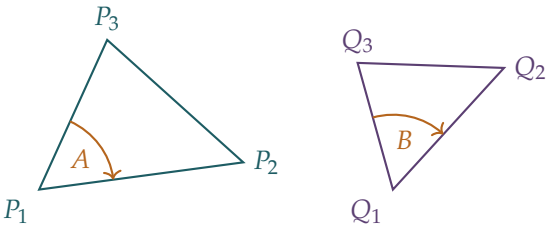
The similarity condition

Write A for the angle of triangle $P_1P_2P_3$ at P_1, so $0 < A < \pi$. In polar form,

$$\frac{z_1 - z_2}{z_1 - z_3} = \frac{z_2 - z_1}{z_3 - z_1} = \frac{P_1P_2}{P_1P_3}\,\mathrm{e}^{\mathrm{i}\theta},$$

where θ is the angle we turn through to get from the direction of $\overrightarrow{P_1P_3}$ to the direction of $\overrightarrow{P_1P_2}$. So $\theta = -A$ if $P_1 \to P_2 \to P_3$ runs anticlockwise (from P_1P_3 we turn clockwise to reach P_1P_2), and $\theta = +A$ if it runs clockwise. In short, the argument of this quotient

(taken in $(-\pi, \pi)$) has size equal to the angle at P_1 and sign that tells us the sense of the triangle. The same holds for the ws, with B the angle at Q_1.



Similar in the same sense $\Rightarrow (*)$. If the triangles are similar with $P_i \leftrightarrow Q_i$, the sides at P_1 and Q_1 are in the same ratio, $P_1P_2/P_1P_3 = Q_1Q_2/Q_1Q_3$, and the angles there agree, $A = B$. Because the two triangles run in the same sense, the two arguments $\mp A$ and $\mp B$ carry the same sign, so they are equal. Equal moduli and equal arguments give $(*)$.

$(*) \Rightarrow$ *similar in the same sense.* If $(*)$ holds, equating moduli gives $P_1P_2/P_1P_3 = Q_1Q_2/Q_1Q_3$. Equating arguments gives $A = B$, and also the same sign, so the triangles run in the same sense. Two sides in proportion with equal included angles is the SAS condition, so the triangles are similar with $P_i \leftrightarrow Q_i$. (If $(*)$ is real, both triples are collinear, and the degenerate "triangles" are again similar.)

Another way to see the same fact: $(*)$ says precisely that $w_i = az_i + b$ for the constants $a = (w_1 - w_2)/(z_1 - z_2) \neq 0$ and $b = w_1 - az_1$. The map $z \mapsto az + b$ is an enlargement by $|a|$ combined with a rotation by $\arg a$ and a translation, and such a map preserves shape and sense.

The determinant form

Clearing denominators (both are nonzero since the points are distinct), $(*)$ is equivalent to

$$(z_1 - z_2)(w_1 - w_3) - (z_1 - z_3)(w_1 - w_2) = 0.$$

Expanding, the z_1w_1 terms cancel and we are left with

$$z_1(w_2 - w_3) - z_2(w_1 - w_3) + z_3(w_1 - w_2) = 0,$$

which is the expansion of the determinant along its top row. So $(*)$ is the same as $\det = 0$.

For the parts it helps to evaluate the determinant another way. Subtract the first column from the second and third columns, which leaves the determinant unchanged, and then expand along the bottom row, which is now $(1, 0, 0)$:

$$\det \begin{pmatrix} z_1 & z_2 & z_3 \\ w_1 & w_2 & w_3 \\ 1 & 1 & 1 \end{pmatrix} = \det \begin{pmatrix} z_2 - z_1 & z_3 - z_1 \\ w_2 - w_1 & w_3 - w_1 \end{pmatrix}. \qquad (\dagger)$$

Part (i)

With $w_i = z_i^2$, (†) becomes

$$\begin{aligned}(z_2 - z_1)(z_3^2 - z_1^2) - (z_3 - z_1)(z_2^2 - z_1^2) &= (z_2 - z_1)(z_3 - z_1)\big[(z_3 + z_1) - (z_2 + z_1)\big] \\ &= (z_2 - z_1)(z_3 - z_1)(z_3 - z_2).\end{aligned}$$

Every factor is nonzero because the z_i are distinct. So the determinant is never zero, and the triangles are never similar (in the sense above).

Part (ii)

With $w_i = z_i^3$, use $z_j^3 - z_1^3 = (z_j - z_1)(z_j^2 + z_j z_1 + z_1^2)$:

$$\begin{aligned}(z_2 - z_1)(z_3^3 - z_1^3) - (z_3 - z_1)(z_2^3 - z_1^3) &= (z_2 - z_1)(z_3 - z_1)\big[(z_3^2 + z_3 z_1) - (z_2^2 + z_2 z_1)\big] \\ &= (z_2 - z_1)(z_3 - z_1)(z_3 - z_2)(z_1 + z_2 + z_3),\end{aligned}$$

since $z_3^2 - z_2^2 + z_1(z_3 - z_2) = (z_3 - z_2)(z_3 + z_2 + z_1)$. The first three factors are nonzero, so the determinant vanishes if and only if $z_1 + z_2 + z_3 = 0$. That is, the triangles are similar if and only if the centroid $\frac{1}{3}(z_1 + z_2 + z_3)$ is the origin.

Part (iii)

The trick is to compare the triangle with itself, relabelled. Take Q_1, Q_2, Q_3 to be P_2, P_3, P_1, that is

$$w_1 = z_2, \qquad w_2 = z_3, \qquad w_3 = z_1.$$

Ask yourself Why should "equilateral" have anything to do with this relabelling?

Similarity with $P_1 \leftrightarrow P_2$, $P_2 \leftrightarrow P_3$, $P_3 \leftrightarrow P_1$ forces corresponding sides into a common ratio $k > 0$:

$$\frac{P_2P_3}{P_1P_2} = \frac{P_3P_1}{P_2P_3} = \frac{P_1P_2}{P_3P_1} = k.$$

Multiplying the three gives $k^3 = 1$, so $k = 1$ and all three sides are equal. Conversely, if the triangle is equilateral, the rotation through 120° about its centre (in the right direction) sends $P_1 \to P_2 \to P_3 \to P_1$. A rotation preserves shape and sense, so the triangle is similar to its relabelled self in the same sense. Hence

$$P_1P_2P_3 \text{ is equilateral} \iff \text{it is similar, in the same sense, to } P_2P_3P_1.$$

By the first part, the right-hand side holds if and only if

$$0 = \det\begin{pmatrix} z_1 & z_2 & z_3 \\ z_2 & z_3 & z_1 \\ 1 & 1 & 1 \end{pmatrix} = z_1(z_3 - z_1) - z_2(z_2 - z_1) + z_3(z_2 - z_3),$$

that is,

$$z_1z_3 - z_1^2 - z_2^2 + z_1z_2 + z_2z_3 - z_3^2 = 0 \iff z_2z_3 + z_3z_1 + z_1z_2 = z_1^2 + z_2^2 + z_3^2,$$

as required.

Checks

- Part (ii): $z_1, z_2, z_3 = 1, \omega, \omega^2$ (with $\omega = e^{2\pi i/3}$) have centroid 0, but their cubes all equal 1 and so aren't distinct. Try instead $z = 1,\ i,\ -1-i$. The centroid is 0 and the cubes are $1,\ -i,\ 2-2i$. Both triangles have sides $\sqrt{2}, \sqrt{5}, \sqrt{5}$ in the order P_1P_2, P_2P_3, P_3P_1 (here they are even congruent), and both run anticlockwise.
- Part (iii): for $z = 0, 1, e^{i\pi/3}$ both sides equal $e^{i\pi/3}$, since $1 + e^{2i\pi/3} = e^{i\pi/3}$.
- A computer test on 10^4 random triangles confirmed that $w_i = az_i + b$ always satisfies $(*)$ while the reflected triangle $w_i = a\bar{z}_i + b$ never does. It also confirmed both factorisations of $(\dagger)$ and the equilateral criterion. The condition as printed (with $z_2 - z_3$) failed in every one of the genuinely similar cases.

Common slips

- Ignoring the sense of the triangles. Moduli alone would only give similarity up to reflection. Mirror-image triangles have arguments of opposite sign, so they fail $(*)$.
- In part (iii), forgetting the converse: the relabelled-similarity argument has to run in both directions, and the step $k^3 = 1 \Rightarrow k = 1$ uses k being a positive real.

Beyond STEP

The maps $z \mapsto az + b$ are the orientation-preserving similarities of the plane. They are the simplest Möbius transformations, and the cross-ratio of four points generalises $(*)$. The equilateral condition also factorises as $(z_1 + \omega z_2 + \omega^2 z_3)(z_1 + \omega^2 z_2 + \omega z_3) = 0$, which says that z_1, z_2, z_3 run around an equilateral triangle in one sense or the other. Needham's *Visual Complex Analysis* develops this geometric way of thinking about complex numbers beautifully.

WORKED SOLUTION

STEP III 1991, Question 3

Pure • Piecewise Functions and Inverses

STEP III 1991, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q3

The key idea

A modulus is just a function that changes formula when its inside changes sign. The three moduli here change sign only at $x = -1$, 0 and 1, so on each of the four intervals they cut out of $x < 2$ the function is an ordinary quadratic. Once we have those four quadratics, every question (turning points, inflections, continuity, smoothness, inverse) is a question about simple parabolas and how they join.

Solution

Removing the moduli

Factorise the insides: $x^2 - x = x(x-1)$, $x^2 - 1 = (x-1)(x+1)$ and $x^2 + x = x(x+1)$. Their signs are:

interval	$x^2 - x$	$x^2 - 1$	$x^2 + x$	$f(x)$
$x \le -1$	$+$	$+$	$+$	$x^2 - 4x - 1$
$-1 \le x \le 0$	$+$	$-$	$-$	$3x^2 + 1$
$0 \le x \le 1$	$-$	$-$	$+$	$1 - 5x^2$
$1 \le x < 2$	$+$	$+$	$+$	$x^2 - 4x - 1$

For example, on $-1 \le x \le 0$,

$$f(x) = 2(x^2 - x) + (1 - x^2) + 2(x^2 + x) = 3x^2 + 1,$$

and on $0 \le x \le 1$, $f(x) = -2(x^2 - x) + (1 - x^2) - 2(x^2 + x) = 1 - 5x^2$. On the two outer intervals all three insides are positive, and $2(x^2 - x) + (x^2 - 1) - 2(x^2 + x) = x^2 - 4x - 1$. At the endpoints an inside is zero, so either formula is correct there.

Continuity and differentiability

f is built from $|\cdot|$ and polynomials, all continuous, so f is **continuous everywhere** on $x < 2$. As a check, the neighbouring formulas agree at the joins: $f(-1) = 4$, $f(0) = 1$,

$f(1) = -4$.

For differentiability, compare one-sided derivatives at the joins. The derivatives of the four pieces are $2x - 4$, $6x$, $-10x$ and $2x - 4$.

- At $x = -1$: from the left $2(-1) - 4 = -6$, from the right $6(-1) = -6$. These agree, so $f'(-1) = -6$.
- At $x = 0$: both sides give 0, so $f'(0) = 0$.
- At $x = 1$: from the left -10, from the right $2 - 4 = -2$. These differ, so the graph has a **corner** at $(1, -4)$.

So f is **not differentiable everywhere**: it fails exactly at $x = 1$.

Turning points and inflections

On every piece $f' < 0$, except at the single point $x = 0$:

- $x < -1$ and $1 < x < 2$: $f'(x) = 2x - 4 < 0$ since $x < 2$;
- $-1 < x < 0$: $f'(x) = 6x < 0$;
- $0 < x < 1$: $f'(x) = -10x < 0$.

Since f is continuous and decreases strictly on each piece, it is **strictly decreasing on the whole of** $x < 2$. In particular there are **no maximum or minimum points**, not even local ones.

? **Ask yourself** $f'(0) = 0$, so isn't $(0, 1)$ a turning point?

No. The left piece $3x^2 + 1$ has its minimum at 0 and the right piece $1 - 5x^2$ has its maximum there, but on the interval where each formula applies, both are decreasing. So f passes through $x = 0$ still going down, with a horizontal tangent.

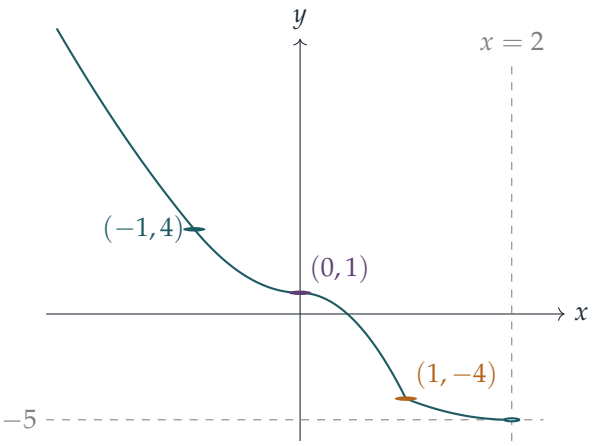
Now concavity. The second derivatives of the pieces are

$$f'' = 2 \ (x < -1), \qquad 6 \ (-1 < x < 0), \qquad -10 \ (0 < x < 1), \qquad 2 \ (1 < x < 2).$$

- At $x = 0$ the graph changes from convex to concave and has a tangent there, so $(0, 1)$ is a **stationary point of inflection**.
- At $x = -1$, f'' jumps from 2 to 6 but stays positive, so there is no inflection.
- At $x = 1$ the curve changes from concave back to convex, but it has a corner there and no tangent. By the usual definition (a tangent that crosses the curve) $(1, -4)$ is not a point of inflection. It's worth saying this explicitly in an answer.

Sketch

As $x \to -\infty$, $f(x) \to +\infty$. As $x \to 2^-$, $f(x) \to 4 - 8 - 1 = -5$, a value that is not attained.



The curve falls throughout. It flattens to a horizontal inflection at $(0,1)$ and has a corner at $(1,-4)$, where the slope jumps from -10 to -2.

The inverse

Because f is continuous and strictly decreasing on $x<2$, with $f\to\infty$ on the left and $f\to -5$ on the right, it is a bijection from $(-\infty,2)$ onto $(-5,\infty)$. So f^{-1} is defined on $x>-5$. We invert each piece on the range it covers, using $y=f(x)$.

- $x\le -1$, range $y\ge 4$: $y=x^2-4x-1=(x-2)^2-5$, so $(x-2)^2=y+5$. Here $x-2<0$, so we take the negative root: $x=2-\sqrt{y+5}$.
- $-1\le x\le 0$, range $1\le y\le 4$: $x^2=(y-1)/3$, and $x\le 0$, so $x=-\sqrt{(y-1)/3}$.
- $0\le x\le 1$, range $-4\le y\le 1$: $x^2=(1-y)/5$, and $x\ge 0$, so $x=\sqrt{(1-y)/5}$.
- $1\le x<2$, range $-5<y\le -4$: again $(x-2)^2=y+5$ with $x-2<0$, so $x=2-\sqrt{y+5}$.

Renaming the variable as x:

$$f^{-1}(x)=\begin{cases}2-\sqrt{x+5}, & -5<x\le -4,\\ \sqrt{\frac{1-x}{5}}, & -4\le x\le 1,\\ -\sqrt{\frac{x-1}{3}}, & 1\le x\le 4,\\ 2-\sqrt{x+5}, & x\ge 4.\end{cases}$$

The formulas agree at the shared endpoints. For example, at $x=4$ both give -1, and at $x=-4$ both give 1.

Checks

- Spot values: $f(\frac{1}{2})=2\cdot\frac{1}{4}+\frac{3}{4}-2\cdot\frac{3}{4}=-\frac{1}{4}$, and $1-5\cdot\frac{1}{4}=-\frac{1}{4}$. Also $f^{-1}(-\frac{1}{4})=\sqrt{1.25/5}=\frac{1}{2}$.

- Directly, $f(-3) = 2 \cdot 12 + 8 - 2 \cdot 6 = 20$. The formula gives $9 + 12 - 1 = 20$, and $f^{-1}(20) = 2 - \sqrt{25} = -3$.
- Numerically, $f^{-1}(f(x)) = x$ for 10^5 random $x < 2$, the function decreases at every point of a fine grid, and the one-sided slopes at $x = 1$ come out as -10 and -2.

Common slips

- Calling $(0, 1)$ a maximum or minimum because $f'(0) = 0$. Each parabola's own turning point is irrelevant if it sits at the edge of the interval where that formula holds.
- Picking the wrong square root in the inverse. The sign of x (or of $x - 2$) on each interval decides it.
- Assuming that a join between two formulas must be a non-differentiable point. At $x = -1$ and $x = 0$ the slopes match.

Beyond STEP

A piecewise-defined function can be continuous without being differentiable, and differentiable without having a continuous second derivative (here f'' jumps at $x = -1$ and $x = 0$). The inverse function theorem says that a strictly monotonic continuous function has a continuous inverse, and that the inverse is differentiable wherever $f' \neq 0$. Notice that f^{-1} has a vertical tangent at $x = 1$, exactly where $f'(0) = 0$. Abbott's *Understanding Analysis* develops these ideas carefully.

WORKED SOLUTION

STEP III 1991, Question 4

Pure • Lines and Planes in 3D

STEP III 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q4

The key idea

Two different lines of sight from the same observer O_1 both pass through O_1, so they span a plane, and the path of P meets both of them. A line that meets two lines through O_1, without passing through O_1 itself, must lie in their plane. The same goes for O_2, so the path is the line where two planes meet. That turns a problem about four skew-looking lines into a cross product and a pair of simultaneous equations.

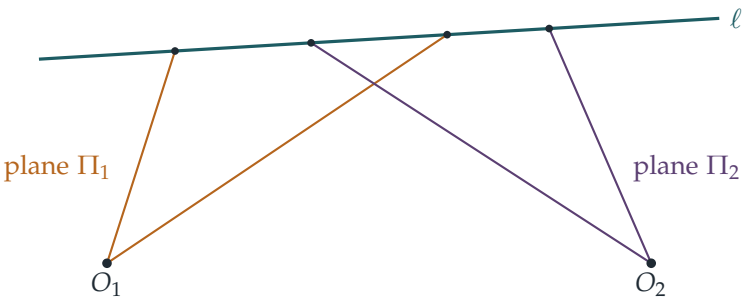
Solution

Write ℓ for the line on which P moves. The four lines of sight have directions

$$\mathbf{d}_1 = (2,0,3), \qquad \mathbf{d}_1' = (0,3,4) \qquad \text{(through } O_1\text{)},$$

$$\mathbf{d}_2 = (-3,1,3), \qquad \mathbf{d}_2' = (-4,2,5) \qquad \text{(through } O_2\text{)}.$$

Each line of sight joins an observer to a position of P, so ℓ meets all four of them.



(A schematic only: the two amber sight lines and ℓ lie in one plane Π_1, the two plum ones and ℓ in another plane Π_2.)

Why ℓ lies in a plane through each observer

First, ℓ cannot pass through O_1. If it did, then whenever $P \neq O_1$ the line of sight from O_1 would be ℓ itself, so there could not be two different lines of sight from O_1. Likewise ℓ does not pass through O_2.

The two sight lines from O_1 are different lines through O_1, so they meet only at O_1, and they span a plane Π_1. The line ℓ meets them at two points, and these are different, since the only common point is O_1 and ℓ misses O_1. Both points lie in Π_1, so the whole of ℓ lies in Π_1. In the same way ℓ lies in the plane Π_2 through O_2 containing $\mathbf{d}_2$ and $\mathbf{d}_2'$.

> **Ask yourself** Couldn't ℓ just cross each sight line at a point, without lying in any particular plane?

For a single line, yes. But here two of the crossing points are on lines through O_1, and two distinct points determine ℓ. Both points are in Π_1, so ℓ has no choice: it is the line through them, which lies in Π_1.

The plane Π_1

A normal to Π_1 is

$$\mathbf{d}_1 \times \mathbf{d}_1' = (2,0,3) \times (0,3,4) = (0-9,\ 0-8,\ 6-0) = (-9,-8,6),$$

and Π_1 passes through the origin, so

$$\Pi_1 : \quad 9x + 8y - 6z = 0.$$

The plane Π_2

A normal to Π_2 is

$$\mathbf{d}_2 \times \mathbf{d}_2' = (-3,1,3) \times (-4,2,5) = (5-6,\ -12+15,\ -6+4) = (-1,3,-2),$$

and Π_2 passes through $O_2(8a,0,0)$, so $-(x-8a) + 3y - 2z = 0$, that is,

$$\Pi_2 : \quad x - 3y + 2z = 8a.$$

(Quick check: $(-3,1,3)\cdot(-1,3,-2) = 3+3-6 = 0$ and $(-4,2,5)\cdot(-1,3,-2) = 4+6-10 = 0$.)

We assume $a \neq 0$, otherwise the two observers coincide. Then O_2 is not in Π_1 (since $9 \cdot 8a \neq 0$) and O_1 is not in Π_2, which matches what we said above.

The path

The normals $(-9,-8,6)$ and $(-1,3,-2)$ are not parallel, so Π_1 and Π_2 meet in a single line, and that line must be ℓ. Its direction is perpendicular to both normals:

$$(-9,-8,6) \times (-1,3,-2) = (16-18,\ -6-18,\ -27-8) = (-2,-24,-35),$$

so we can take the direction $(2, 24, 35)$. For a point on it, put $y = 0$. The equations become $9x = 6z$ and $x + 2z = 8a$, so $x = \frac{2}{3}z$ and $\frac{8}{3}z = 8a$. This gives $z = 3a$ and $x = 2a$. So $(2a, 0, 3a)$ is on ℓ. It is also on the first sight line from O_1, as it should be.

Therefore the path of P is

$$\boxed{\frac{x - 2a}{2} = \frac{y}{24} = \frac{z - 3a}{35}}$$

or, equivalently, the pair of equations $9x + 8y - 6z = 0$ and $x - 3y + 2z = 8a$.

Finally, ℓ really does meet all four sight lines, at points other than the observers. Writing ℓ as $(2a + 2\lambda,\ 24\lambda,\ 3a + 35\lambda)$, it meets them at

$$(2a, 0, 3a), \qquad (0, -24a, -32a), \qquad \left(\tfrac{80a}{37}, \tfrac{72a}{37}, \tfrac{216a}{37}\right), \qquad \left(\tfrac{56a}{25}, \tfrac{72a}{25}, \tfrac{36a}{5}\right)$$

respectively.

> **Ask yourself** The second point has y and z negative for $a > 0$. Is that a problem?

The sight lines are given as whole lines, and $(0, -24a, -32a)$ satisfies $x = 0$, $y/3 = z/4$. We only used the lines, not which way the observer was facing, so the answer stands. If you read "line of sight" as a ray from the observer, this observation lies behind O_1. That is a quirk of the data, not an error in the method.

Checks

- Substitute the point $(2a, 0, 3a)$ and the direction $(2, 24, 35)$ into both plane equations: $9(2) + 8(24) - 6(35) = 18 + 192 - 210 = 0$ and $2 - 72 + 70 = 0$, so the direction lies in both planes.
- With $a = 1$, the point $\left(\frac{80}{37}, \frac{72}{37}, \frac{216}{37}\right)$ equals $O_2 + \frac{72}{37}(-3, 1, 3)$, and $\frac{80/37 - 2}{2} = \frac{3}{37} = \frac{72/37}{24}$. So it is on both the sight line and the path.
- We checked all four intersection points in exact rational arithmetic for several values of a.

Common slips

- Trying to find where pairs of sight lines meet: lines of sight from different observers at different times need not meet at all.
- Forgetting that Π_2 passes through O_2, not the origin, which gives $x - 3y + 2z = 0$ instead of $= 8a$.
- Sign errors in the cross product. Always dot the normal back with both direction vectors.

Beyond STEP

How many lines meet four given lines in space? For four lines in general position the answer is two (counted over the complex numbers; over the reals it can be zero). That is a classic result of enumerative geometry, and the four lines here are special because they pass through two observers in pairs. The same plane-of-sight idea underlies triangulation and the epipolar geometry of stereo vision, where two cameras recover the positions of points in 3D. For the linear algebra of lines, planes and intersections, read Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 1991, Question 5

Pure • Polar Curves and ODEs

STEP III 1991, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q5

The key idea

Forget the polar coordinates for a moment: the equation is just a linear second-order ODE with constant coefficients and θ as the variable. Solve it in the usual way (complementary function plus particular integral), fix the constants from the conditions at $\theta = \frac{\pi}{2}$, and then turn back to geometry. A curve given by $r = f(\theta)$ is a closed loop when r starts and finishes at 0 and stays positive in between, and its area is $\frac{1}{2}\int r^2 \,\mathrm{d}\theta$.

Solution

Solving the differential equation

The auxiliary equation $m^2 + 4 = 0$ has roots $m = \pm 2\mathrm{i}$, so the complementary function is

$$r = A\cos 2\theta + B\sin 2\theta.$$

For a particular integral, $\sin 3\theta$ is not in the complementary function, so we try $r = k\sin 3\theta$. Then $r'' = -9k\sin 3\theta$, and

$$-9k\sin 3\theta + 4k\sin 3\theta = 5\sin 3\theta \quad\Longrightarrow\quad k = -1.$$

So the general solution is

$$r = A\cos 2\theta + B\sin 2\theta - \sin 3\theta, \qquad \frac{\mathrm{d}r}{\mathrm{d}\theta} = -2A\sin 2\theta + 2B\cos 2\theta - 3\cos 3\theta.$$

At $\theta = \frac{\pi}{2}$ we have $\cos 2\theta = -1$, $\sin 2\theta = 0$, $\sin 3\theta = -1$ and $\cos 3\theta = 0$. The two conditions become

$$r = -A + 1 = 1 \quad\Longrightarrow\quad A = 0, \qquad \frac{\mathrm{d}r}{\mathrm{d}\theta} = -2B = -2 \quad\Longrightarrow\quad B = 1.$$

Hence

$$r = \sin 2\theta - \sin 3\theta.$$

The curve is a closed loop

To see where r vanishes and what sign it has, factorise with the identity $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$:

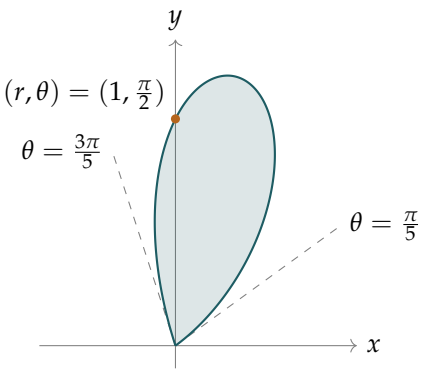
$$r = 2\cos\frac{5\theta}{2}\sin\left(-\frac{\theta}{2}\right) = -2\sin\frac{\theta}{2}\cos\frac{5\theta}{2}.$$

Ask yourself Why does the factorised form settle the "closed loop" question?

Because both factors are easy to read off on the given range. For $\frac{\pi}{5} \le \theta \le \frac{3\pi}{5}$:

- $\frac{\theta}{2}$ lies in $\left[\frac{\pi}{10}, \frac{3\pi}{10}\right]$, so $\sin\frac{\theta}{2} > 0$;
- $\frac{5\theta}{2}$ lies in $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$, so $\cos\frac{5\theta}{2} \le 0$, with equality only at the two ends.

So $r = 0$ at $\theta = \frac{\pi}{5}$ and at $\theta = \frac{3\pi}{5}$, and $r > 0$ for every θ strictly between. The curve therefore leaves the origin along the ray $\theta = \frac{\pi}{5}$, sweeps round with positive r, and returns to the origin along the ray $\theta = \frac{3\pi}{5}$. Each ray $\theta = \text{const}$ in between meets the curve exactly once (at the single point with that $r > 0$), so the curve does not cross itself. It is a single closed loop, bounding the region $0 \le r \le \sin 2\theta - \sin 3\theta$, $\frac{\pi}{5} \le \theta \le \frac{3\pi}{5}$.



The area

The area enclosed is

$$\mathcal{A} = \frac{1}{2}\int_{\pi/5}^{3\pi/5} r^2\,\mathrm{d}\theta = \frac{1}{2}\int_{\pi/5}^{3\pi/5}\left(\sin^2 2\theta - 2\sin 2\theta\sin 3\theta + \sin^2 3\theta\right)\mathrm{d}\theta.$$

Using $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and $2\sin P\sin Q = \cos(P-Q) - \cos(P+Q)$,

$$r^2 = 1 - \tfrac{1}{2}\cos 4\theta - \tfrac{1}{2}\cos 6\theta - \cos\theta + \cos 5\theta.$$

Write $D = \sin\frac{2\pi}{5} - \sin\frac{\pi}{5}$. We integrate term by term over $\left[\frac{\pi}{5}, \frac{3\pi}{5}\right]$, reducing each sine to $\pm\sin\frac{\pi}{5}$ or $\pm\sin\frac{2\pi}{5}$ using $\sin(x+2\pi) = \sin x$ and $\sin(\pi - x) = \sin x$:

$$\begin{aligned}
\int 1\,\mathrm{d}\theta &= \tfrac{2\pi}{5},\\
\int -\tfrac{1}{2}\cos 4\theta\,\mathrm{d}\theta &= -\tfrac{1}{8}\left[\sin\tfrac{12\pi}{5} - \sin\tfrac{4\pi}{5}\right] = -\tfrac{1}{8}\left[\sin\tfrac{2\pi}{5} - \sin\tfrac{\pi}{5}\right] = -\tfrac{1}{8}D,\\
\int -\tfrac{1}{2}\cos 6\theta\,\mathrm{d}\theta &= -\tfrac{1}{12}\left[\sin\tfrac{18\pi}{5} - \sin\tfrac{6\pi}{5}\right] = -\tfrac{1}{12}\left[-\sin\tfrac{2\pi}{5} + \sin\tfrac{\pi}{5}\right] = \tfrac{1}{12}D,\\
\int -\cos\theta\,\mathrm{d}\theta &= -\left[\sin\tfrac{3\pi}{5} - \sin\tfrac{\pi}{5}\right] = -D,\\
\int \cos 5\theta\,\mathrm{d}\theta &= \tfrac{1}{5}\left[\sin 3\pi - \sin\pi\right] = 0.
\end{aligned}$$

(For the third line, $\sin\frac{18\pi}{5} = \sin\left(-\frac{2\pi}{5}\right) = -\sin\frac{2\pi}{5}$ and $\sin\frac{6\pi}{5} = -\sin\frac{\pi}{5}$.) Adding,

$$\int_{\pi/5}^{3\pi/5} r^2\,\mathrm{d}\theta = \frac{2\pi}{5} + \left(-\frac{1}{8} + \frac{1}{12} - 1\right)D = \frac{2\pi}{5} - \frac{25}{24}D,$$

and halving,

$$\mathcal{A} = \frac{\pi}{5} - \frac{25}{48}D = \frac{\pi}{5} + \frac{25}{48}\left[\sin\frac{\pi}{5} - \sin\frac{2\pi}{5}\right],$$

which is the required result.

Checks

- At $\theta = \frac{\pi}{2}$: $r = \sin\pi - \sin\frac{3\pi}{2} = 1$ and $r' = 2\cos\pi - 3\cos\frac{3\pi}{2} = -2$, as required.
- Numerically, $\mathcal{A} \approx 0.6283 - \frac{25}{48}(0.3633) \approx 0.4391$. Simpson's rule on $\frac{1}{2}\int r^2\,\mathrm{d}\theta$, and the shoelace formula on the traced loop, both give 0.43911. Stepping the differential equation numerically from $\theta = \frac{\pi}{2}$ reproduces $r = \sin 2\theta - \sin 3\theta$ and confirms that r returns to 0 at both ends.
- The answer is plausible: the loop reaches $r \approx 1.22$ and spans an angle of $\frac{2\pi}{5}$. A sector of radius 1.22 and that angle has area about 0.93, and the loop fills a bit under half of it.

Common slips

- Getting the particular integral's sign wrong: $(-9+4)k = 5$ gives $k = -1$, not $+1$.
- Forgetting to show $r > 0$ between the endpoints. Two zeros of r alone don't make a loop: r could change sign in between, and then $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ would not be the enclosed area.
- Reducing $\sin\frac{12\pi}{5}$ or $\sin\frac{18\pi}{5}$ carelessly: each is $\pm\sin\frac{2\pi}{5}$, and the sign matters.

Beyond STEP

The method here works because the equation is linear: every solution is a particular one plus a combination of the two independent solutions $\cos 2\theta$ and $\sin 2\theta$, and two initial conditions pick out exactly one solution. That is the existence and uniqueness theorem for linear ODEs, which first-year courses prove and use constantly. An attractive related fact is that $u = 1/r$ satisfies $u'' + u = \text{const}$ for orbits under an inverse-square force, which is how Newton's laws give conic-section orbits. Boyce and DiPrima, *Elementary Differential Equations*, covers the linear theory in detail.

WORKED SOLUTION

STEP III 1991, Question 6

Pure • Hyperbolic Linear Transformations

STEP III 1991, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 7.6, *Hyperbolic Rotations and Relativity*: https://beyondstep.keithmcnulty.org/ch07/hyperbolic-rotations.html

The key idea

The matrix behaves like a rotation matrix, but with hyperbolic functions in place of circular ones, and the identities $\cosh^2 u - \sinh^2 u = 1$ and the addition formulae for cosh and sinh do all the work. Writing a direction of gradient $\tanh v$ as the vector $(\cosh v, \sinh v)$ makes the effect of T on gradients obvious: it simply adds u to the "hyperbolic angle" v. For the angle condition, we express the tangent of the angle between two lines in terms of v_1 and v_2 and see what has to stay fixed.

Solution

Throughout, write $c = \cosh u$ and $s = \sinh u$, so that $c^2 - s^2 = 1$, and $s > 0$ because $u > 0$. The transformation is

$$x' = cx + sy, \qquad y' = sx + cy.$$

The hyperbola

For any point,

$$\begin{aligned} x'^2 - y'^2 &= (cx + sy)^2 - (sx + cy)^2 \\ &= (c^2 - s^2)x^2 + 2csxy - 2csxy - (c^2 - s^2)y^2 \\ &= x^2 - y^2. \end{aligned}$$

So T preserves $x^2 - y^2$, and every point of $x^2 - y^2 = 1$ is sent to a point of $x^2 - y^2 = 1$. The curve is covered completely, not just mapped into itself: the matrix with $-u$ in place of u is the inverse of T (multiply out and use $c^2 - s^2 = 1$), and it also preserves

$x^2 - y^2$. So every point of the hyperbola is the image of a point of the hyperbola. The curve transforms into itself.

(In fact each branch goes to itself: if $x \geq 1$ then $|y| < x$, so $x' = cx + sy > (c - s)x = \mathrm{e}^{-u}x > 0$.)

Invariant lines through the origin

Since T is linear, it sends the origin to itself and the line through the origin in direction $\mathbf{d}$ to the line through the origin in direction $T\mathbf{d}$. So we need $T\mathbf{d}$ parallel to $\mathbf{d}$, that is, $\mathbf{d}$ an eigenvector. Try $\mathbf{d} = (1, m)$:

$$T\begin{pmatrix}1\\m\end{pmatrix} = \begin{pmatrix}c+sm\\s+cm\end{pmatrix},$$

which is parallel to $(1, m)$ exactly when $s + cm = m(c + sm)$, that is, $s(1 - m^2) = 0$. Since $s \neq 0$, $m = \pm 1$. (The vertical direction $(0, 1)$ goes to (s, c), which is not vertical, so it gives nothing new.) The two lines are

$$\boxed{y = x \quad \text{and} \quad y = -x}$$

and indeed $T(1, 1) = \mathrm{e}^u(1, 1)$ and $T(1, -1) = \mathrm{e}^{-u}(1, -1)$. These are the asymptotes of the hyperbola, which fits with the hyperbola mapping to itself.

Gradients: $v' = v + u$

Take a line through a point $\mathbf{p}$ with direction $\mathbf{d}$, so its points are $\mathbf{p} + t\mathbf{d}$. By linearity its image is $T\mathbf{p} + tT\mathbf{d}$: a line through $T\mathbf{p}$ with direction $T\mathbf{d}$. So the position of the line is irrelevant and only the direction matters.

A line of gradient $\tanh v$ has direction $(1, \tanh v)$, or, multiplying by $\cosh v > 0$, direction $(\cosh v, \sinh v)$. Then

$$T\begin{pmatrix}\cosh v\\\sinh v\end{pmatrix} = \begin{pmatrix}\cosh u\cosh v + \sinh u\sinh v\\\sinh u\cosh v + \cosh u\sinh v\end{pmatrix} = \begin{pmatrix}\cosh(u+v)\\\sinh(u+v)\end{pmatrix}$$

by the addition formulae. The image line therefore has gradient $\tanh(v + u)$. Since $\tanh$ is one-to-one, $\tanh v' = \tanh(v + u)$ gives

$$v' = v + u,$$

as required. (Only lines with gradient strictly between -1 and 1 can be written this way, which is why the question uses $\tanh v$.)

Equal angles

If two lines have gradients m_1 and m_2 and $1 + m_1m_2 \neq 0$, the acute angle θ between them satisfies

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right|.$$

Here $|m_1|, |m_2| < 1$, so $1 + m_1 m_2 > 0$: the lines are never perpendicular, $\theta \in [0, \frac{\pi}{2})$, and θ is fixed by $\tan\theta$.

Ask yourself The formula looks like $\tanh(v_1 - v_2)$, but the sign in the denominator is wrong. What does it simplify to instead?

Put $m_i = \tanh v_i$ and multiply the top and bottom by $\cosh v_1 \cosh v_2 > 0$:

$$\tan\theta = \frac{|\sinh v_1 \cosh v_2 - \cosh v_1 \sinh v_2|}{\cosh v_1 \cosh v_2 + \sinh v_1 \sinh v_2} = \frac{|\sinh(v_1 - v_2)|}{\cosh(v_1 + v_2)}.$$

The transformed lines have $v_i' = v_i + u$. Their difference is unchanged, but their sum goes up by $2u$, so the angle θ' between them satisfies

$$\tan\theta' = \frac{|\sinh(v_1 - v_2)|}{\cosh(v_1 + v_2 + 2u)}.$$

For ℓ_1 and ℓ_2 to intersect at all we need $v_1 \neq v_2$, so the numerator is not zero. Hence

$$\theta = \theta' \iff \cosh(v_1 + v_2) = \cosh(v_1 + v_2 + 2u).$$

Since $\cosh a = \cosh b$ exactly when $a = \pm b$, this means either $v_1 + v_2 = v_1 + v_2 + 2u$, which is impossible as $u > 0$, or $v_1 + v_2 = -(v_1 + v_2 + 2u)$. So the necessary and sufficient condition is

$$\boxed{v_1 + v_2 = -u}$$

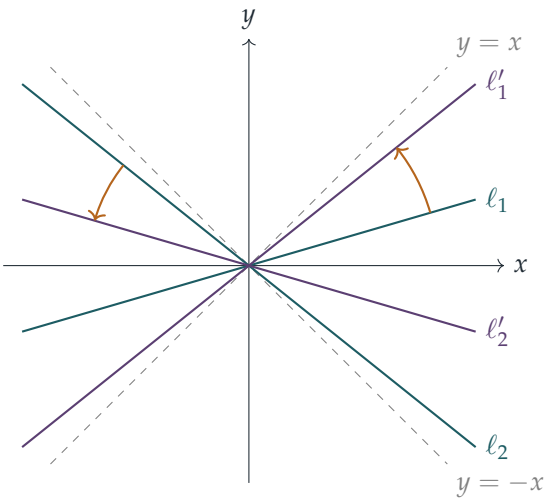
or equivalently $v_1' + v_2' = u$.

The picture when the lines meet at the origin

When $v_1 + v_2 = -u$ we have

$$v_1' = v_1 + u = -v_2, \qquad v_2' = v_2 + u = -v_1.$$

So ℓ_1' has gradient $-\tanh v_2$: it is the reflection of ℓ_2 in the x-axis. Likewise ℓ_2' is the reflection of ℓ_1. If the lines pass through the origin, the transformed pair is just the mirror image of the original pair in the x-axis, and that is why the angle between them is the same. The diagram uses $u = 0.8$, $v_1 = 0.3$, $v_2 = -1.1$: the gradients are about 0.29 and -0.80 for ℓ_1, ℓ_2, and 0.80 and -0.29 for ℓ_1', ℓ_2'. Each line turns towards $y = x$, the invariant line that T stretches.



Teal lines are ℓ_1, ℓ_2, plum lines their images; the amber arrows show $\ell_1 \to \ell_1'$ and $\ell_2 \to \ell_2'$. The plum pair is the reflection of the teal pair in the x-axis.

Checks

- With $u = 0.8$, $v_1 = 0.3$, $v_2 = -1.1$ the angle between ℓ_1 and ℓ_2 and between their images both come to 54.92° numerically. Changing v_2 slightly makes the two angles differ.
- As $u \to 0$, T becomes the identity, and the condition becomes $v_1 + v_2 = 0$: two lines symmetric about the x-axis, which an identity map trivially keeps symmetric.
- $T(1,1) = \mathrm{e}^u(1,1)$ is easy to check by hand, since $\cosh u + \sinh u = \mathrm{e}^u$.

Common slips

- Showing only that the hyperbola maps *into* itself; the inverse transformation shows nothing is missed.
- Writing the angle formula as $\tanh(v_1 - v_2)$; the $+$ sign in $1 + m_1 m_2$ gives $\cosh(v_1 + v_2)$ in the denominator instead, and it is the *sum* that changes under T.
- Forgetting the root $v_1 + v_2 = v_1 + v_2 + 2u$ of $\cosh a = \cosh b$ and not saying why it is impossible.

Beyond STEP

This matrix is a Lorentz boost from special relativity, with x and y playing the roles of time and space (in units where the speed of light is 1). The quantity $x^2 - y^2$ is the invariant interval, $\tanh v$ is a velocity, and v is its *rapidity*. The result $v' = v + u$ says that

rapidities add, which is the relativistic velocity-addition formula in disguise. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops special relativity this way. For the eigenvector view of the invariant lines, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP III 1991, Question 7

Pure • Integration Using Symmetry

STEP III 1991, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q7

The key idea

The integral of $\ln(\sin x)$ can't be found from an antiderivative, but symmetry does all the work. The substitution $x \mapsto \frac{1}{2}\pi - x$ swaps sin and cos, and $x \mapsto \pi - x$ shows that $\sin x$ on $[0, \pi]$ is two copies of its values on $[0, \frac{1}{2}\pi]$. Writing $\sin x \cos x = \frac{1}{2}\sin 2x$ then gives an equation that the unknown integral must satisfy. Part (iii) is an integration by parts that turns $x \cot x$ into $\ln(\sin x)$, and part (ii) deals with the boundary term.

Solution

Throughout, write

$$I = \int_0^{\frac{1}{2}\pi} \ln(\sin x)\,\mathrm{d}x.$$

Part (i)

The sine and cosine integrals agree. Substitute $x = \frac{1}{2}\pi - t$, so $\mathrm{d}x = -\mathrm{d}t$ and $\sin x = \cos t$. The limits $0, \frac{1}{2}\pi$ become $\frac{1}{2}\pi, 0$, and the minus sign reverses them back:

$$I = \int_{\frac{1}{2}\pi}^{0} \ln(\cos t)\,(-\mathrm{d}t) = \int_0^{\frac{1}{2}\pi} \ln(\cos t)\,\mathrm{d}t.$$

Relating to $\sin 2x$**.** Adding the two equal integrals and using $\ln a + \ln b = \ln(ab)$ (valid since $\sin x, \cos x > 0$ on the open interval):

$$\begin{aligned} 2I &= \int_0^{\frac{1}{2}\pi} \ln(\sin x \cos x)\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \ln\left(\tfrac{1}{2}\sin 2x\right)\mathrm{d}x \\ &= \int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x - \tfrac{1}{2}\pi \ln 2. \end{aligned}$$

Halving gives $I = \frac{1}{2}\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x - \frac{1}{4}\pi \ln 2$, which completes the first chain of equalities.

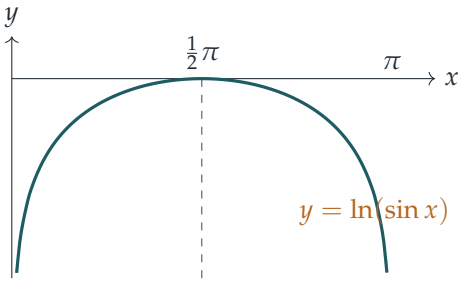
The second chain. With $u = 2x$, $\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$, and the limits become 0 and π:

$$\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x = \tfrac{1}{2}\int_0^{\pi} \ln(\sin u)\,\mathrm{d}u.$$

Now split $[0, \pi]$ at $\frac{1}{2}\pi$. On the second half substitute $u = \pi - v$. Since $\sin(\pi - v) = \sin v$ and the limits $\frac{1}{2}\pi, \pi$ become $\frac{1}{2}\pi, 0$,

$$\int_{\frac{1}{2}\pi}^{\pi} \ln(\sin u)\,\mathrm{d}u = \int_0^{\frac{1}{2}\pi} \ln(\sin v)\,\mathrm{d}v = I.$$

The graph below shows this symmetry: the curve $y = \ln(\sin x)$ is symmetric about $x = \frac{1}{2}\pi$.



So $\int_0^{\pi} \ln(\sin u)\,\mathrm{d}u = 2I$, and therefore

$$\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x = \tfrac{1}{2}\int_0^{\pi} \ln(\sin x)\,\mathrm{d}x = \tfrac{1}{2}(2I) = I,$$

as required.

Evaluating I. Putting the second result into the first gives

$$I = \tfrac{1}{2}I - \tfrac{1}{4}\pi\ln 2 \quad\Longrightarrow\quad I = \int_0^{\frac{1}{2}\pi} \ln(\sin x)\,\mathrm{d}x = -\tfrac{1}{2}\pi\ln 2.$$

This step needs I to be a finite number, which is why the question lets us assume convergence. Without that, "$I = \frac{1}{2}I + c$" would prove nothing.

Part (ii)

Let $x \geq 1$ and put $u = \sqrt{x}$. Then $u \geq 1$, so the given inequality applies:

$$\ln\sqrt{x} < \sqrt{x}, \qquad \text{that is,} \qquad \tfrac{1}{2}\ln x < \sqrt{x}.$$

First limit. For $x \geq 1$ we have $\ln x \geq 0$, and dividing $\ln x < 2\sqrt{x}$ by $x > 0$ gives

$$0 \leq \frac{\ln x}{x} < \frac{2}{\sqrt{x}}.$$

The right-hand side tends to 0 as $x \to \infty$, so by the squeeze (sandwich) principle $\dfrac{\ln x}{x} \to 0$.

Second limit. For $0 < x < 1$ put $x = 1/t$, so $t > 1$ and $t \to \infty$ as $x \to 0^+$. Then

$$x \ln x = \frac{1}{t} \ln \frac{1}{t} = -\frac{\ln t}{t} \to 0$$

by the first limit. So $x \ln x \to 0$ as $x \to 0$ through positive values.

Part (iii)

Note first that the integrand is harmless at 0: $x \cot x = \dfrac{x}{\sin x} \cos x \to 1$. Since $\cot x$ is the derivative of $\ln(\sin x)$ on $(0, \frac{1}{2}\pi)$, we integrate by parts with $u = x$ and $\mathrm{d}v = \cot x \, \mathrm{d}x$, $v = \ln(\sin x)$. The logarithm blows up at 0, so we work on $[\varepsilon, \frac{1}{2}\pi]$ and then let $\varepsilon \to 0^+$:

$$\int_\varepsilon^{\frac{1}{2}\pi} x \cot x \, \mathrm{d}x = \Big[x \ln(\sin x) \Big]_\varepsilon^{\frac{1}{2}\pi} - \int_\varepsilon^{\frac{1}{2}\pi} \ln(\sin x) \, \mathrm{d}x.$$

At the top limit $\ln(\sin \frac{1}{2}\pi) = \ln 1 = 0$. So the boundary term is just $-\varepsilon \ln(\sin \varepsilon)$.

> **Ask yourself** Why does $\varepsilon \ln(\sin \varepsilon)$ tend to 0? Part (ii) is about $x \ln x$, not $x \ln(\sin x)$.

Write $s = \sin \varepsilon$, which tends to 0 through positive values as $\varepsilon \to 0^+$. Then

$$\varepsilon \ln(\sin \varepsilon) = \frac{\varepsilon}{\sin \varepsilon} \cdot s \ln s.$$

The first factor tends to 1 (the standard limit $\sin \varepsilon / \varepsilon \to 1$), and $s \ln s \to 0$ by part (ii). So the boundary term tends to 0.

Letting $\varepsilon \to 0^+$ and using part (i):

$$\int_0^{\frac{1}{2}\pi} x \cot x \, \mathrm{d}x = 0 - I = \tfrac{1}{2}\pi \ln 2.$$

Checks

- Sign: $\ln(\sin x) \le 0$ on the interval, so I must be negative, and $-\frac{1}{2}\pi \ln 2 \approx -1.0888$ is. On the other hand $x \cot x > 0$ on $(0, \frac{1}{2}\pi)$, and our answer is positive.
- Size: $x \cot x$ falls from 1 to 0 across an interval of length $\frac{1}{2}\pi \approx 1.571$, so an integral of about 1.09 is plausible.
- Numerically, midpoint quadrature gives $\int_0^{\frac{1}{2}\pi} x \cot x \, \mathrm{d}x \approx 1.0887930$, matching $\frac{1}{2}\pi \ln 2 = 1.0887930\ldots$, and $\int_0^{\frac{1}{2}\pi} \ln(\sin x) \, \mathrm{d}x \approx -1.08879$.

Common slips

- Forgetting the factor $\frac{1}{2}$ from $\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$ when changing to $u = 2x$, which gives $\int_0^{\frac{1}{2}\pi} \ln(\sin 2x)\,\mathrm{d}x = 2I$ and leads nowhere.
- In part (iii), writing $[x\ln(\sin x)]_0^{\frac{1}{2}\pi} = 0$ without justifying the limit at 0. That limit is exactly what part (ii) is for.

Beyond STEP

Both integrals here are *improper*: their integrands, or the antiderivatives used, are unbounded at an endpoint. They are defined as limits, as we did with ε. A first analysis course makes this precise and gives tests for when such limits exist. These are the tests behind the question's "you may assume" clause. The same symmetry tricks evaluate many more integrals, such as $\int_0^{\pi/2} \ln(\tan x)\,\mathrm{d}x = 0$. Abbott's *Understanding Analysis* covers the theory of limits and integration that underpins all of this.

WORKED SOLUTION

STEP III 1991, Question 8

Pure • Reduction Formulae and Sums

STEP III 1991, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q8

The key idea

Part (i) is a reduction formula, so integration by parts is the natural first move. The leftover integrand contains $\sin x \sin kx$, and the compound-angle formula for $\cos(k-1)x$ turns it back into integrals of the same family. Part (ii) is the real part of a binomial expansion of $(1+\mathrm{e}^{2\mathrm{i}\theta})^m$. Part (iii) then comes from integrating (ii) and unwinding the recurrence from (i) by telescoping.

Solution

Part (i)

Take $k \geq 1$ and integrate by parts with $u = \cos^k x$ and $\mathrm{d}v = \cos kx\,\mathrm{d}x$, so that $v = \frac{1}{k}\sin kx$ and $\mathrm{d}u = -k\cos^{k-1}x\sin x\,\mathrm{d}x$:

$$\begin{aligned} I_k &= \left[\tfrac{1}{k}\cos^k x \sin kx\right]_0^\theta + \int_0^\theta \cos^{k-1}x\,\sin x\,\sin kx\,\mathrm{d}x \\ &= \tfrac{1}{k}\cos^k\theta\sin k\theta + \int_0^\theta \cos^{k-1}x\,\sin x\,\sin kx\,\mathrm{d}x, \end{aligned}$$

since the bracket vanishes at $x = 0$.

Ask yourself How can $\sin x \sin kx$ be rewritten in terms of cosines of multiples of x?

Writing $(k-1)x = kx - x$, the compound-angle formula gives

$$\cos(k-1)x = \cos kx\cos x + \sin kx\sin x,$$

so

$$\sin x\sin kx = \cos(k-1)x - \cos x\cos kx.$$

Therefore

$$\begin{aligned}\int_0^\theta \cos^{k-1} x \,\sin x\, \sin kx \,\mathrm{d}x &= \int_0^\theta \cos^{k-1} x \cos(k-1)x \,\mathrm{d}x \\ &\quad - \int_0^\theta \cos^k x \cos kx \,\mathrm{d}x \\ &= I_{k-1} - I_k.\end{aligned}$$

Substituting back and multiplying by k,

$$kI_k = \cos^k\theta \sin k\theta + kI_{k-1} - kI_k,$$

that is,

$$2kI_k = kI_{k-1} + \cos^k\theta \sin k\theta,$$

as required.

Part (ii)

Let $z = \mathrm{e}^{2\mathrm{i}\theta}$. By the binomial theorem,

$$\begin{aligned}(1+z)^m &= \sum_{r=0}^m \binom{m}{r} z^r \\ &= \sum_{r=0}^m \binom{m}{r} \big(\cos 2r\theta + \mathrm{i}\sin 2r\theta\big).\end{aligned}$$

Now factor out the "half angle":

$$1 + \mathrm{e}^{2\mathrm{i}\theta} = \mathrm{e}^{\mathrm{i}\theta}\big(\mathrm{e}^{-\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}\theta}\big) = 2\cos\theta\,\mathrm{e}^{\mathrm{i}\theta},$$

so

$$(1+z)^m = 2^m \cos^m\theta\,\mathrm{e}^{\mathrm{i}m\theta} = 2^m\cos^m\theta\,(\cos m\theta + \mathrm{i}\sin m\theta).$$

Since $2^m\cos^m\theta$ is real, equating real parts of the two expressions gives

$$\sum_{r=0}^m \binom{m}{r}\cos 2r\theta = 2^m\cos^m\theta\cos m\theta,$$

which is the required identity (the terms $r = 0$ and $r = m$ are 1 and $\cos 2m\theta$).

Part (iii)

Call the first expression S. The right-hand side of (ii) is 2^m times the integrand of I_m, so we integrate (ii) from 0 to θ. The $r = 0$ term integrates to θ, and for $r \geq 1$, $\int_0^\theta \cos 2rx\,\mathrm{d}x = \frac{\sin 2r\theta}{2r}$. Hence

$$\theta + \sum_{r=1}^m \binom{m}{r}\frac{\sin 2r\theta}{2r} = 2^m I_m, \qquad \text{so} \qquad S = 2^m I_m - \theta.$$

Now we evaluate $2^m I_m$ using (i). Dividing the recurrence by $2k$ and multiplying by 2^k gives, for $k \geq 1$,

$$2^k I_k = 2^{k-1} I_{k-1} + \frac{2^{k-1}}{k} \cos^k \theta \sin k\theta.$$

Ask yourself Why multiply by 2^k?

Written as $I_k = \frac{1}{2} I_{k-1} + \ldots$, the recurrence has a factor $\frac{1}{2}$ at each step. Using $J_k = 2^k I_k$ removes it: $J_k - J_{k-1}$ is then a simple explicit term, and the sum telescopes. Adding the equations for $k = 1, 2, \ldots, m$, the intermediate terms cancel:

$$2^m I_m = 2^0 I_0 + \sum_{k=1}^{m} \frac{2^{k-1}}{k} \cos^k \theta \sin k\theta.$$

Finally $I_0 = \int_0^\theta 1 \, \mathrm{d}x = \theta$, so

$$\begin{aligned} S = 2^m I_m - \theta &= \sum_{k=1}^{m} \frac{1}{k} 2^{k-1} \cos^k \theta \sin k\theta \\ &= \cos\theta \sin\theta + \cos^2\theta \sin 2\theta + \cdots + \frac{1}{m} 2^{m-1} \cos^m \theta \sin m\theta, \end{aligned}$$

which is the required result.

Checks

- $m = 1$: the claim is $\frac{1}{2} \sin 2\theta = \cos\theta \sin\theta$, which is the double-angle formula.
- $m = 2$: the left side is $\sin 2\theta + \frac{1}{4} \sin 4\theta$, and the right side is $\cos\theta \sin\theta + \cos^2\theta \sin 2\theta = \frac{1}{2} \sin 2\theta + \frac{1}{2} \sin 2\theta (1 + \cos 2\theta)$, which is the same, since $\sin 2\theta \cos 2\theta = \frac{1}{2} \sin 4\theta$.
- Numerically, with $m = 5$ and $\theta = 0.7$ both expressions equal 1.51963 (to 5 decimal places). The reduction formula also checks against Simpson's-rule values of I_k for random k and θ.

Common slips

- Sign errors in the integration by parts: $\frac{\mathrm{d}}{\mathrm{d}x} \cos^k x = -k \cos^{k-1} x \sin x$, and that minus sign cancels the minus in front of $\int v \, \mathrm{d}u$.
- Forgetting the $r = 0$ term in (ii). It integrates to θ, and it cancels the $I_0 = \theta$ at the bottom of the telescoping sum.

Beyond STEP

Writing $\cos^m\theta\cos m\theta$ as a combination of $\cos 2r\theta$ amounts to computing a Fourier series. Here it is a finite one, and integrating term by term is exactly how Fourier series are handled in a first-year methods course. Reduction formulae of the kind in (i) are also the standard tool for integrals of the form $\int \cos^n x\,\mathrm{d}x$ and for the Wallis product. Needham's *Visual Complex Analysis* gives a geometric picture of the trick $1+\mathrm{e}^{2\mathrm{i}\theta}=2\cos\theta\,\mathrm{e}^{\mathrm{i}\theta}$ used in (ii).

WORKED SOLUTION

STEP III 1991, Question 9

Pure • Ellipse Arc Length

STEP III 1991, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q9

The key idea

E_1 is the ellipse in its centre-based form, and E_2 is the same curve in polar form with the pole at a focus. Turning E_2 into Cartesian form and completing the square shows where its centre and axes are, and matching semi-axes gives k and b. For the integrals, the perimeter is one number, but each parametrisation gives a different-looking arc-length integral for it.

Solution

The axes, k and b

Write (x, y) for the coordinates of E_2 and put $r = \sqrt{x^2+y^2}$, $\theta = \theta_2$. Then E_2 says $r = \dfrac{k}{1+e\cos\theta}$, that is $r + e\,r\cos\theta = k$, so

$$r = k - ex.$$

Squaring (this loses nothing, since $r = k/(1+e\cos\theta) > 0$ on the curve),

$$x^2+y^2 = k^2 - 2kex + e^2x^2 \quad\Longrightarrow\quad (1-e^2)x^2 + 2kex + y^2 = k^2.$$

Completing the square in x:

$$(1-e^2)\Big(x + \frac{ke}{1-e^2}\Big)^2 + y^2 = k^2 + \frac{k^2e^2}{1-e^2} = \frac{k^2}{1-e^2},$$

so

$$\frac{\left(x + \frac{ke}{1-e^2}\right)^2}{k^2/(1-e^2)^2} + \frac{y^2}{k^2/(1-e^2)} = 1.$$

This is an ellipse with centre $\left(-\frac{ke}{1-e^2}, 0\right)$ in the E_2 axes, axes of symmetry parallel to the E_2 axes, and semi-axes

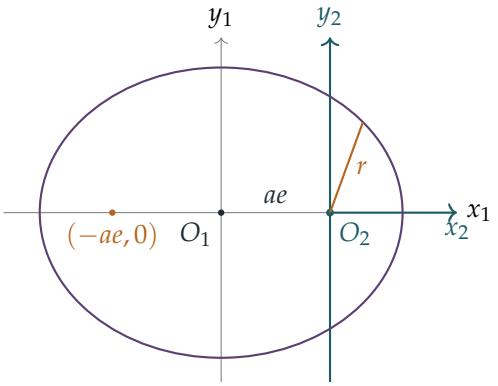
$$A = \frac{k}{1-e^2} \text{ (along the } x\text{-direction)}, \qquad B = \frac{k}{\sqrt{1-e^2}} \text{ (along the } y\text{-direction)}.$$

Since $0 < 1 - e^2 < 1$ we have $A > B$, so the major axis lies along the E_2 x-axis. For E_1, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $a > b$, so its major axis is along the E_1 x-axis. The curves are the same, so the major semi-axes agree and so do the minor ones:

$$a = \frac{k}{1-e^2}, \quad \text{i.e.} \quad k = a(1-e^2), \qquad b^2 = \frac{k^2}{1-e^2} = \frac{a^2(1-e^2)^2}{1-e^2} = a^2(1-e^2),$$

which are the two required results.

The centre, which is the E_1 origin, is at $x = -\frac{ke}{1-e^2} = -ae$ on the E_2 x-axis. So the E_2 axes are the E_1 axes translated a distance ae along the major axis: the E_2 origin is the point $(ae, 0)$ of E_1, which is a focus, since $a^2 - b^2 = a^2e^2$. The two x-axes lie along the same line (the major axis) and the y-axes are parallel; in E_1 coordinates, $x_1 = x_2 + ae$, $y_1 = y_2$. The ellipse is symmetric in both its axes, so we could equally take the E_2 origin at the other focus $(-ae, 0)$ with the x-axis reversed. Either way, the point $\theta_2 = 0$ is the vertex nearest the focus.



The two integrals

We compute the length of the upper half of the ellipse ($y \geq 0$) in two ways.

Using E_1. As θ_1 runs from 0 to π the point runs over the upper half once. With $\dot{x} = -a\sin\theta_1$, $\dot{y} = b\cos\theta_1$,

$$\dot{x}^2 + \dot{y}^2 = a^2\sin^2\theta_1 + b^2\cos^2\theta_1 = a^2 - (a^2 - b^2)\cos^2\theta_1 = a^2\left(1 - e^2\cos^2\theta_1\right),$$

using $a^2 - b^2 = a^2e^2$. So the half-perimeter is

$$\tfrac{1}{2}P = a\int_0^\pi \sqrt{1 - e^2\cos^2\theta}\,\mathrm{d}\theta.$$

Using E_2. Here $y_2 = r\sin\theta_2 \geq 0$ exactly when $0 \leq \theta_2 \leq \pi$, since $r > 0$. So $\theta_2 \in [0, \pi]$ also traces the upper half once. In polar coordinates $\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)^2 = r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2$. With $r = k(1 + e\cos\theta)^{-1}$,

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{ke\sin\theta}{(1 + e\cos\theta)^2},$$

so

$$r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 = \frac{k^2\left[(1+e\cos\theta)^2 + e^2\sin^2\theta\right]}{(1+e\cos\theta)^4} = \frac{k^2\left(1+e^2+2e\cos\theta\right)}{(1+e\cos\theta)^4}.$$

Hence, using $k = a(1-e^2)$,

$$\tfrac{1}{2}P = a(1-e^2)\int_0^{\pi} \frac{\sqrt{1+e^2+2e\cos\theta}}{(1+e\cos\theta)^2}\,\mathrm{d}\theta.$$

Equating the two expressions for $\frac{1}{2}P$ and dividing by $a > 0$ gives the required identity.

Ask yourself Why must both parameters run over $[0, \pi]$, rather than one of them over a different range?

The arc lengths must be of the same piece of curve. θ_1 and θ_2 are different angles (one is the eccentric angle measured from the centre, the other the polar angle measured from the focus), but each of them, on $[0, \pi]$, sweeps the upper half from the right-hand vertex $(a, 0)$ to the left-hand vertex $(-a, 0)$. So both integrals measure the same arc.

The approximation

Since $0 \le e^2\cos^2\theta < 1$, the binomial series applies:

$$\sqrt{1-u} = 1 - \tfrac{1}{2}u - \tfrac{1}{8}u^2 + O(u^3), \qquad u = e^2\cos^2\theta.$$

The $O(u^3)$ term is $O(e^6)$, which we neglect. Using

$$\int_0^{\pi}\cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{2}, \qquad \int_0^{\pi}\cos^4\theta\,\mathrm{d}\theta = \frac{3\pi}{8}$$

(the second from $\cos^4\theta = \frac{3}{8} + \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta$), we get

$$\begin{aligned}\int_0^{\pi}\sqrt{1-e^2\cos^2\theta}\,\mathrm{d}\theta &\approx \pi - \frac{e^2}{2}\cdot\frac{\pi}{2} - \frac{e^4}{8}\cdot\frac{3\pi}{8} = \pi\left(1 - \frac{e^2}{4} - \frac{3e^4}{64}\right)\\ &= \tfrac{1}{64}\pi\left(64 - 16e^2 - 3e^4\right).\end{aligned}$$

By the identity just proved, the second integral has the same value. That completes the question.

Checks

- At $e = 0$ both integrals equal π, and E_2 becomes the circle $r = k$ with $k = a$: correct.
- With $a = 2$, $b = 1$ (so $e^2 = \frac{3}{4}$), a 200 000-sided polygon gives perimeter 9.6884482, matching $2a$ times the first integral.
- Numerically, at $e = 0.2$ both integrals are 3.1099371 and the approximation gives 3.1099411. The difference is about $-0.063e^6$, consistent with the next term of the series, $-\frac{5\pi}{256}e^6 \approx -0.061e^6$.

Common slips

- Putting the E_2 origin at the centre. The polar form with $1 + e\cos\theta$ in the denominator always has its pole at a focus.
- Forgetting the $(\mathrm{d}r/\mathrm{d}\theta)^2$ term in the polar arc length, or integrating over $[0, 2\pi]$ on one side and $[0, \pi]$ on the other.

Beyond STEP

The integral $\int_0^{\pi/2} \sqrt{1 - e^2\sin^2\theta}\,\mathrm{d}\theta$ is the complete elliptic integral of the second kind, $E(e)$, so the perimeter of an ellipse is $4aE(e)$. It cannot be written in elementary functions, which is why series like the one here (and Ramanujan's remarkably accurate approximations) matter. Elliptic integrals lead on to elliptic functions and elliptic curves, a central topic of modern number theory. For more questions in this style, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1991, Question 10

Pure • Newton's Identities

STEP III 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q10

The key idea

The polynomial has only three non-zero coefficients, so almost all of the symmetric functions Σ_m vanish and the given identity (Newton's identity) collapses to a two-term recurrence. Beyond degree n the identity no longer applies, but something even simpler does: every root satisfies the equation itself, so multiplying it by α^k and summing over the roots links S_{n+k} to earlier power sums.

Solution

The power sums up to S_{n-1}

Comparing $x^n - qx^{n-1} + r$ with $\prod_i (x - \alpha_i) = x^n - \Sigma_1 x^{n-1} + \Sigma_2 x^{n-2} - \cdots + (-1)^n \Sigma_n$, we read off

$$\Sigma_1 = q, \qquad \Sigma_2 = \Sigma_3 = \cdots = \Sigma_{n-1} = 0, \qquad (-1)^n \Sigma_n = r.$$

Take $1 \le p \le n-1$ in the given identity. Every term containing $\Sigma_2, \ldots, \Sigma_p$ vanishes, because $p \le n-1$. (When $p = 1$ the identity is just $S_1 - \Sigma_1 = 0$.) What survives is

$$S_p - q\,S_{p-1} = 0 \quad (2 \le p \le n-1), \qquad S_1 = q.$$

So $S_1 = q$ and each power sum is q times the one before. By induction on p,

$$S_p = q^p \qquad \text{for } 1 \le p \le n-1,$$

as required. It is convenient to note $S_0 = n$ too, since each $\alpha_i^0 = 1$.

S_n, S_{n+1} and S_{n+2}

For $p = n$ the identity still applies, and now the last term is present:

$$S_n - q\,S_{n-1} + (-1)^n n\,\Sigma_n = 0.$$

Since $(-1)^n \Sigma_n = r$, this gives

$$S_n = q \cdot q^{n-1} - nr = q^n - nr.$$

Ask yourself The identity we were given stops at $p = n$. How do we get beyond it?

We use the equation itself. Each root satisfies $\alpha_i^n = q\alpha_i^{n-1} - r$. Multiplying by α_i^k (with $k \geq 0$) and summing over i gives

$$S_{n+k} = q\, S_{n+k-1} - r\, S_k \qquad (k \geq 0). \tag{$*$}$$

With $k = 0$ this gives $S_n = q^n - nr$ again, a reassuring cross-check. Now

$$S_{n+1} = q\, S_n - r\, S_1 = q^{n+1} - nqr - qr = q^{n+1} - (n+1)qr,$$
$$S_{n+2} = q\, S_{n+1} - r\, S_2 = q^{n+2} - (n+1)q^2 r - q^2 r = q^{n+2} - (n+2)q^2 r.$$

Here we used $S_1 = q$ and $S_2 = q^2$, which hold because $n \geq 5$ makes $2 \leq n - 1$.

The general formula

The pattern $S_n = q^n - nr$, $S_{n+1} = q^{n+1} - (n+1)qr$, $S_{n+2} = q^{n+2} - (n+2)q^2 r$ suggests

$$S_{n+m} = q^{n+m} - (n+m)\, q^m r \qquad \text{for } 0 \leq m \leq n-1.$$

We prove this by induction on m. The case $m = 0$ is $S_n = q^n - nr$, shown above.

Suppose the formula holds for some $m - 1$, where $1 \leq m \leq n - 1$. By $(*)$ with $k = m$,

$$\begin{aligned} S_{n+m} &= q\, S_{n+m-1} - r\, S_m \\ &= q\big(q^{n+m-1} - (n+m-1)q^{m-1} r\big) - r\, q^m \\ &= q^{n+m} - (n+m)\, q^m r, \end{aligned}$$

where we used $S_m = q^m$, valid because $1 \leq m \leq n - 1$. This is the formula for m. By induction, it holds for all $0 \leq m \leq n - 1$, that is, for $m < n$.

Ask yourself Where exactly does the condition $m < n$ come in?

Only in the step $S_m = q^m$. At $m = n$ we would need S_n, which is $q^n - nr$, not q^n, so the extra $-nr$ feeds in and the pattern breaks. In fact $(*)$ gives $S_{2n} = q^{2n} - 2nq^n r + nr^2$.

Checks

- With $r = 0$ the equation is $x^{n-1}(x - q) = 0$, with roots q and 0 ($n - 1$ times), so every $S_k = q^k$. All our formulae reduce to this.
- With $q = 0$ the roots are the nth roots of $-r$. Then $S_p = 0$ for $1 \leq p \leq n - 1$, and $S_n = n \cdot (-r) = -nr$, matching $q^n - nr$.

- Numerically, we found all roots for random q, r and $5 \leq n \leq 9$ and summed their powers. $S_p = q^p$ and the formula for S_{n+m} $(0 \leq m \leq n-1)$ agreed to about 10^{-12}, while at $m = n$ the formula visibly fails.

Common slips

- Forgetting the factor p on the last term of the identity, which gives $S_n = q^n - r$ instead of $q^n - nr$.
- Getting the sign of Σ_n wrong: the constant term is $(-1)^n\Sigma_n$, so $(-1)^n n\Sigma_n = nr$ whatever the parity of n.
- Applying the given identity with $p > n$, where it is not valid as stated.

Beyond STEP

These are Newton's identities, and $(*)$ shows that the power sums satisfy a linear recurrence whose characteristic polynomial is the original equation. In matrix language, S_k is the trace of A^k, where A is the companion matrix of the polynomial. The fact that any symmetric polynomial in the roots can be written in terms of the Σ_m is the fundamental theorem of symmetric polynomials, which leads on to Galois theory. Allenby, *Rings, Fields and Groups*, is a good place to see this algebra developed.

WORKED SOLUTION

STEP III 1991, Question 11

Mechanics • Rigid Body Equilibrium

STEP III 1991, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

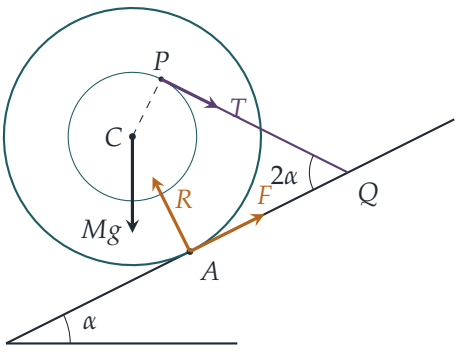
step.maths.org/questions/91-s3-q11

The key idea

The cylinder is a rigid body in equilibrium under four forces: its weight, the normal reaction, friction at the point of contact, and the tension in the string. Two of these, the weight and the normal reaction, pass through the centre of the cylinder, so taking moments about the centre links the friction directly to the tension. Resolving along and perpendicular to the plane then gives the friction and the normal reaction, and the inequality $F \leq \mu R$ does the rest. In the second part the cylinder doesn't know what is holding the string, so the tension is unchanged and we only need to look at the particle.

Solution

We model the cylinder as a uniform rigid body whose centre of mass is its centre C, the string as light and inextensible, and the second body as a particle. Let A be the point of contact with the plane, and let the string's tension be T. Since the string is wound round the groove, the free part PQ is tangent to the groove at P, so its line of action is at perpendicular distance a from C. From the diagram, Q is further up the plane than A, and the string pulls the cylinder from P towards Q: up the plane and in towards it, at angle 2α to the plane.



The dashed radius CP has length a and is perpendicular to the string. We take F as the friction on the cylinder, positive up the plane.

The cylinder: showing $\mu \geq \frac{1}{3}\tan\alpha$

Moments about C. The weight acts at C and the normal reaction acts along AC, so neither has a moment about C. Friction acts at distance $2a$ from C and the tension at perpendicular distance a, turning the cylinder in opposite senses, so

$$F \cdot 2a = T \cdot a, \qquad \text{that is,} \qquad T = 2F.$$

Ask yourself Why do the friction and the tension turn the cylinder in opposite senses?

They are the only two forces with a moment about C, so their moments must cancel. On its own, the string's pull at P would turn the cylinder so that it rolls up the plane. Friction at A pointing up the plane turns it the other way. If we had guessed the direction of friction wrongly, the equations below would simply give $F < 0$, so the sign isn't an extra assumption.

Resolving parallel to the plane (up the plane positive):

$$F + T\cos 2\alpha = Mg\sin\alpha \quad \Longrightarrow \quad F(1 + 2\cos 2\alpha) = Mg\sin\alpha.$$

Resolving perpendicular to the plane: the tension has a component $T\sin 2\alpha$ into the plane, so

$$R = Mg\cos\alpha + T\sin 2\alpha = Mg\cos\alpha + 2F\sin 2\alpha.$$

Substituting $F = Mg\sin\alpha/(1 + 2\cos 2\alpha)$,

$$\begin{aligned} R(1 + 2\cos 2\alpha) &= Mg\big(\cos\alpha + 2\cos 2\alpha\cos\alpha + 2\sin 2\alpha\sin\alpha\big) \\ &= Mg\big(\cos\alpha + 2\cos(2\alpha - \alpha)\big) = 3Mg\cos\alpha. \end{aligned}$$

So

$$F = \frac{Mg\sin\alpha}{1 + 2\cos 2\alpha}, \qquad R = \frac{3Mg\cos\alpha}{1 + 2\cos 2\alpha}.$$

The cylinder stays in contact, so $R > 0$. Since α is acute, $\cos\alpha > 0$, and so $1 + 2\cos 2\alpha > 0$, which means $F > 0$ as well. Dividing,

$$\frac{F}{R} = \frac{\sin\alpha}{3\cos\alpha} = \tfrac{1}{3}\tan\alpha.$$

For equilibrium friction must satisfy $F \leq \mu R$, so $\mu \geq F/R = \frac{1}{3}\tan\alpha$, as required.

The particle: the least coefficient of friction

The string stays where it was, so the cylinder is in exactly the same position and its three equations are unchanged. In particular the tension is still

$$T = 2F = \frac{2Mg\sin\alpha}{1 + 2\cos 2\alpha}.$$

With $\tan\alpha = \frac{1}{2}$ we have $\sin\alpha = \frac{1}{\sqrt{5}}$ and $\cos\alpha = \frac{2}{\sqrt{5}}$, so

$$\cos 2\alpha = \frac{1-\tan^2\alpha}{1+\tan^2\alpha} = \frac{3}{5}, \qquad \sin 2\alpha = \frac{2\tan\alpha}{1+\tan^2\alpha} = \frac{4}{5},$$

and $1 + 2\cos 2\alpha = \frac{11}{5}$. Hence

$$T = \frac{2Mg}{\sqrt{5}} \cdot \frac{5}{11} = \frac{2\sqrt{5}}{11}\,Mg.$$

(The cylinder also needs $\mu \geq \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.)

Now look at the particle at Q. The string pulls it towards P, that is, *down* the plane with component $T\cos 2\alpha$ and *away from* the plane with component $T\sin 2\alpha$. Its weight $3Mg$ also has a component down the plane, so friction F' on the particle must act up the plane. Let the normal reaction on it be R'.

Parallel to the plane:

$$F' = 3Mg\sin\alpha + T\cos 2\alpha = \frac{3Mg}{\sqrt{5}} + \frac{3}{5}\cdot\frac{2\sqrt{5}}{11}Mg = \frac{33\sqrt{5}}{55}Mg + \frac{6\sqrt{5}}{55}Mg = \frac{39\sqrt{5}}{55}Mg.$$

Perpendicular to the plane:

$$R' = 3Mg\cos\alpha - T\sin 2\alpha = \frac{6Mg}{\sqrt{5}} - \frac{4}{5}\cdot\frac{2\sqrt{5}}{11}Mg = \frac{66\sqrt{5}}{55}Mg - \frac{8\sqrt{5}}{55}Mg = \frac{58\sqrt{5}}{55}Mg.$$

This is positive, so the particle does stay on the plane. Equilibrium needs $F' \leq \mu' R'$, where μ' is the coefficient of friction between the particle and the plane, so

$$\mu' \geq \frac{F'}{R'} = \frac{39}{58}.$$

The least value of the coefficient of friction is $\dfrac{39}{58}$.

Checks

- Taking moments about A instead gives $T\,a(1+2\cos 2\alpha) = 2a\,Mg\sin\alpha$, the same tension. (The perpendicular distance from A to the string is $a + 2a\cos 2\alpha$.)
- As $\alpha \to 0$ the plane is horizontal, the tension and friction vanish, and the bound $\frac{1}{3}\tan\alpha \to 0$, as it should.
- A numerical check builds the geometry from coordinates for 10 000 random acute α below 60°, solves the equilibrium equations, and confirms $F/R = \frac{1}{3}\tan\alpha$. For $\tan\alpha = \frac{1}{2}$ it gives $T \approx 0.4066\,Mg$, $F' \approx 1.5856\,Mg$, $R' \approx 2.3580\,Mg$, and $F'/R' \approx 0.67241 = 39/58$.

Common slips

- Forgetting that the tension pushes the cylinder *into* the plane, which increases R. Leaving out $T\sin 2\alpha$ gives the wrong ratio.
- Getting the direction of the string's pull on the particle wrong. It pulls the particle towards P, so it *reduces* the normal reaction on the particle and adds to the down-slope force.
- Recomputing the tension from scratch for the second part. The cylinder's equilibrium hasn't changed, so neither has T.

Beyond STEP

Statics of rigid bodies comes down to two vector equations: the total force is zero and the total moment about *any* point is zero. Choosing that point well, here the centre, through which two unknown forces pass, is the main skill. At university this becomes the study of systems of forces and their equivalent "wrenches". Friction problems also lead on to questions about *how* equilibrium is lost, whether by slipping, rolling or toppling. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and rotation carefully.

WORKED SOLUTION

STEP III 1991, Question 12

Mechanics • Motion on a Curve

STEP III 1991, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q12

The key idea

The particle is tied to the ellipse, so its position is completely described by the single parameter θ. Differentiate the position vector twice and project the acceleration onto the inward unit normal: the $\ddot{\theta}$ terms drop out, because they point along the tangent. For the second part, the particle leaves the surface when the normal reaction falls to zero, and the tube is smooth, so energy conservation gives the speed at every point.

Solution

The normal acceleration

We model the particle as a point moving on a smooth curve, so the only forces are its weight mg downwards and a normal reaction R from the tube.

The position is $\mathbf{r} = (a\cos\theta,\ b\sin\theta)$. Differentiating with respect to time, using the chain rule,

$$\dot{x} = -a\sin\theta\,\dot{\theta}, \qquad \dot{y} = b\cos\theta\,\dot{\theta},$$
$$\ddot{x} = -a\sin\theta\,\ddot{\theta} - a\cos\theta\,\dot{\theta}^2, \qquad \ddot{y} = b\cos\theta\,\ddot{\theta} - b\sin\theta\,\dot{\theta}^2.$$

The tangent direction is $(-a\sin\theta,\ b\cos\theta)$, so a normal direction is $(b\cos\theta,\ a\sin\theta)$ (their dot product is 0). At $\theta = 0$ this is $(b, 0)$, which points out of the ellipse, so the inward unit normal is

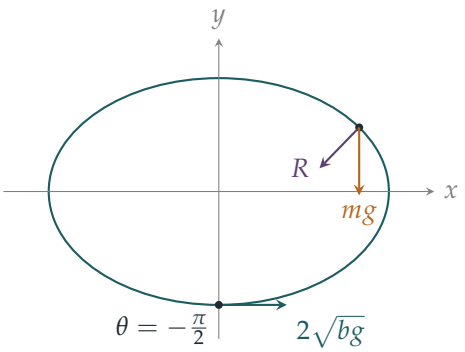
$$\mathbf{n} = -\frac{(b\cos\theta,\ a\sin\theta)}{D}, \qquad D = \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}.$$

The acceleration along the inward normal is therefore

$$\ddot{\mathbf{r}}\cdot\mathbf{n} = -\frac{1}{D}\Big[b\cos\theta\big(-a\sin\theta\,\ddot{\theta} - a\cos\theta\,\dot{\theta}^2\big) + a\sin\theta\big(b\cos\theta\,\ddot{\theta} - b\sin\theta\,\dot{\theta}^2\big)\Big]$$
$$= -\frac{1}{D}\Big[-ab\,\dot{\theta}^2\big(\cos^2\theta + \sin^2\theta\big)\Big] = \frac{ab\,\dot{\theta}^2}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}},$$

since the two $\ddot{\theta}$ terms cancel. This is the required result.

Leaving the surface



The lowest point has $\theta = -\frac{\pi}{2}$, where $y = -b$. The tube is smooth, so the reaction does no work and energy is conserved. With speed v at the point with parameter θ,

$$\tfrac{1}{2}mv^2 + mgb\sin\theta = \tfrac{1}{2}m(4bg) - mgb \quad\Longrightarrow\quad v^2 = 2gb(1-\sin\theta).$$

From the first part, $v^2 = \dot{x}^2 + \dot{y}^2 = D^2\dot{\theta}^2$, so $\dot{\theta}^2 = v^2/D^2$.

Now resolve along the inward normal $\mathbf{n}$. The reaction R acts along $\mathbf{n}$ (the particle is on the inside, so the tube pushes inwards and $R \geq 0$ while there is contact). The weight is $(0, -mg)$, whose component along $\mathbf{n}$ is $(0,-mg)\cdot\mathbf{n} = mga\sin\theta/D$. Newton's second law gives

$$R + \frac{mga\sin\theta}{D} = \frac{m\,ab\,\dot{\theta}^2}{D} = \frac{m\,ab\,v^2}{D^3}, \qquad \text{so} \qquad R = \frac{ma}{D^3}\left(bv^2 - g\sin\theta\, D^2\right).$$

Ask yourself Why is the weight's component $+mga\sin\theta/D$, and not with a minus sign?

In the upper half ($\sin\theta > 0$) the inward normal has a downward component, so gravity helps to pull the particle inwards there, just as at the top of a vertical circle. Both R and this part of the weight supply the centripetal acceleration.

Now put $a = \frac{3}{2}b$ and $v^2 = 2gb(1-\sin\theta)$, and write $s = \sin\theta$. Then

$$D^2 = \tfrac{9}{4}b^2s^2 + b^2(1-s^2) = b^2\left(1 + \tfrac{5}{4}s^2\right),$$

so

$$bv^2 - g\sin\theta\, D^2 = gb^2\left[2(1-s) - s\left(1+\tfrac{5}{4}s^2\right)\right] = -\tfrac{1}{4}gb^2\left(5s^3 + 12s - 8\right).$$

Since $ma/D^3 > 0$, the sign of R is the sign of $-(5s^3 + 12s - 8)$.

The particle leaves the surface where R first becomes zero. The function $p(s) = 5s^3 + 12s - 8$ is strictly increasing, because $p'(s) = 15s^2 + 12 > 0$, and $p(-1) = -25 < 0$, $p(1) = 9 > 0$. So p has exactly one root in $(-1, 1)$. As the particle rises from the lowest point, s increases from -1, and $R > 0$ until s reaches that root. Also $v^2 = 2gb(1-s) > 0$

for all $s < 1$, so the particle is still moving up the tube when it gets there. Hence R first vanishes, and the particle leaves the surface, at the point where

$$5\sin^3\theta + 12\sin\theta - 8 = 0,$$

as required. Numerically, $\sin\theta \approx 0.584$, so it leaves a little above the level of the axis.

Checks

- If $a = b = r$ the normal acceleration becomes $r\dot{\theta}^2 = v^2/r$, the familiar circular result.
- At the lowest point, $R = m\left(\frac{bv^2}{a^2} + g\right)$ with $v^2 = 4bg$, which is positive (the radius of curvature there is a^2/b), and $p(-1) = -25 < 0$ agrees.
- With $b = 1$, $a = 1.5$ and $g = 9.81$ we integrated the equation of motion in θ step by step from the lowest point. The reaction first vanished at $\sin\theta \approx 0.58379$, matching the root 0.58377 of the cubic ($\theta \approx 35.7^\circ$).

Common slips

- Treating θ as the polar angle. The point $(a\cos\theta, b\sin\theta)$ is not at angle θ from the x-axis unless $a = b$, and the normal is not along the radius.
- Using v^2/b or v^2/a for the normal acceleration. On an ellipse the radius of curvature changes from point to point, which is exactly what the first part computes.
- Getting the sign of the weight's normal component wrong, which turns the cubic into $5s^3 - 4s + \ldots$ and gives no sensible root.

Beyond STEP

The first part is a special case of the formula v^2/ρ for normal acceleration, where ρ is the radius of curvature. Indeed, $\rho = D^3/(ab)$ for this ellipse. Splitting acceleration into tangential and normal parts (the Frenet frame) is standard in first-year mechanics and differential geometry. Treating the constraint with one coordinate θ leads naturally to Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers constrained motion and polar and intrinsic coordinates well.

WORKED SOLUTION

STEP III 1991, Question 13

Mechanics • Projectile Impacts with Ceiling

STEP III 1991, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

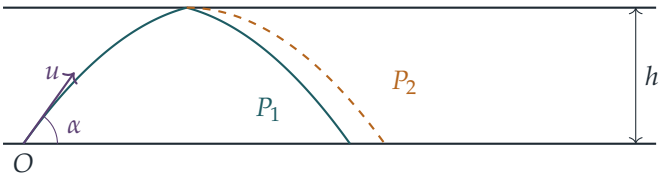
step.maths.org/questions/91-s3-q13

The key idea

The ceiling is smooth, so the horizontal velocity $u\cos\alpha$ never changes and every range is just $u\cos\alpha$ times a flight time. All the work is in the vertical motion, and the energy equation gives the vertical speed at the ceiling straight away. After an elastic impact the particle comes down exactly as it went up, played backwards; after an inelastic one it simply drops from rest.

Solution

Model both particles as point masses moving freely under gravity, with no air resistance. Write $u = \sqrt{8gh}$, $s = \sin\alpha$ and $c = \cos\alpha$. Take x horizontal and y vertically up from O.



Reaching the ceiling

The initial vertical velocity is us, and while rising

$$\dot{y}^2 = u^2s^2 - 2gy = 8gh\,s^2 - 2gy.$$

The particle reaches height h only if $\dot{y}^2 \geq 0$ there, that is $8ghs^2 \geq 2gh$, so $s^2 \geq \frac{1}{4}$. Since α is acute, $s > 0$ and this is $\sin\alpha \geq \frac{1}{2}$, i.e. $\alpha \geq \frac{1}{6}\pi$. (At $\alpha = \frac{1}{6}\pi$ exactly it just grazes the ceiling.) When it does arrive, its upward speed there is

$$w = \sqrt{8ghs^2 - 2gh} = \sqrt{2gh}\,\sqrt{4s^2 - 1}.$$

Range of P_1

On the way up, the vertical velocity falls steadily from us to w at rate g, so the time to the ceiling is

$$t_1 = \frac{us - w}{g}.$$

Ask yourself How long does P_1 take to come back down from the ceiling?

The impact is perfectly elastic and the ceiling is smooth, so the vertical velocity becomes w downwards and the horizontal velocity is unchanged. The vertical motion from the ceiling down is then the vertical motion on the way up run backwards in time (start at height h with speed w towards the floor, accelerate at g). So it arrives at the floor with vertical speed us, taking another time t_1. Directly: the vertical speed grows from w to $\sqrt{w^2 + 2gh} = us$ at rate g.

The total flight time is $2t_1$, so the range is

$$\begin{aligned} R_1 &= uc \cdot \frac{2(us - w)}{g} = \frac{2\sqrt{8gh}\,c}{g}\left(2\sqrt{2gh}\,s - \sqrt{2gh}\sqrt{4s^2 - 1}\right) \\ &= \frac{2\sqrt{8gh}\sqrt{2gh}}{g}\,c\left(2s - \sqrt{4s^2 - 1}\right) = 8h\cos\alpha\left(2\sin\alpha - \sqrt{4\sin^2\alpha - 1}\right), \end{aligned}$$

using $us = \sqrt{8gh}\,s = 2\sqrt{2gh}\,s$ and $\sqrt{8gh}\sqrt{2gh} = 4gh$. This is the required range.

Range of P_2 and the difference D

P_2 follows the same path up to the ceiling, taking time t_1. The impact is perfectly inelastic, so its vertical velocity becomes zero; the ceiling is smooth, so the horizontal velocity is still uc. It then falls a height h from rest, taking time $\sqrt{2h/g}$. Hence

$$R_2 = uc\left(t_1 + \sqrt{\tfrac{2h}{g}}\right), \qquad R_2 - R_1 = uc\left(\sqrt{\tfrac{2h}{g}} - t_1\right).$$

Now

$$t_1 = \frac{\sqrt{2gh}\,(2s - \sqrt{4s^2 - 1})}{g} = \sqrt{\frac{2h}{g}}\,(2s - \sqrt{4s^2 - 1})$$

and $uc\sqrt{2h/g} = \sqrt{8gh}\sqrt{2h/g}\,c = 4hc$, so

$$R_2 - R_1 = 4h\cos\alpha\left(1 - 2\sin\alpha + \sqrt{4\sin^2\alpha - 1}\right).$$

This is positive for $\frac{1}{6}\pi < \alpha < \frac{1}{2}\pi$: for $s > \frac{1}{2}$ we have $(2s-1)^2 = 4s^2 - 1 - 2(2s-1) < 4s^2 - 1$, so $2s - 1 < \sqrt{4s^2 - 1}$. So P_2 always lands further away (the elastic bounce throws P_1 down faster), and

$$D = 4h\cos\alpha\left(1 - 2\sin\alpha + \sqrt{4\sin^2\alpha - 1}\right).$$

Maximising D

Let $f(\alpha) = D/(4h) = c(1-2s) + cr$, where $r = \sqrt{4s^2-1}$, on $\frac{1}{6}\pi \le \alpha < \frac{1}{2}\pi$. Note $f(\frac{1}{6}\pi) = 0$, $f \to 0$ as $\alpha \to \frac{1}{2}\pi$ (since $c \to 0$), and $f > 0$ in between. Differentiating, with $\frac{dr}{d\alpha} = \frac{4sc}{r}$,

$$f'(\alpha) = -s(1-2s) - 2c^2 - sr + \frac{4sc^2}{r}.$$

Putting $c^2 = 1 - s^2$ and multiplying by $r > 0$ (for $\alpha > \frac{1}{6}\pi$):

$$\begin{aligned} rf'(\alpha) &= (4s^2 - s - 2)\,r - sr^2 + 4s(1-s^2) \\ &= (4s^2 - s - 2)\,r - s(8s^2 - 5). \end{aligned}$$

At $\alpha = \frac{1}{4}\pi$, $s = \frac{1}{\sqrt{2}}$ and $r = 1$, giving $(2 - \frac{1}{\sqrt{2}} - 2) - \frac{1}{\sqrt{2}}(4-5) = 0$. So $\alpha = \frac{1}{4}\pi$ is a stationary point.

> **Ask yourself** How do we know it's the maximum, not some other stationary point?

At any stationary point, $(4s^2 - s - 2)r = s(8s^2 - 5)$. Squaring and using $r^2 = 4s^2 - 1$, expanding and factorising gives

$$(4s^2 - s - 2)^2(4s^2 - 1) - s^2(8s^2 - 5)^2 = -2(2s^2 - 1)(8s^3 - s^2 - 2s - 2),$$

so every stationary point has $s^2 = \frac{1}{2}$ or $q(s) = 8s^3 - s^2 - 2s - 2 = 0$. On $[\frac{1}{2}, 1]$, $q'(s) = 24s^2 - 2s - 2 > 0$, so q is increasing; since $q(0.8) = -0.144 < 0 < 0.0985 = q(0.82)$, its only root there lies in $(0.8, 0.82)$. But for such s,

$$4s^2 - s - 2 < 4(0.82)^2 - 0.8 - 2 < 0, \qquad 8s^2 - 5 > 8(0.8)^2 - 5 > 0,$$

so the left side $(4s^2 - s - 2)r$ is negative and the right side $s(8s^2 - 5)$ is positive: this root came from squaring and is not a stationary point. Hence $\alpha = \frac{1}{4}\pi$ is the only stationary point in $(\frac{1}{6}\pi, \frac{1}{2}\pi)$. Since f is continuous, positive inside the interval and zero (or tending to zero) at both ends, it has a maximum inside, which must be at this stationary point. So D is greatest when $\alpha = \frac{1}{4}\pi$, with

$$D_{\max} = 4h \cdot \tfrac{1}{\sqrt{2}}(1 - \sqrt{2} + 1) = 4h(\sqrt{2} - 1).$$

Checks

- At $\alpha = \frac{1}{6}\pi$ the particle just touches the ceiling, so the two particles should agree: indeed $D = 0$, and $R_1 = 8h \cdot \frac{\sqrt{3}}{2} = 4\sqrt{3}\,h$ is the usual range $u^2 \sin 2\alpha / g$.
- With $h = 1.7\,\text{m}$, stepping both motions numerically at $\alpha = 1$ gives $R_1 \approx 2.420\,\text{m}$ and $D \approx 2.464\,\text{m}$, matching the formulae. A fine search over α puts the maximum of $D/(4h)$ at $\alpha = 0.78540 \approx \frac{1}{4}\pi$ with value $0.41421 \approx \sqrt{2} - 1$.

Common slips

- Solving $h = ust - \frac{1}{2}gt^2$ for t_1 and taking the larger root, which is the time the particle would pass height h on the way down if there were no ceiling.
- Writing $D = R_1 - R_2$, which is negative: the inelastic particle lands further away.
- Stopping at $f'(\frac{1}{4}\pi) = 0$ without showing that this stationary point is a maximum.

Beyond STEP

The time-reversal argument for P_1 is a first glimpse of a deep fact: Newton's laws with conservative forces look the same when time runs backwards, and a perfectly elastic impact preserves that symmetry, while an inelastic one (which loses energy) breaks it. Coefficients of restitution between 0 and 1 would interpolate between the two ranges found here. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and energy conservation in this spirit.

WORKED SOLUTION

STEP III 1991, Question 14

Mechanics • Circular Motion with Springs

STEP III 1991, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

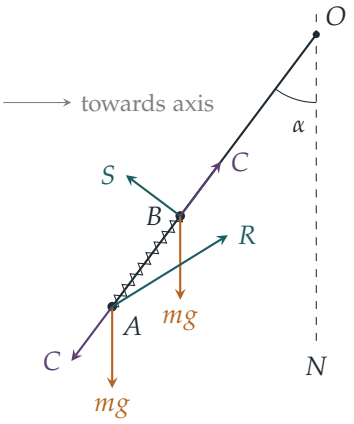
step.maths.org/questions/91-s3-q14

The key idea

Everything moves in a horizontal circle about ON at the same angular speed, so each mass has a known acceleration $\omega^2 r$ towards the axis and nothing else. For the ring, which sits on a *smooth* rod, the cleanest equation is the one along the rod, because the unknown normal reaction drops out of it. Then the spring force links the ring to the particle at A, where the direction of the rod's reaction pins down k.

Solution

Model. The rod is light and smooth, A and B are particles of mass m, and the spring lies along the rod between B and A. Write $\alpha = \sin^{-1}\frac{3}{5}$, so $\sin\alpha = \frac{3}{5}$, $\cos\alpha = \frac{4}{5}$, and $\omega^2 = \frac{5g}{2a}$. Let $OB = x$; then $AB = 2a - x$.



Here C is the thrust in the spring (we shall find that the spring is *compressed*, so it pushes B towards O and pushes A away from O), S is the normal reaction of the smooth rod on B (perpendicular to the rod, in the vertical plane, sense to be found), and R is

the reaction of the rod on A. Since ω is constant and B is at rest relative to the rod, there is no acceleration across the vertical plane, so all the forces lie in that plane.

The length OB

B moves in a horizontal circle of radius $x\sin\alpha$, so its acceleration is $\omega^2 x\sin\alpha$ horizontally towards ON. Resolve along the rod in the direction $O\to A$ (down and away from the axis). The acceleration component in this direction is $-\omega^2 x\sin\alpha\cdot\sin\alpha$, since the acceleration points towards the axis and the rod makes angle α with the vertical.

It is simplest to let the spring force on B be a *tension* T (pulling B towards A) and let the sign sort itself out. By Hooke's law,

$$T=\frac{kmg\left((2a-x)-a\right)}{a}=\frac{kmg(a-x)}{a}.$$

The reaction S has no component along the rod because the rod is smooth. So, along OA,

$$mg\cos\alpha+\frac{kmg(a-x)}{a}=-m\omega^2 x\sin^2\alpha.$$

With $\cos\alpha=\frac{4}{5}$, $\sin^2\alpha=\frac{9}{25}$ and $\omega^2=\frac{5g}{2a}$, the right-hand side is $-\frac{9gx}{10a}\,m$. Dividing by mg and multiplying by $10a$:

$$\begin{aligned}8a+10k(a-x)&=-9x\\(10k+8)a&=(10k-9)x,\end{aligned}$$

so

$$OB=x=\frac{(10k+8)a}{10k-9},$$

as required. Notice that $x>a$ (we need $10k>9$ for a positive answer), so $AB=2a-x<a$: the spring is compressed, with thrust

$$C=-T=\frac{kmg(x-a)}{a}=\frac{17k\,mg}{10k-9},$$

using $x-a=\dfrac{17a}{10k-9}$.

Finding k

> **Ask yourself** Why isn't the reaction at A perpendicular to the rod, like the one at B?

The particle at A is *attached* to the rod, not threaded on it, so the rod can push or pull it in any direction. Smoothness only restricts the force on the sliding ring. So we write R in components: R_h horizontally towards the axis and R_v vertically upwards.

A moves in a circle of radius $2a\sin\alpha=\frac{6a}{5}$, so its acceleration is $\omega^2\cdot\frac{6a}{5}=3g$ towards the axis. The thrust C acts on A along OA away from O, so it has components $C\sin\alpha$

away from the axis and $C\cos\alpha$ downwards. Hence

$$\text{horizontal (towards axis):}\quad R_h - \tfrac{3}{5}C = 3mg,$$
$$\text{vertical:}\quad R_v - mg - \tfrac{4}{5}C = 0.$$

Put $C = c\,mg$. The reaction makes angle $\tan^{-1}\frac{13}{21}$ with the horizontal, so

$$\frac{R_v}{R_h} = \frac{1+\frac{4}{5}c}{3+\frac{3}{5}c} = \frac{13}{21} \implies 21 + \tfrac{84}{5}c = 39 + \tfrac{39}{5}c \implies 9c = 18, \quad c = 2.$$

(Both R_h and R_v are positive for $c > 0$, so the reaction really does point upwards and inwards and there is no sign ambiguity.) Then

$$\frac{17k}{10k-9} = 2 \implies 17k = 20k - 18 \implies k = 6.$$

So $k = 6$, giving $OB = \frac{68a}{51} = \frac{4}{3}a$, which is safely between a and $2a$, and a thrust $C = 2mg$. (The reaction on A is then $R_h = \frac{21}{5}mg$, $R_v = \frac{13}{5}mg$.)

The reaction on the ring

Ask yourself Which direction should we resolve in to get S with the least work?

Perpendicular to the rod: the spring force then drops out. Take the unit direction perpendicular to OA in the vertical plane pointing *upwards and away from the axis*; it makes angle α with the horizontal, so it has components $\cos\alpha$ outwards and $\sin\alpha$ upwards. Let S be the component of the reaction on B in this direction. The weight contributes $-mg\sin\alpha$, and the acceleration ($\omega^2 x\sin\alpha$ towards the axis) contributes $-\omega^2 x\sin\alpha\cos\alpha$. So

$$S - mg\sin\alpha = -m\omega^2 x\sin\alpha\cos\alpha.$$

With $x = \frac{4}{3}a$:

$$m\omega^2 x\sin\alpha\cos\alpha = m\cdot\frac{5g}{2a}\cdot\frac{4a}{3}\cdot\frac{12}{25} = \frac{8}{5}mg, \qquad \text{so} \qquad S = \frac{3}{5}mg - \frac{8}{5}mg = -mg.$$

The reaction between the rod and the ring therefore has magnitude

$$\boxed{mg},$$

and the minus sign tells us it acts perpendicular to the rod, *downwards and towards the axis*.

Checks

- Full vector check on B (horizontal components towards the axis, vertical upwards), with $C = 2mg$ acting along AO and $S = mg$ as found. Horizontal: $\frac{3}{5}(2mg) + \frac{4}{5}(mg) = 2mg$, and $m\omega^2 x\sin\alpha = m\cdot\frac{5g}{2a}\cdot\frac{4a}{3}\cdot\frac{3}{5} = 2mg$. Vertical: $\frac{4}{5}(2mg) - \frac{3}{5}mg - mg = 0$. Both balance.

- As $k \to \infty$ the spring becomes rigid and $OB \to a$, its natural length, as it should.
- Numerically, with $m = g = a = 1$, we checked the OB formula exactly for 200 random values of k, found the angle $\tan^{-1}\frac{13}{21}$ only at $k = 6$, and got $|S| = 1.0$. Stepping B's motion along the rod from a small displacement keeps it near $x \approx 1.3333$.

Common slips

- Assuming the spring is stretched and then being puzzled by $x > a$. Let the sign of the tension come out of the algebra.
- Taking the reaction at A to be perpendicular to the rod. Only the sliding ring has that restriction.
- Using radius x instead of $x\sin\alpha$ for the circular motion.

Beyond STEP

In a frame rotating with the rod, B feels a "centrifugal" force $m\omega^2 r$ outwards, and the along-rod equation becomes an ordinary equilibrium problem. Taking that viewpoint further gives effective potentials, the Coriolis force (which here is balanced by the rod's reaction as soon as B moves along the rod), and the stability of the equilibrium: is $x = \frac{4}{3}a$ stable? Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames and rotating systems at exactly this level.

WORKED SOLUTION

STEP III 1991, Question 15

Probability and Statistics • Counting Pairs in Samples

STEP III 1991, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q15

The key idea

Every set of k cards is equally likely, so each probability is a count of hands divided by $\binom{2n}{k}$. To count hands with a given pair structure, build them in stages. First choose which sorts appear as complete pairs. Then choose which other sorts appear once. Finally choose which of the two cards of each single sort we hold. Once that count is in place, every part of the question is a special case of one formula.

Solution

Think of the $2n$ cards as all distinguishable (the two cards of a sort might be numbered 1 and 2). Drawing k cards without replacement then gives each of the $\binom{2n}{k}$ subsets of size k the same probability. The hand contains a *pair* of a sort when both cards of that sort are drawn.

Four cards, no pairs

A hand with no pair contains four different sorts, one card of each. There are $\binom{n}{4}$ ways to choose the sorts and 2 ways to choose the card of each sort, which gives $2^4\binom{n}{4}$ hands. Since

$$\binom{2n}{4} = \frac{2n(2n-1)(2n-2)(2n-3)}{24} = \frac{4n(n-1)(2n-1)(2n-3)}{24},$$

we get

$$\mathrm{P}(\text{no pairs}) = \frac{16 \cdot n(n-1)(n-2)(n-3)/24}{4n(n-1)(2n-1)(2n-3)/24} = \frac{4(n-2)(n-3)}{(2n-1)(2n-3)},$$

as required. (Here $n \geq 4$ makes sure four different sorts exist.)

Four cards, exactly one pair

Choose the sort that appears as a pair (n ways). The other two cards must come from two different sorts out of the remaining $n-1$, which can be chosen in $\binom{n-1}{2}$ ways, with 2 choices of card for each. The number of hands is

$$n \cdot \frac{(n-1)(n-2)}{2} \cdot 4 = 2n(n-1)(n-2),$$

so

$$\mathrm{P}(\text{exactly one pair}) = \frac{24 \cdot 2n(n-1)(n-2)}{4n(n-1)(2n-1)(2n-3)} = \frac{12(n-2)}{(2n-1)(2n-3)}.$$

k cards, exactly r pairs

Suppose the hand has exactly r pairs. They use $2r$ cards. The remaining $k-2r$ cards all belong to different sorts, none of which is one of the r paired sorts, since otherwise we would have more pairs. So we count:

- the paired sorts: $\binom{n}{r}$ ways;
- the $k-2r$ sorts appearing once, chosen from the other $n-r$ sorts: $\binom{n-r}{k-2r}$ ways;
- which of the two cards we hold of each single sort: 2^{k-2r} ways.

Each hand with exactly r pairs arises from exactly one such sequence of choices. Therefore

$$\boxed{\mathrm{P}(\text{exactly } r \text{ pairs}) = \frac{2^{k-2r}\binom{n}{r}\binom{n-r}{k-2r}}{\binom{2n}{k}}}$$

? **Ask yourself** Is this ever zero, and does it agree with the four-card answers?

The single sorts need $k-2r \le n-r$, that is, $r \ge k-n$. If $k > n$, any k cards must contain at least $k-n$ pairs, and $\binom{n-r}{k-2r} = 0$ handles this automatically. With $k=4$, the case $r=0$ gives $2^4\binom{n}{4}/\binom{2n}{4}$ and $r=1$ gives $2^2\, n\binom{n-1}{2}/\binom{2n}{4}$, which are exactly the two counts above.

k even, all cards in pairs

Put $k = 2m$. The hand consists of m pairs exactly when it is the union of m complete sorts. There are $\binom{n}{m}$ such hands (this is the general formula with $r = m$). So

$$\mathrm{P} = \frac{\binom{n}{m}}{\binom{2n}{2m}} = \frac{n!}{m!\,(n-m)!} \cdot \frac{(2m)!\,(2n-2m)!}{(2n)!}.$$

To simplify, pair up the even and odd factors of a factorial:

$$(2j)! = \big(2 \cdot 4 \cdots 2j\big)\big(1 \cdot 3 \cdots (2j-1)\big) = 2^j\, j!\,\big(1 \cdot 3 \cdots (2j-1)\big).$$

Write $O_j = 1 \cdot 3 \cdots (2j-1)$, with $O_0 = 1$. Then $(2j)!/j! = 2^j O_j$, so

$$\mathrm{P} = \frac{2^m O_m \cdot 2^{n-m} O_{n-m}}{2^n O_n} = \frac{O_m\, O_{n-m}}{O_n}.$$

Dividing O_n by O_{n-m} leaves the odd numbers from $2n-2m+1 = 2n-k+1$ up to $2n-1$, so

$$\mathrm{P}(\tfrac{1}{2}k \text{ pairs}) = \frac{1 \times 3 \times 5 \times \cdots \times (k-1)}{(2n-1)(2n-3)\cdots(2n-k+1)},$$

as required.

There's also a neat direct argument. Shuffle the whole pack and let the hand be the top k cards. Look at the top card. Its partner is equally likely to be in any of the other $2n-1$ positions, and $k-1$ of those are in the hand, so the partner is in the hand with probability $\frac{k-1}{2n-1}$. If it is, remove that pair. What remains is a random arrangement of $2n-2$ cards with the hand being the top $k-2$ of them. Repeating the argument gives $\frac{k-3}{2n-3}$, and so on down to $\frac{1}{2n-k+1}$. Multiplying these conditional probabilities gives the same product.

Checks

- For four cards the three cases are 0, 1 and 2 pairs, and 2 pairs has probability $\binom{n}{2}/\binom{2n}{4} = \frac{3}{(2n-1)(2n-3)}$. Indeed

  $$4(n-2)(n-3) + 12(n-2) + 3 = 4n^2 - 8n + 3 = (2n-1)(2n-3),$$

  so the three probabilities add to 1.
- With $n = 5$ and four cards the probabilities are $\frac{8}{21}$, $\frac{4}{7}$ and $\frac{1}{21}$. A computer enumeration of all hands for $4 \le n \le 8$ and every k from 3 to $2n-1$ agreed with the general formula and with the all-pairs formula.
- With $k = 2$ the all-pairs formula gives $\frac{1}{2n-1}$, the chance that the second card matches the first.

Common slips

- Choosing the single cards from all $2n-2r$ remaining cards with $\binom{2n-2r}{k-2r}$. That allows two of them to be of the same sort, which creates extra pairs.
- Forgetting the factor 2^{k-2r}. Each sort that appears once could be either of its two cards.

Beyond STEP

The number of hands with exactly r pairs is a typical "choose the structure, then fill it in" count, and the product $1 \cdot 3 \cdots (2m-1)$ (the double factorial $(2m-1)!!$) counts the

ways to split $2m$ objects into pairs. Double factorials turn up again as the even moments of the standard normal distribution. The same matching problem, of how many pairs survive in a random sample, appears in the classical "lost socks" and birthday-type problems. Grimmett and Welsh, *Probability: An Introduction*, gives a good grounding in this kind of counting.

WORKED SOLUTION

STEP III 1991, Question 16

Probability and Statistics • Joint and Marginal Distributions

STEP III 1991, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q16

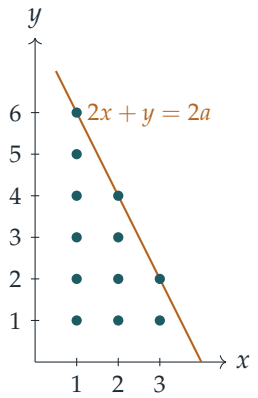
The key idea

Everything here is a finite sum over a triangle of lattice points. Draw the region first: for each x we need to know which y are allowed, and for each y which x are allowed. Once the limits are right, each marginal is the sum of an arithmetic progression. The only subtlety is that the largest allowed x for a given y is $\lfloor (2a-y)/2 \rfloor$, which depends on whether y is even or odd.

Solution

The region

We need $x \geq 1$, $y \geq 1$ and $2x + y \leq 2a$. Since $y \geq 1$, we need $2x \leq 2a - 1$, so $x \leq a - 1$. For a fixed x with $1 \leq x \leq a - 1$, the allowed values are $y = 1, 2, \ldots, 2a - 2x$. Likewise $y \leq 2a - 2$, because $x \geq 1$.



The figure shows the case $a = 4$: column x contains $2a - 2x$ points.

The marginal distribution of X and the value of c

Fix x and write $n = 2a - 2x$. Summing the arithmetic progression $2x+1, \ldots, 2x+n$,

$$\begin{aligned} \mathrm{P}(X = x) &= c \sum_{y=1}^{n} (2x + y) = c\left(2xn + \frac{n(n+1)}{2}\right) = c\,\frac{n}{2}\,(4x + n + 1) \\ &= c\,(a - x)(2a + 2x + 1), \qquad x = 1, 2, \ldots, a-1, \end{aligned}$$

and $\mathrm{P}(X = x) = 0$ otherwise.

The probabilities must add to 1. Put $k = a - x$, which runs from 1 to $a - 1$; then $2a + 2x + 1 = 4a + 1 - 2k$ and

$$\begin{aligned} \sum_{k=1}^{a-1} k(4a + 1 - 2k) &= (4a+1)\frac{(a-1)a}{2} - 2 \cdot \frac{(a-1)a(2a-1)}{6} \\ &= a(a-1)\left(\frac{4a+1}{2} - \frac{2a-1}{3}\right) = \frac{a(a-1)(8a+5)}{6}. \end{aligned}$$

Setting c times this equal to 1 gives

$$c = \frac{6}{a(a-1)(8a+5)},$$

as required.

The marginal distribution of Y

Fix y with $1 \le y \le 2a - 2$. The allowed x are $1, 2, \ldots, N$, where N is the largest integer with $2N \le 2a - y$. For N of these terms,

$$\sum_{x=1}^{N} (2x + y) = N(N+1) + yN = N(N + 1 + y).$$

y **even.** Then $2a - y$ is even and $N = \frac{1}{2}(2a - y)$, so $N + 1 + y = \frac{1}{2}(2a + 2 + y)$ and

$$\mathrm{P}(Y = y) = c \cdot \frac{(2a - y)(2a + 2 + y)}{4} = \frac{3(2a - y)(2a + 2 + y)}{2a(a-1)(8a+5)},$$

using the value of c. This is the required result.

? **Ask yourself** Why can't we use $N = \frac{1}{2}(2a - y)$ when y is odd?

When y is odd, $\frac{1}{2}(2a - y)$ is not an integer. The largest integer x with $2x \le 2a - y$ is $N = \frac{1}{2}(2a - y - 1)$: rounding down drops the half.

y **odd.** With $N = \frac{1}{2}(2a - 1 - y)$ we get $N + 1 + y = \frac{1}{2}(2a + 1 + y)$, so

$$\mathrm{P}(Y = y) = c \cdot \frac{(2a - 1 - y)(2a + 1 + y)}{4} = \frac{3(2a - 1 - y)(2a + 1 + y)}{2a(a-1)(8a+5)}.$$

(The two cases can be written as $3\big((2a+1)^2 - (y+1)^2\big)$ and $3\big(4a^2 - (y+1)^2\big)$ over the same denominator.)

The expectation of Y

Summing $y\,\mathrm{P}(Y=y)$ means handling even and odd y separately. It is cleaner to go back to the joint distribution and sum down each column. For fixed x, with $n = 2a-2x$ as before,

$$\sum_{y=1}^{n} y(2x+y) = 2x\cdot\frac{n(n+1)}{2} + \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)}{6}\big(6x+2n+1\big).$$

Again put $k = a-x$, so $n = 2k$ and $6x+2n+1 = 6a-2k+1$. The column sum becomes

$$\frac{2k(2k+1)(6a+1-2k)}{6} = \frac{k(2k+1)(6a+1-2k)}{3}.$$

Expanding, $k(2k+1)(6a+1-2k) = (6a+1)(2k^2+k) - 4k^3 - 2k^2$. Using the standard sums up to $a-1$,

$$\sum_{k=1}^{a-1}(2k^2+k) = \frac{a(a-1)(2a-1)}{3} + \frac{a(a-1)}{2} = \frac{a(a-1)(4a+1)}{6},$$

$$\sum_{k=1}^{a-1} k^3 = \frac{a^2(a-1)^2}{4}, \qquad \sum_{k=1}^{a-1} k^2 = \frac{a(a-1)(2a-1)}{6}.$$

So

$$\begin{aligned}
&\sum_{k=1}^{a-1} k(2k+1)(6a+1-2k)\\
&\quad= a(a-1)\left(\frac{(6a+1)(4a+1)}{6} - a(a-1) - \frac{2a-1}{3}\right)\\
&\quad= \frac{a(a-1)}{6}\big(24a^2+10a+1-6a^2+6a-4a+2\big)\\
&\quad= \frac{a(a-1)(6a^2+4a+1)}{2}.
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathrm{E}(Y) &= \frac{c}{3}\cdot\frac{a(a-1)(6a^2+4a+1)}{2} = \frac{6}{a(a-1)(8a+5)}\cdot\frac{a(a-1)(6a^2+4a+1)}{6}\\
&= \frac{6a^2+4a+1}{8a+5}.
\end{aligned}$$

Checks

- For $a=2$ there are only two points, $(1,1)$ and $(1,2)$, with weights $3c$ and $4c$, so $c=\frac{1}{7}$. This agrees with $6/(2\cdot 1\cdot 21)$. Then $\mathrm{E}(Y) = \frac{3+8}{7} = \frac{11}{7}$, and the formula gives $\frac{33}{21} = \frac{11}{7}$.
- The marginal formulas for X and for Y (both parities), and $\mathrm{E}(Y)$, were checked by exact enumeration with fractions for every a from 2 to 29. For example, $a=3$ gives $\mathrm{E}(Y) = \frac{67}{29}$.

- For large a, $\mathrm{E}(Y) \approx \frac{3}{4}a$. The weight $2x + y$ favours points near the slanted edge, so Y sits well above the $\frac{2}{3}a$ you would get with equal weights on the triangle.

Common slips

- Letting x run up to a: the point $x = a$ would force $y \leq 0$.
- Using $N = \frac{1}{2}(2a - y)$ for odd y, which isn't an integer.
- Trying to find $\mathrm{E}(Y)$ from the two-case marginal. It works, but summing over columns of the joint distribution avoids the parity split altogether.

Beyond STEP

Getting a marginal from a joint distribution means summing out the other variable, and the hard part is always the limits of the region. With continuous variables the same task becomes choosing the limits in a double integral, and swapping the order of summation (as we did for $\mathrm{E}(Y)$) becomes swapping the order of integration. Grimmett and Welsh, *Probability: An Introduction*, covers joint and marginal distributions for both discrete and continuous variables.

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